

Quantitative quantum soundness for all multipartite compiled nonlocal games

Xiangling Xu 许湘灵, Team PhIQuS, Inria Saclay

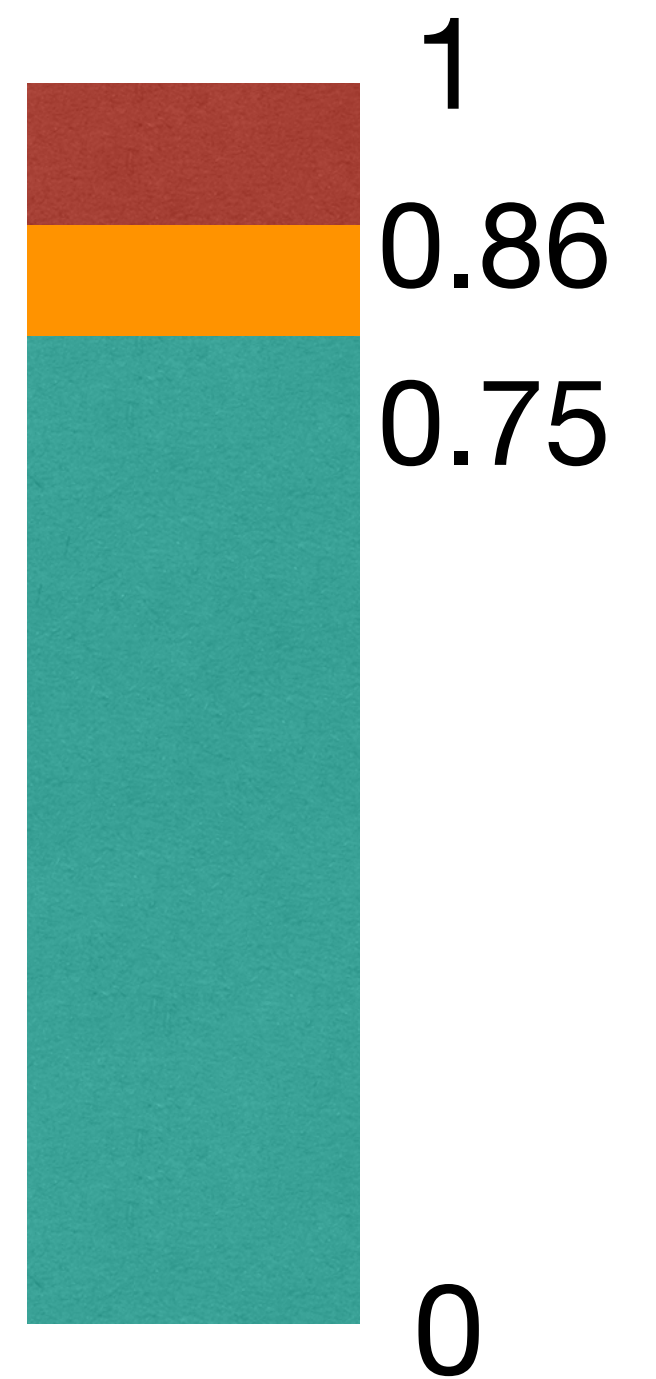
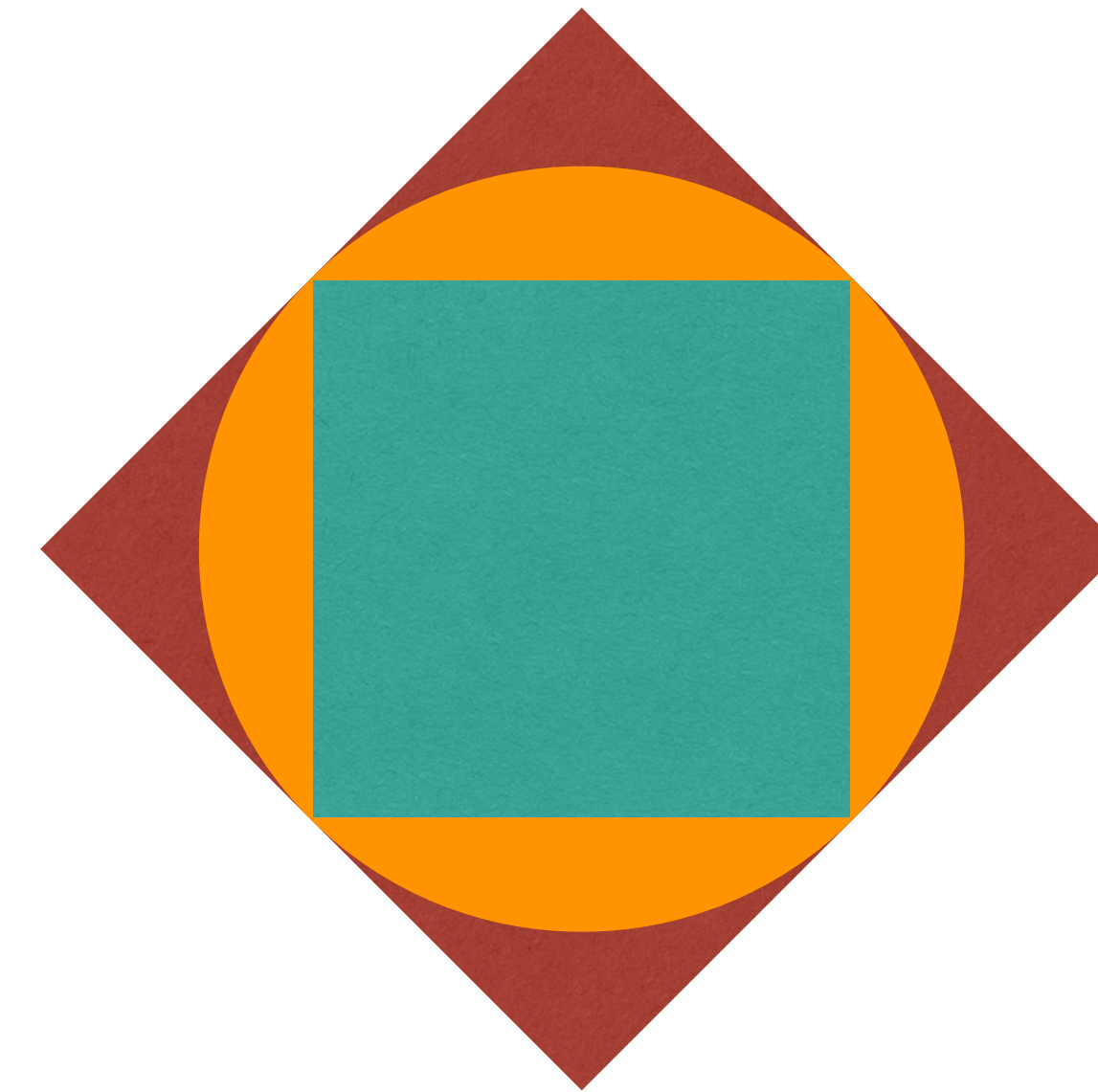
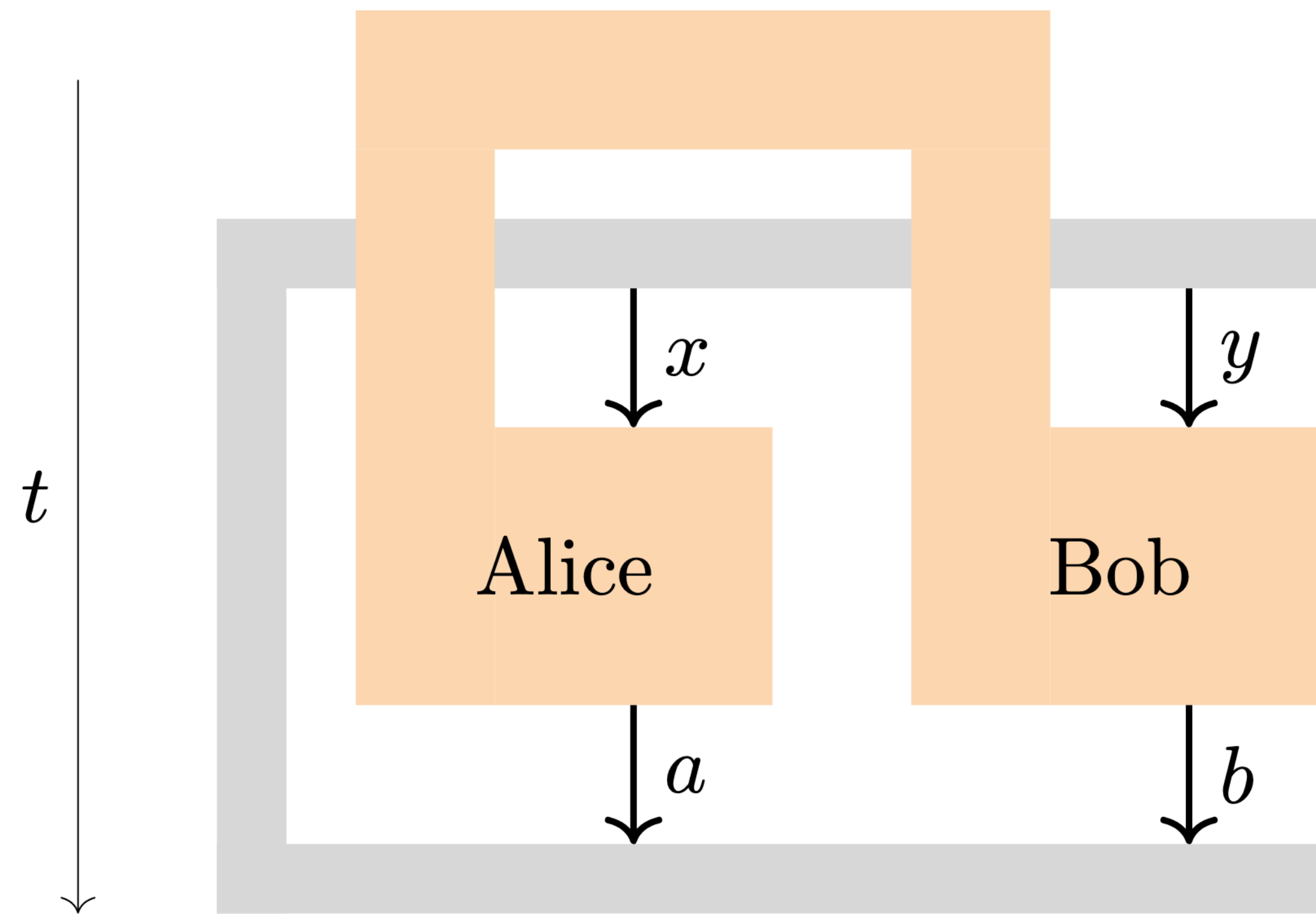
13/11/2025, GdR TeQ, Grenoble



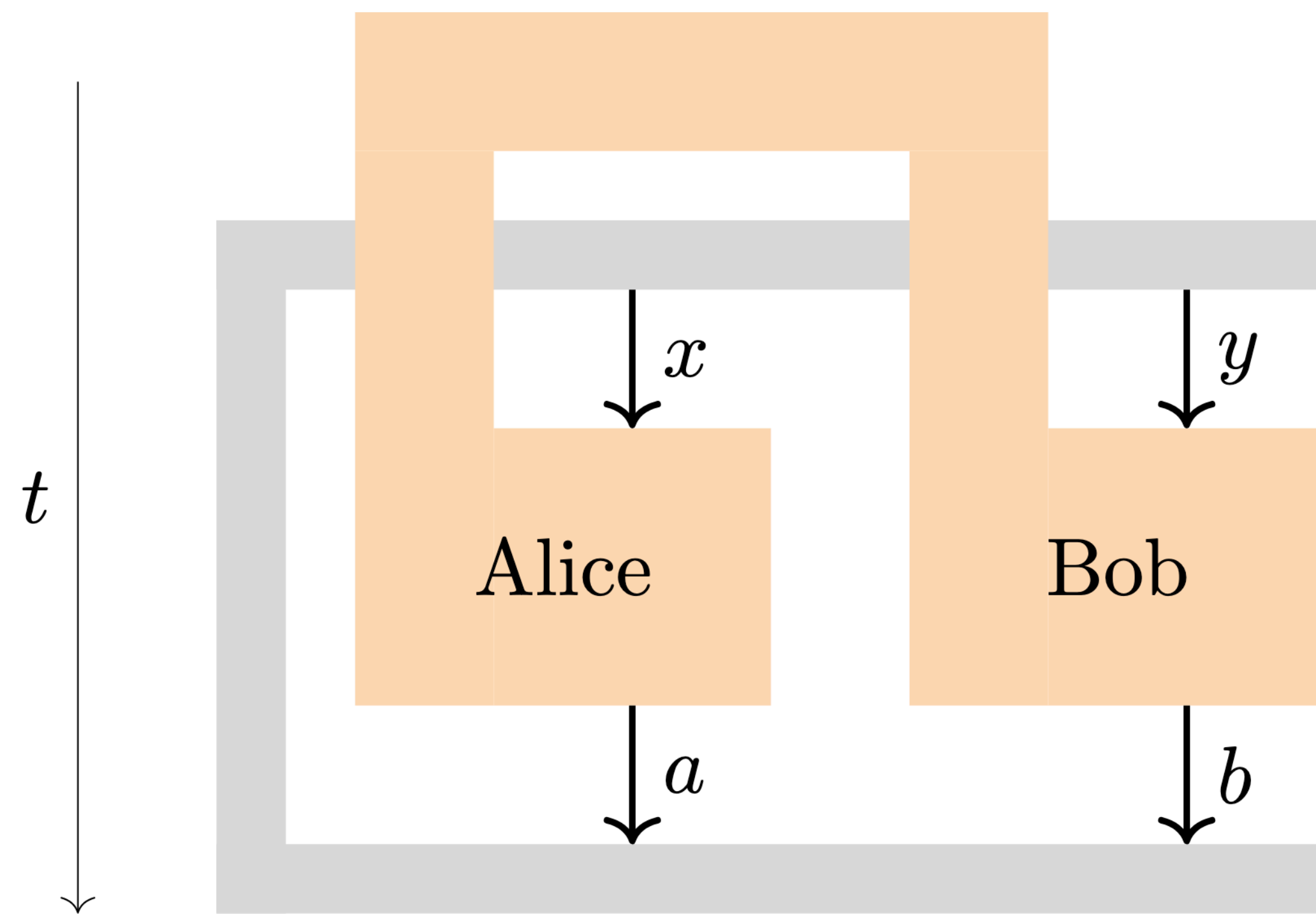
3rd colloquium GdR TeQ
Quantum Technologies

Based on three works, with Matilde Baroni, Siniša Janković, Igor Klep, Dominik Leichtle, Connor Paddock, Marc-Olivier Renou, Simon Schmidt, Ivan Šupić, Lucas Tendick, and Yuming Zhao

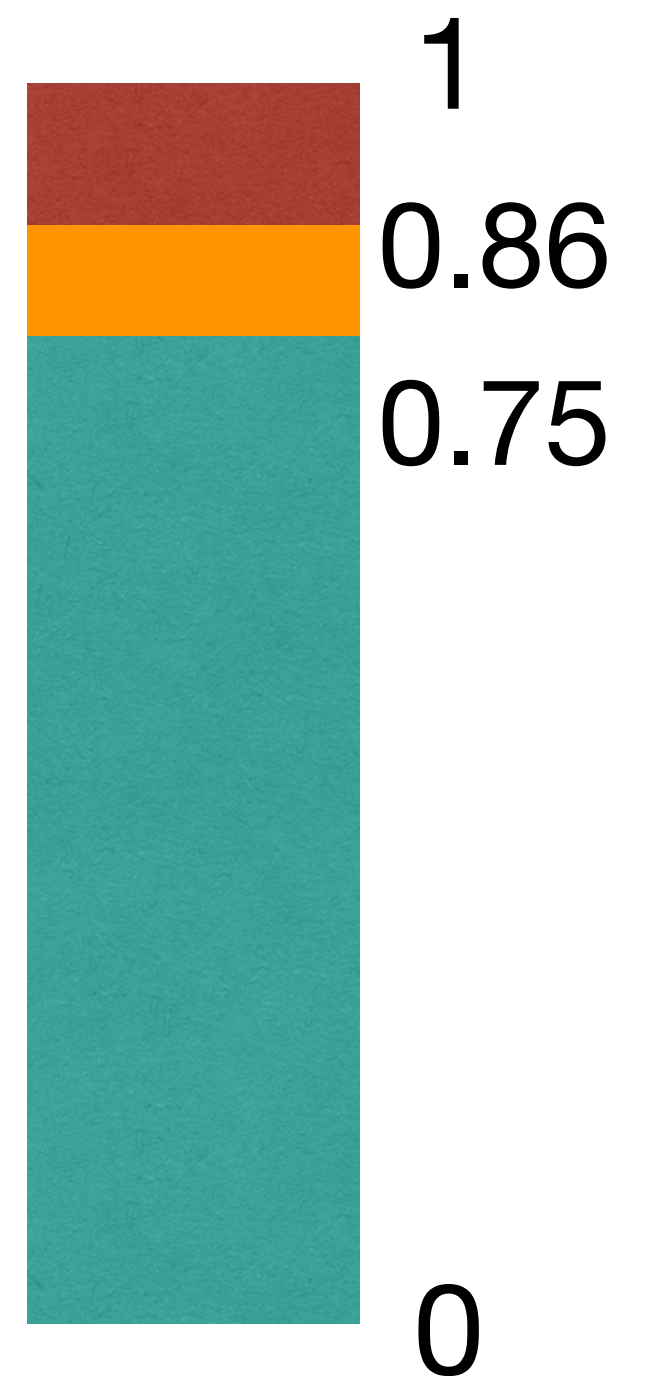
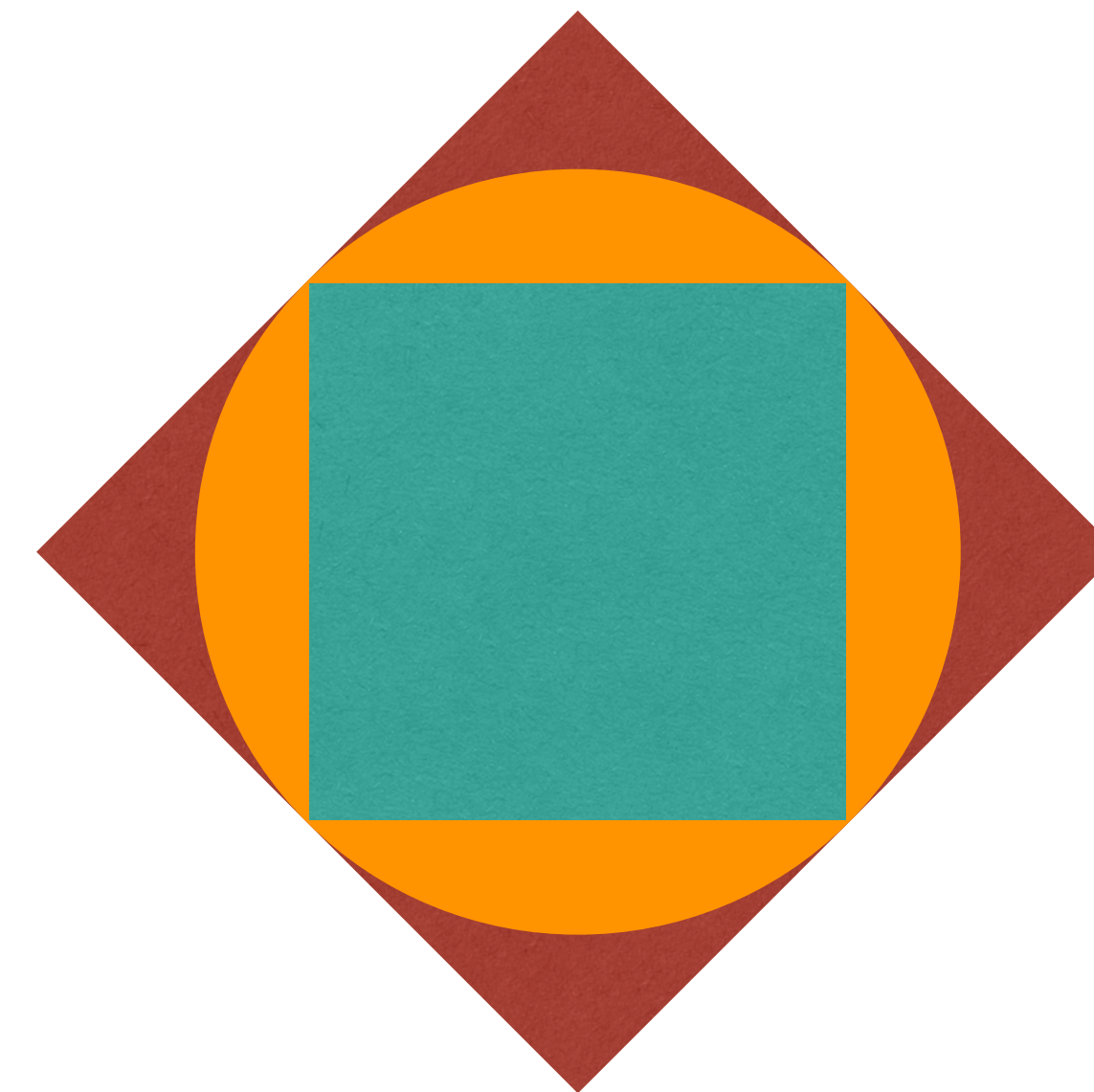
Device-independent non-locality 101



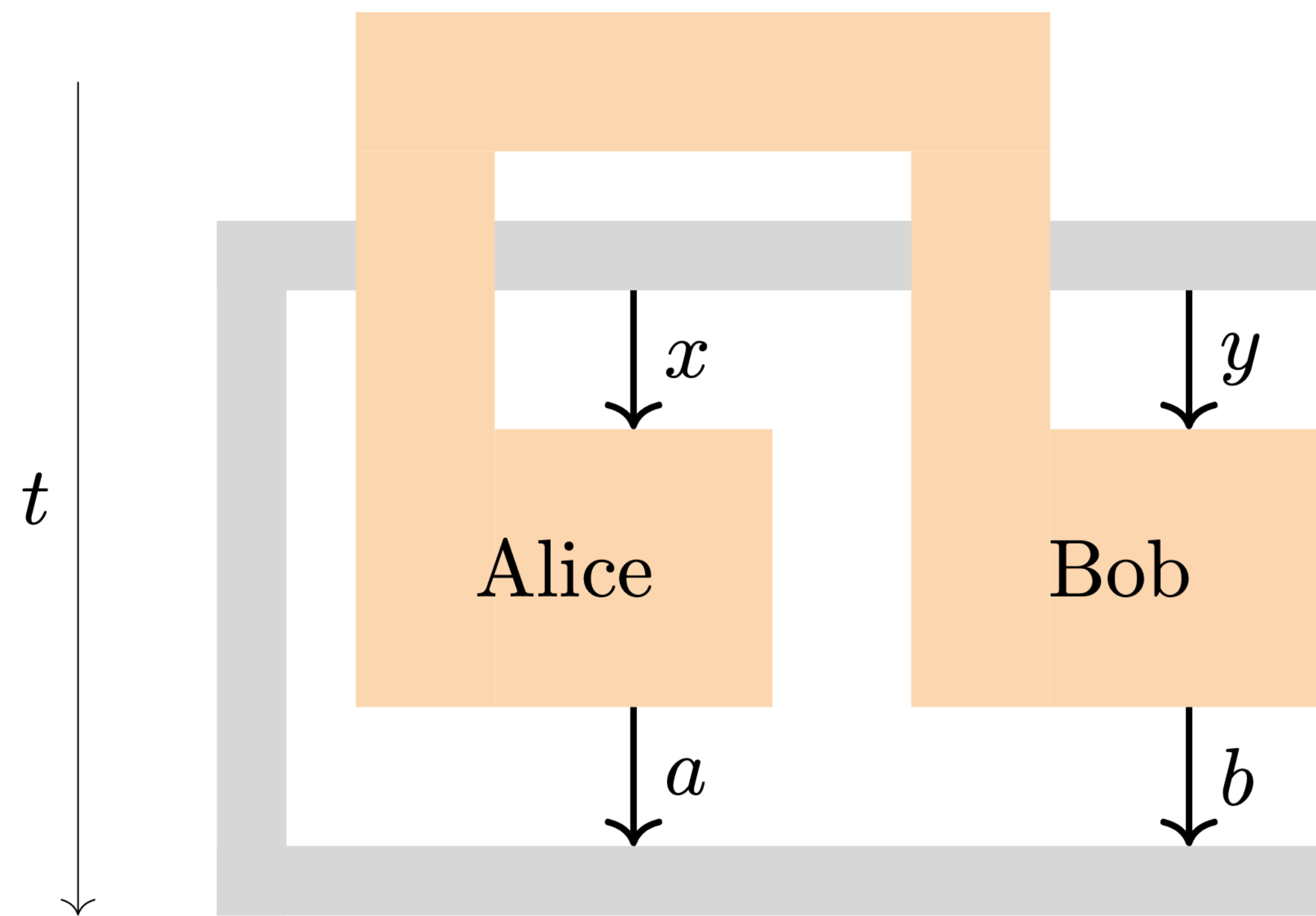
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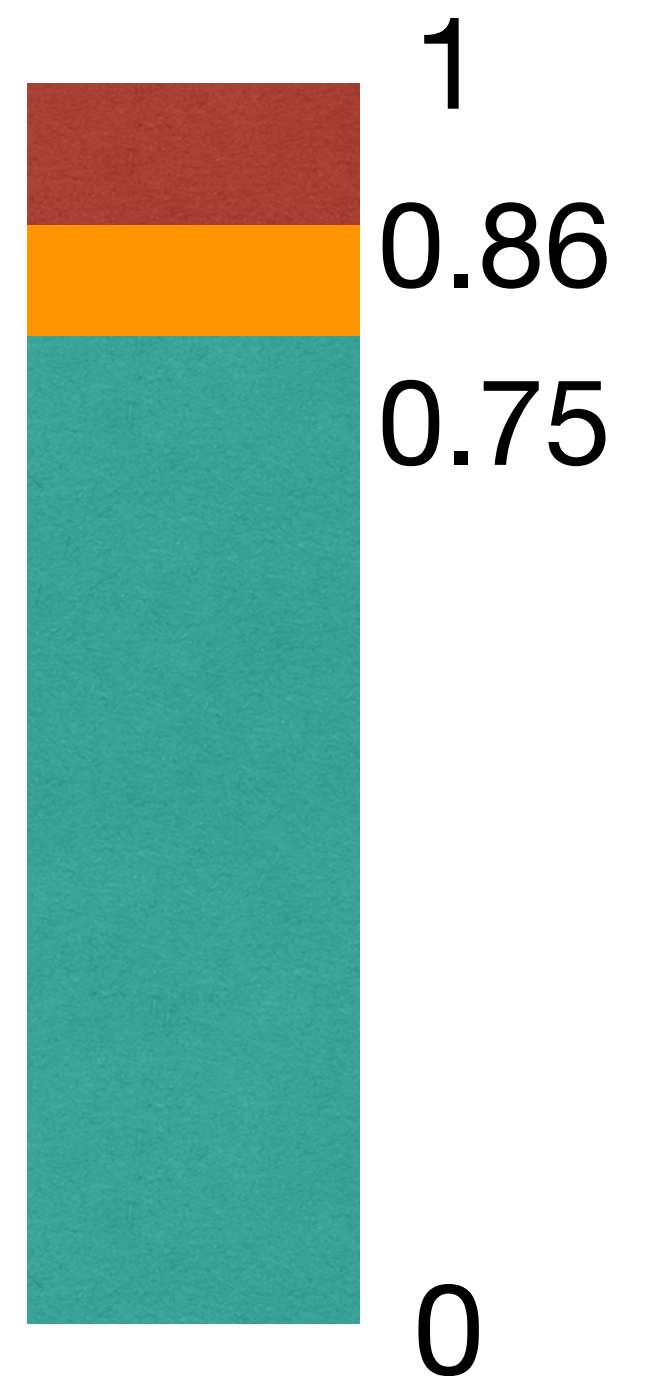
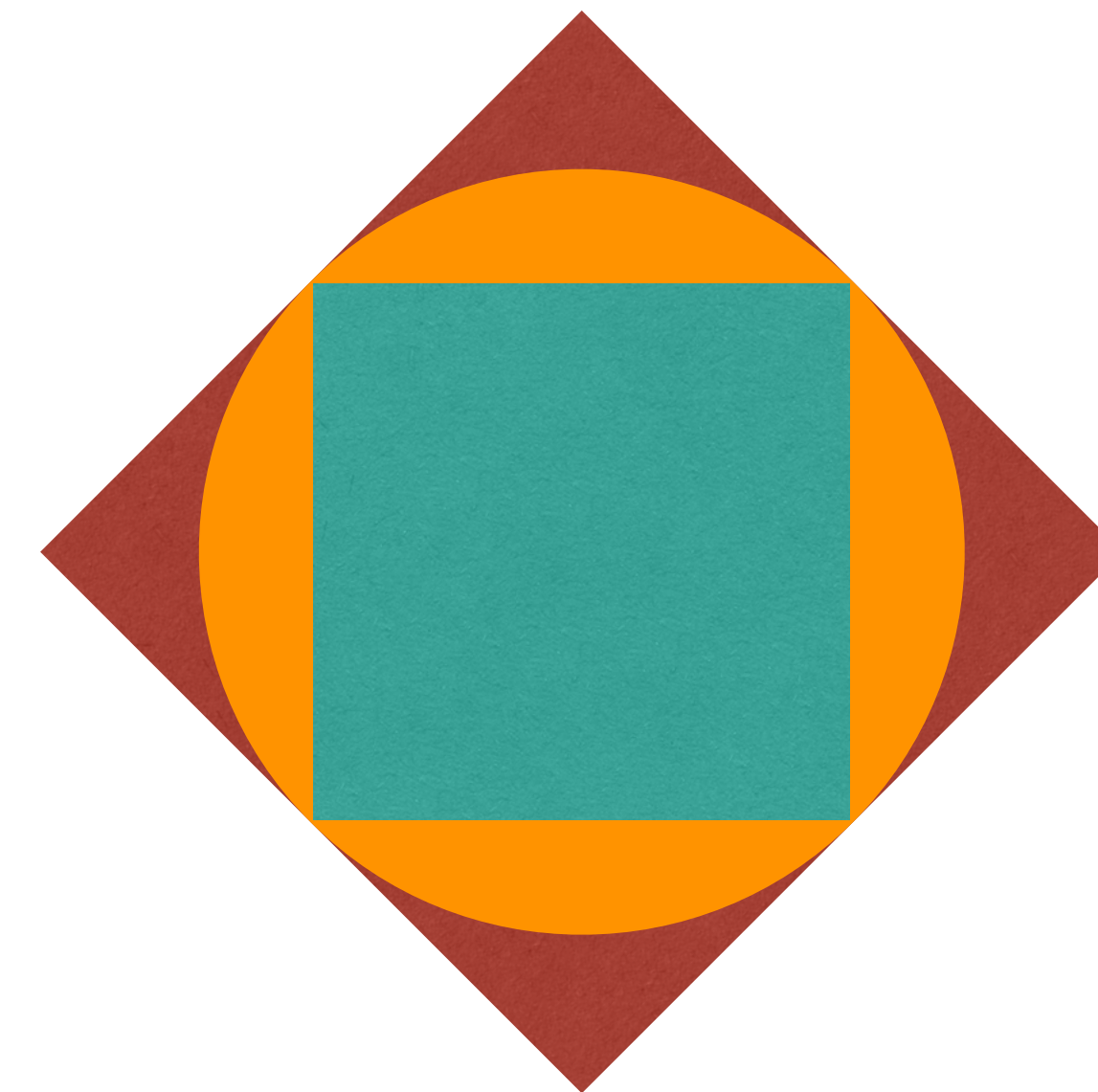
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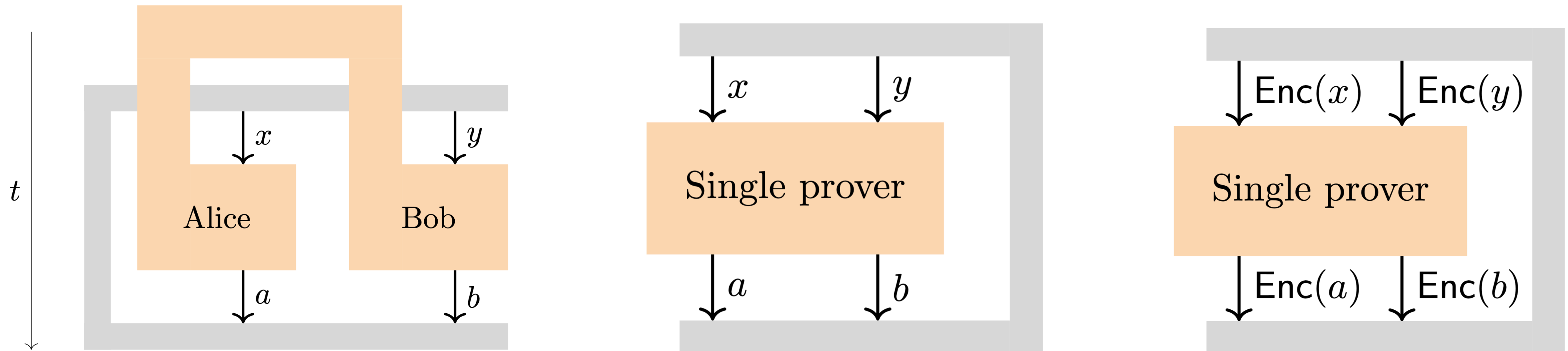


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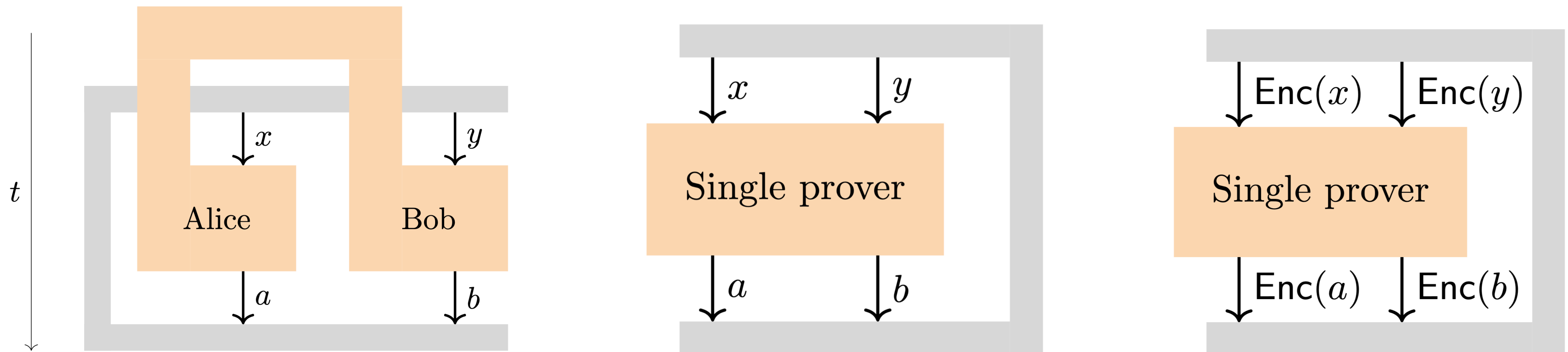


The more nonlocal a physical theory, the higher the score

Removing space-like separation using cryptography

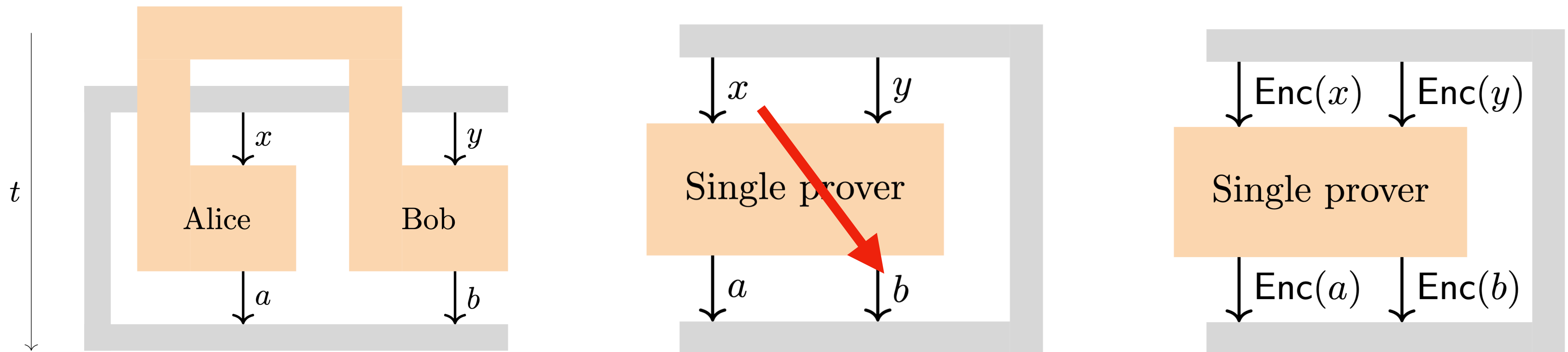


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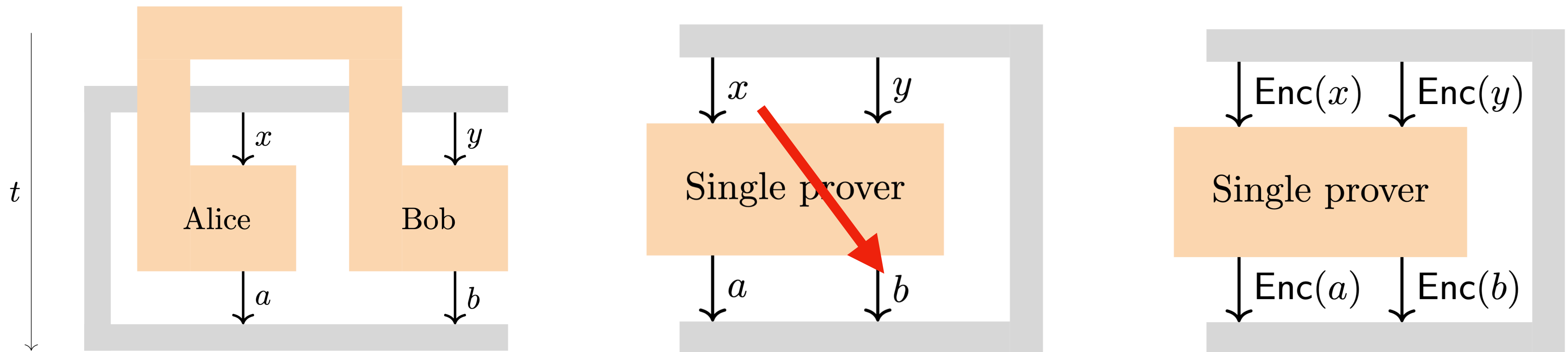
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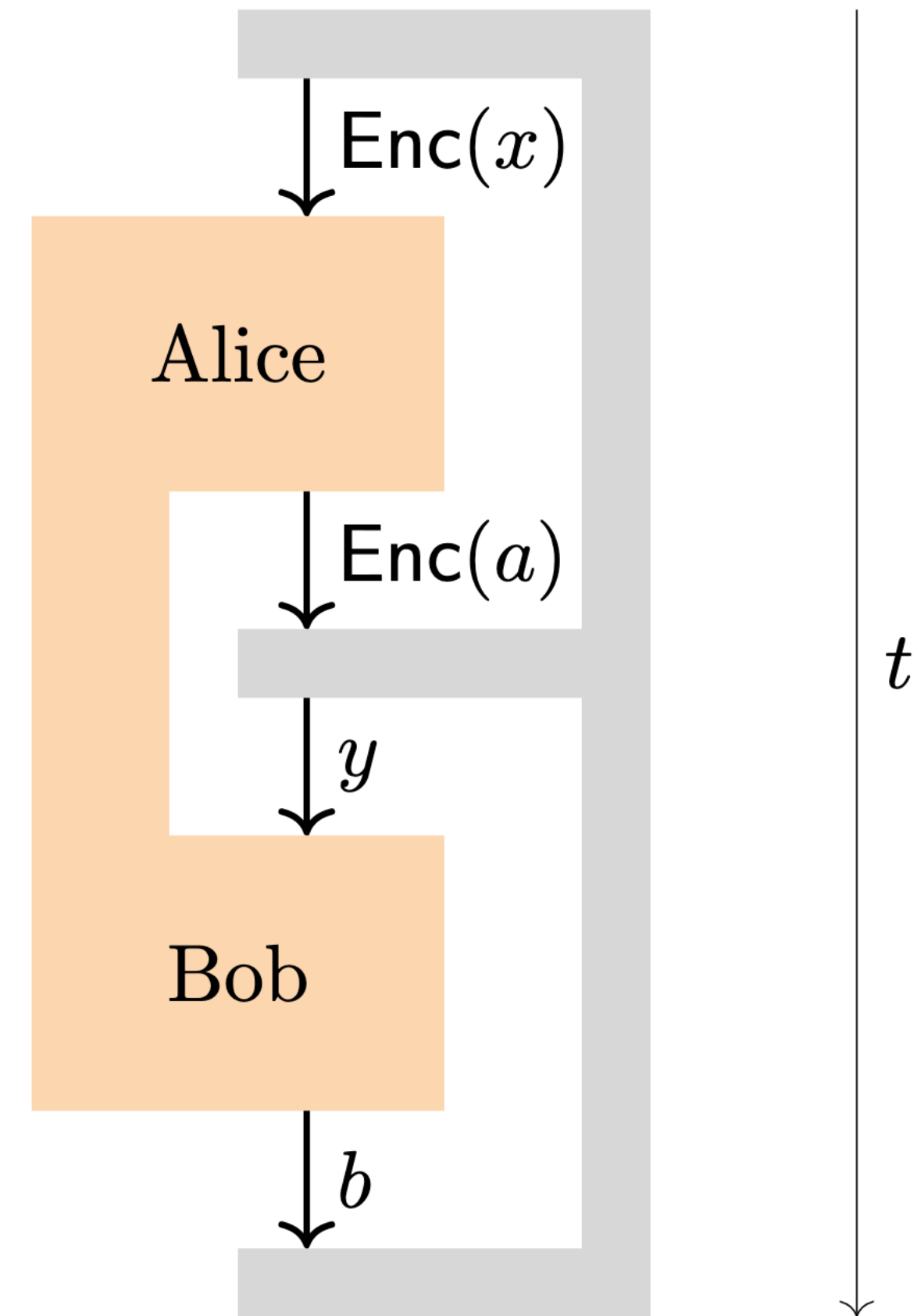
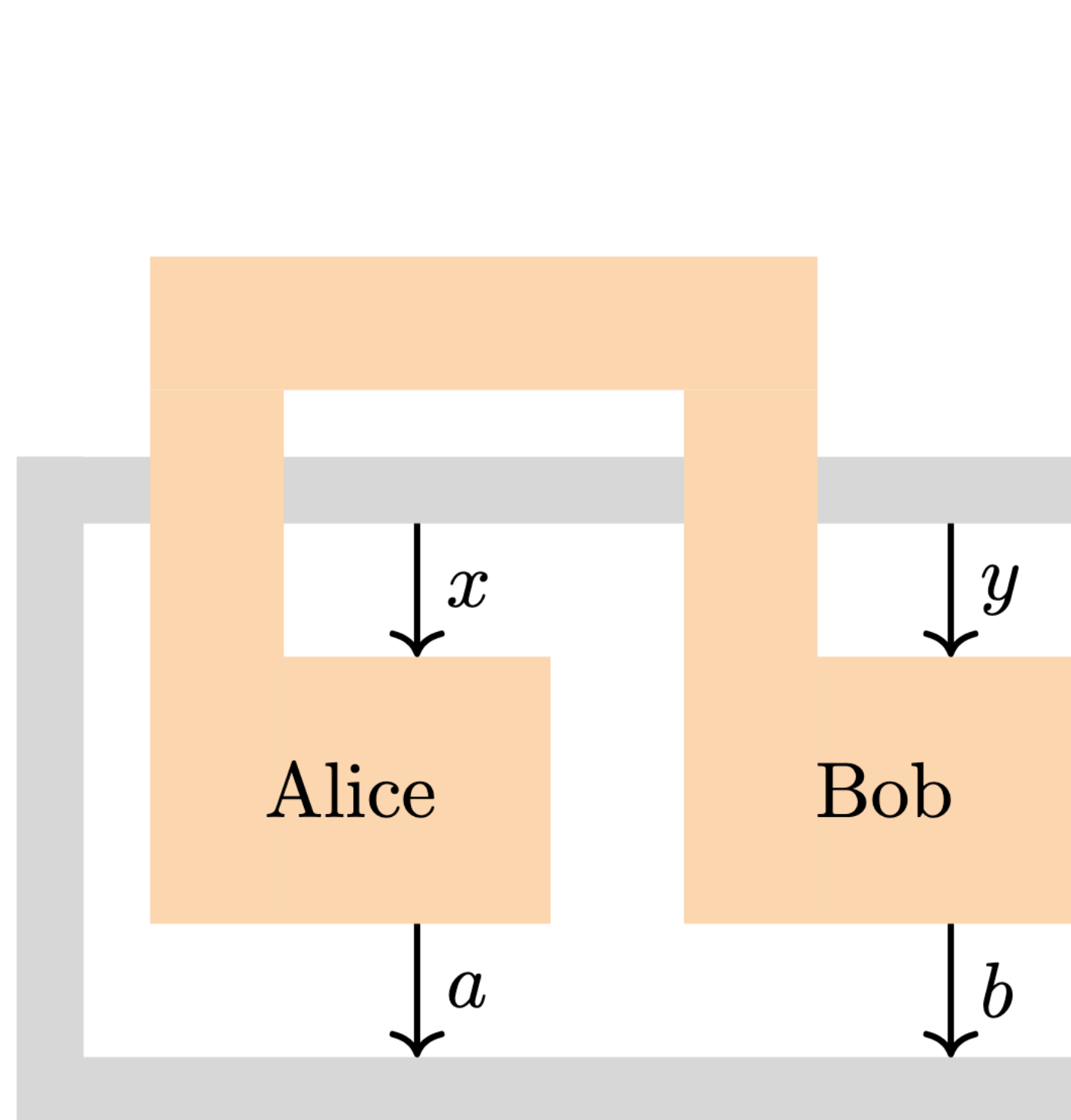
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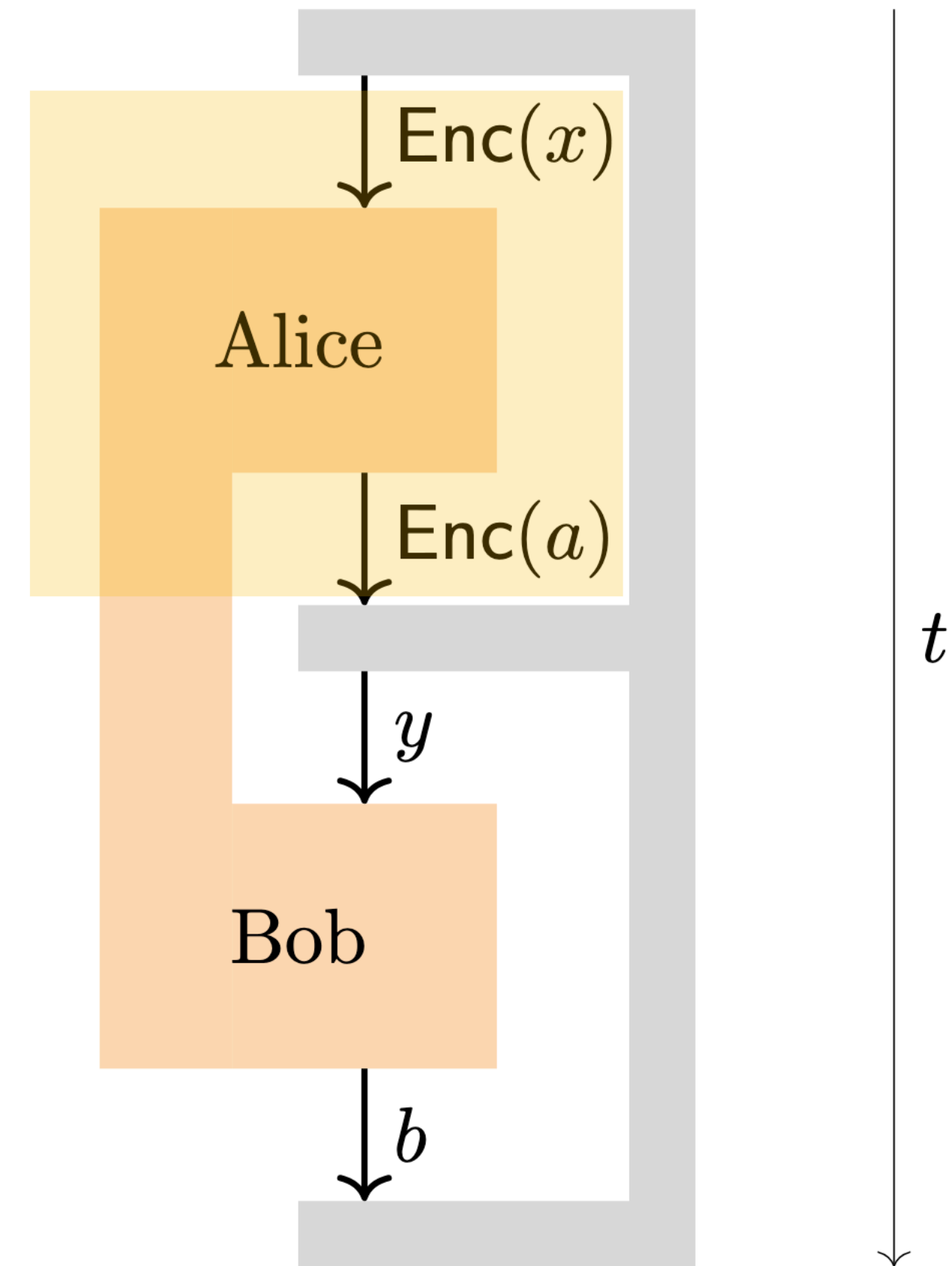
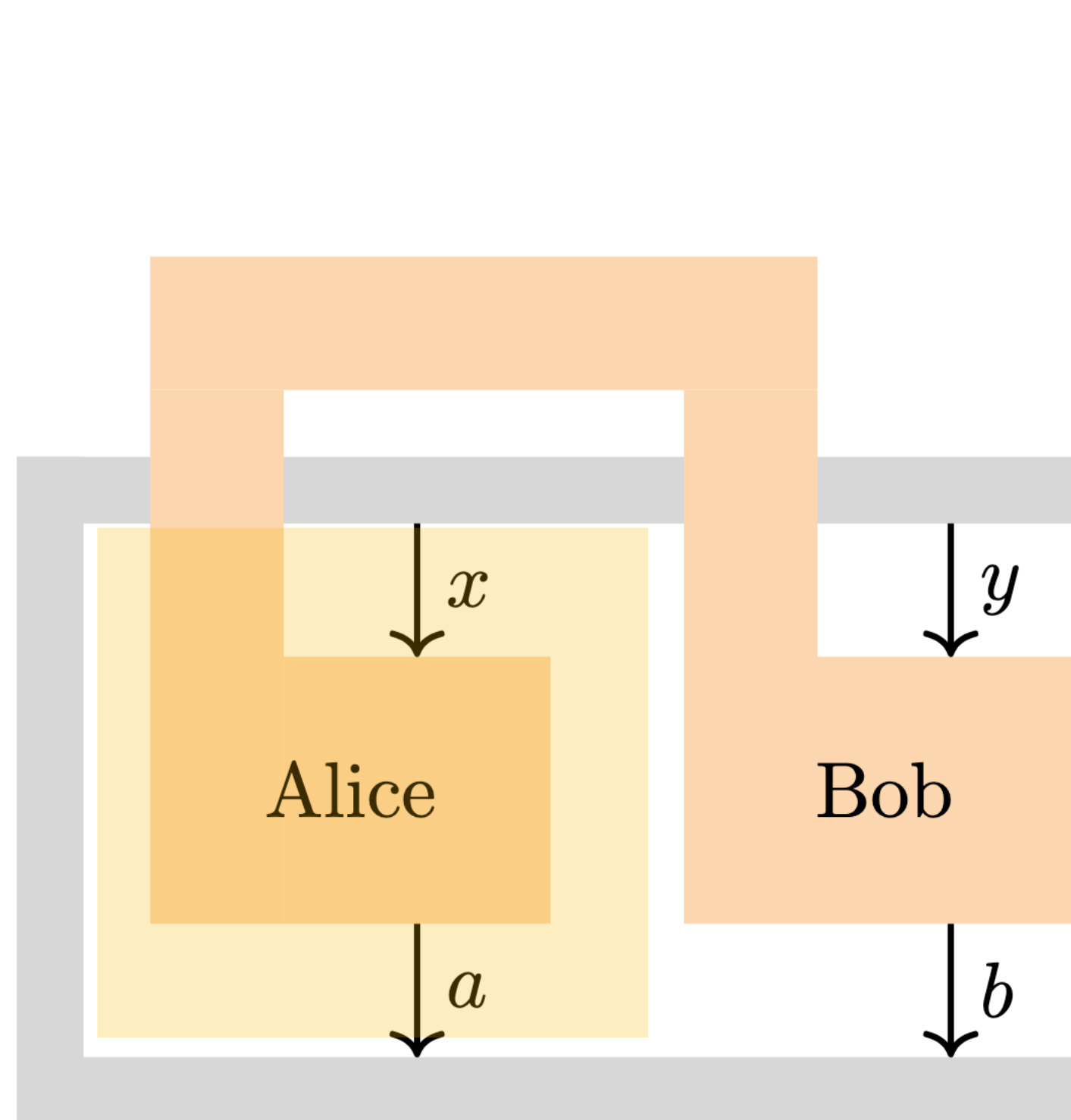


- Goal: performing Bell test on a single untrusted device?
- Idea: Encryption acts as a barrier

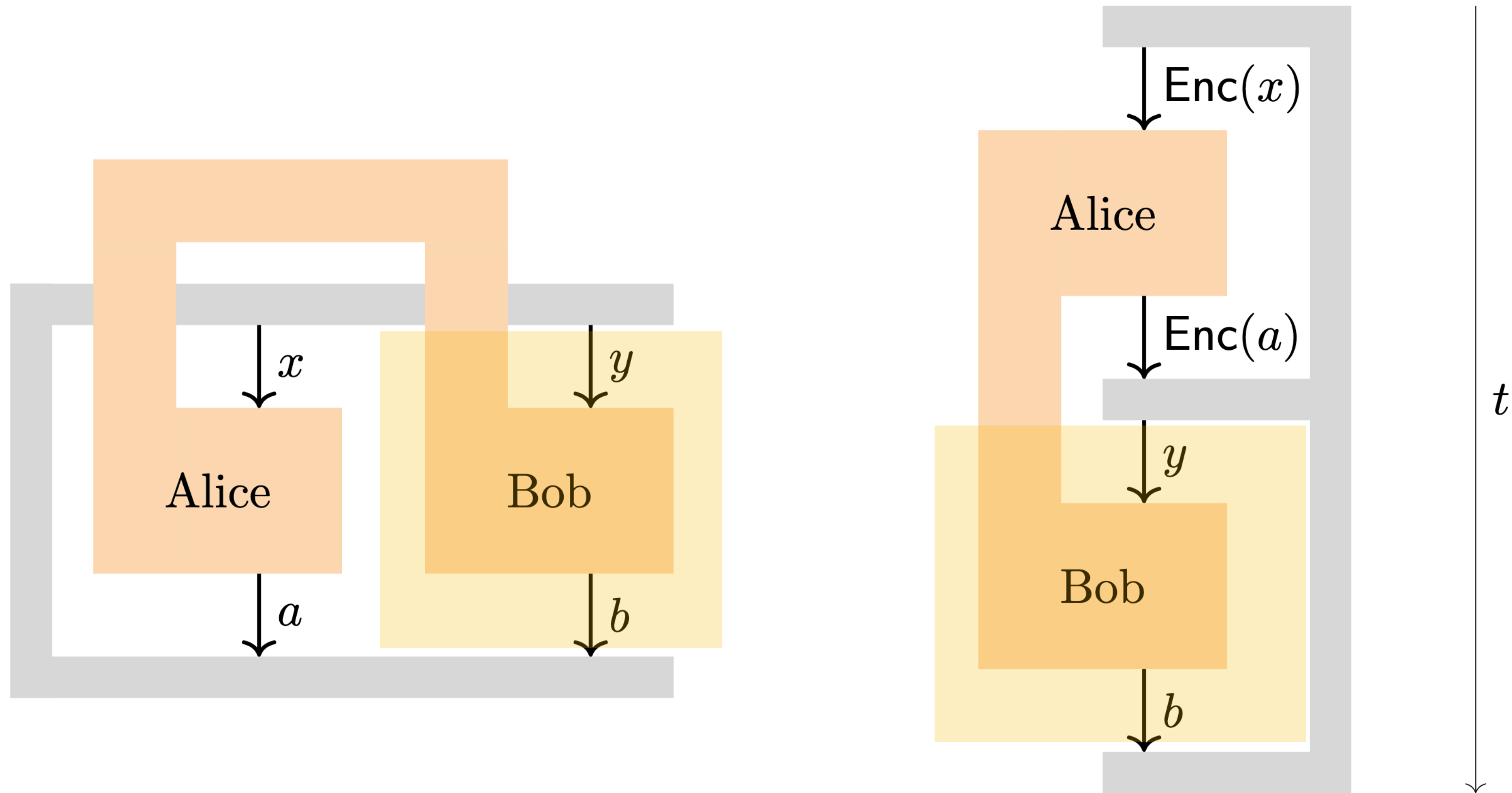
KL VY compiler



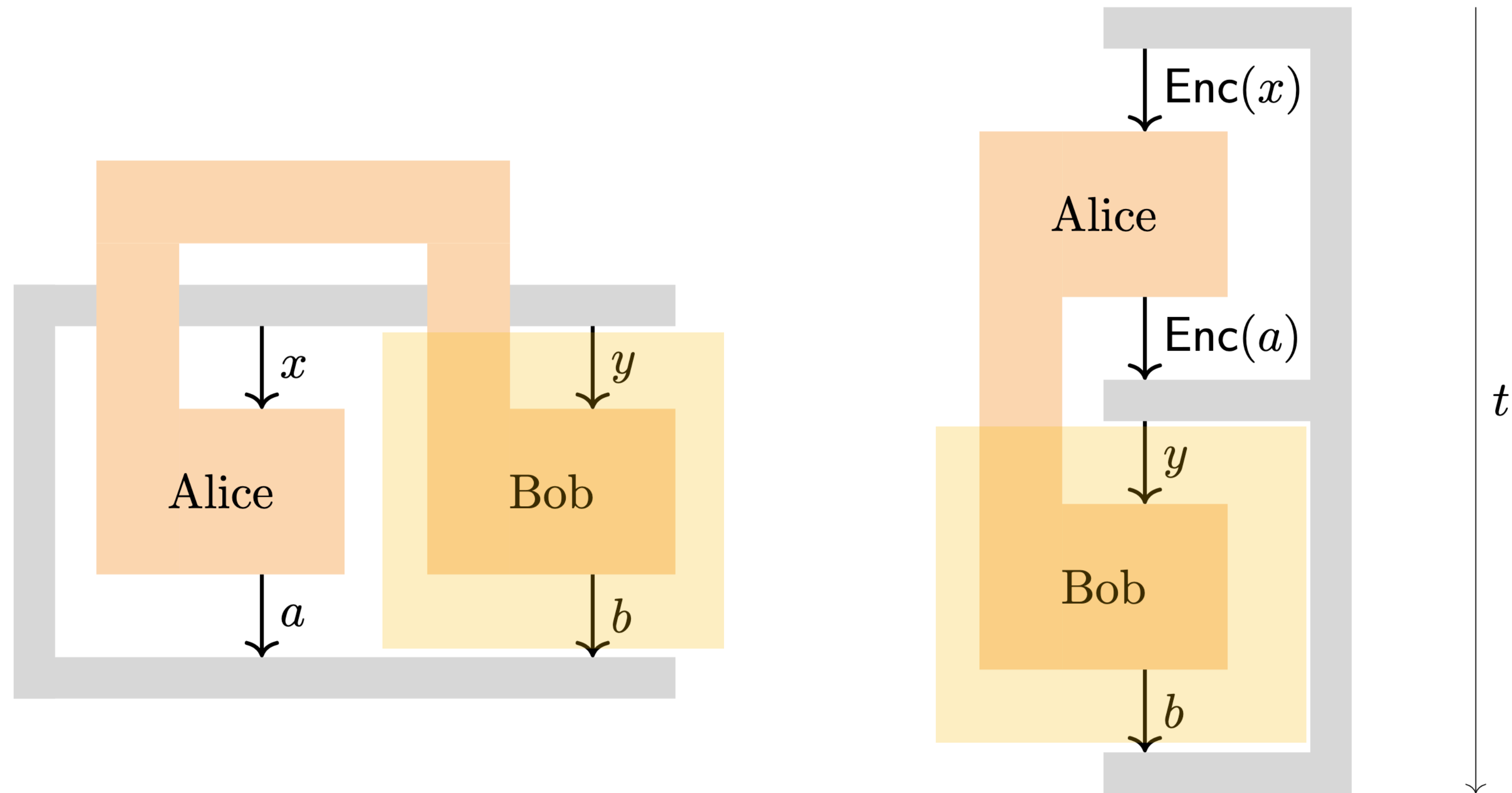
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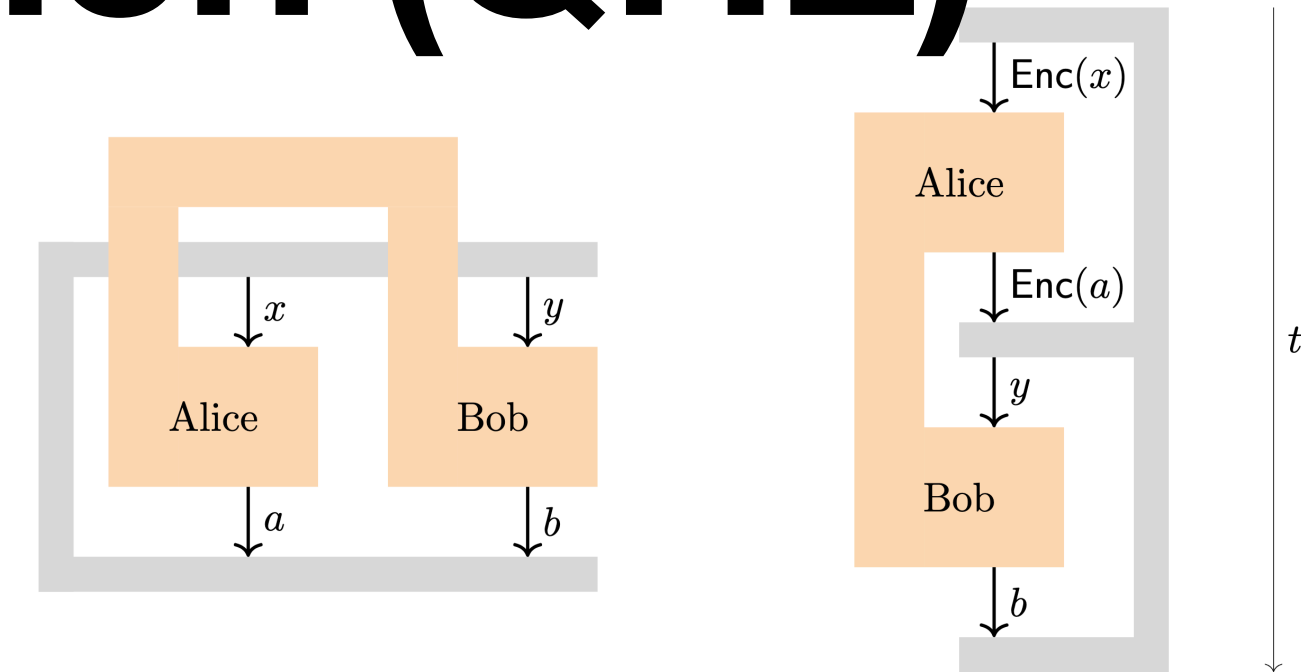


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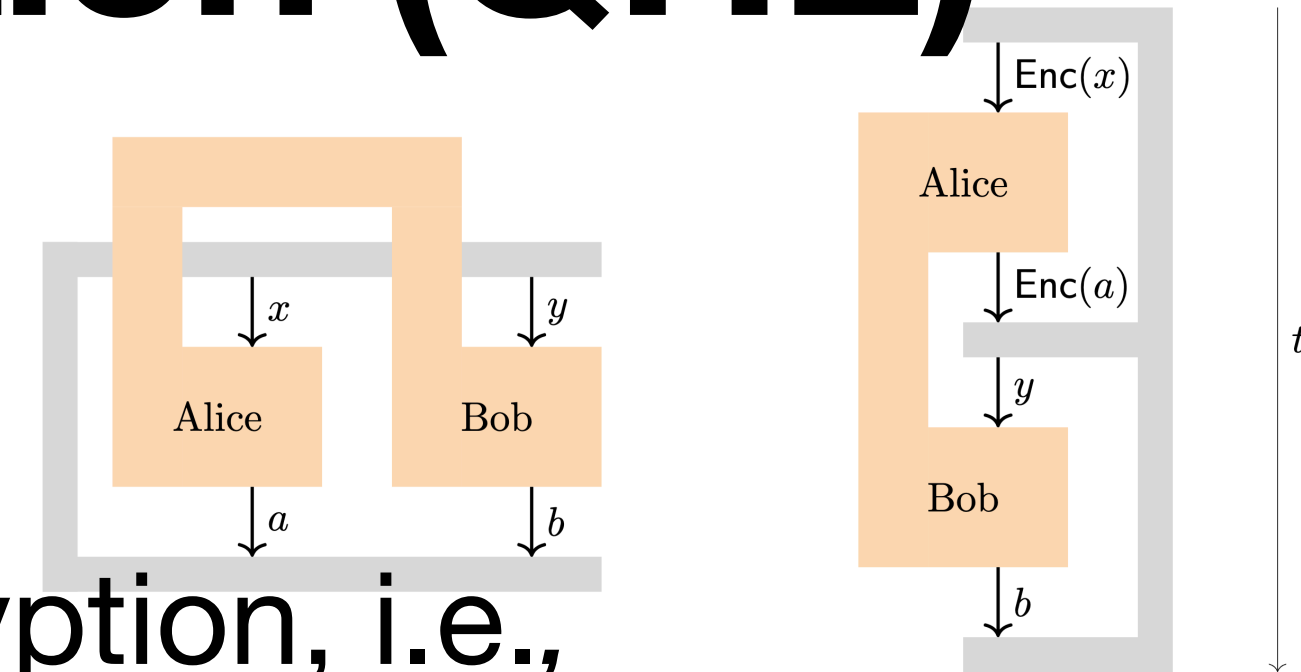


Works for all k players games!

Quantum Homomorphic Encryption (QHE)

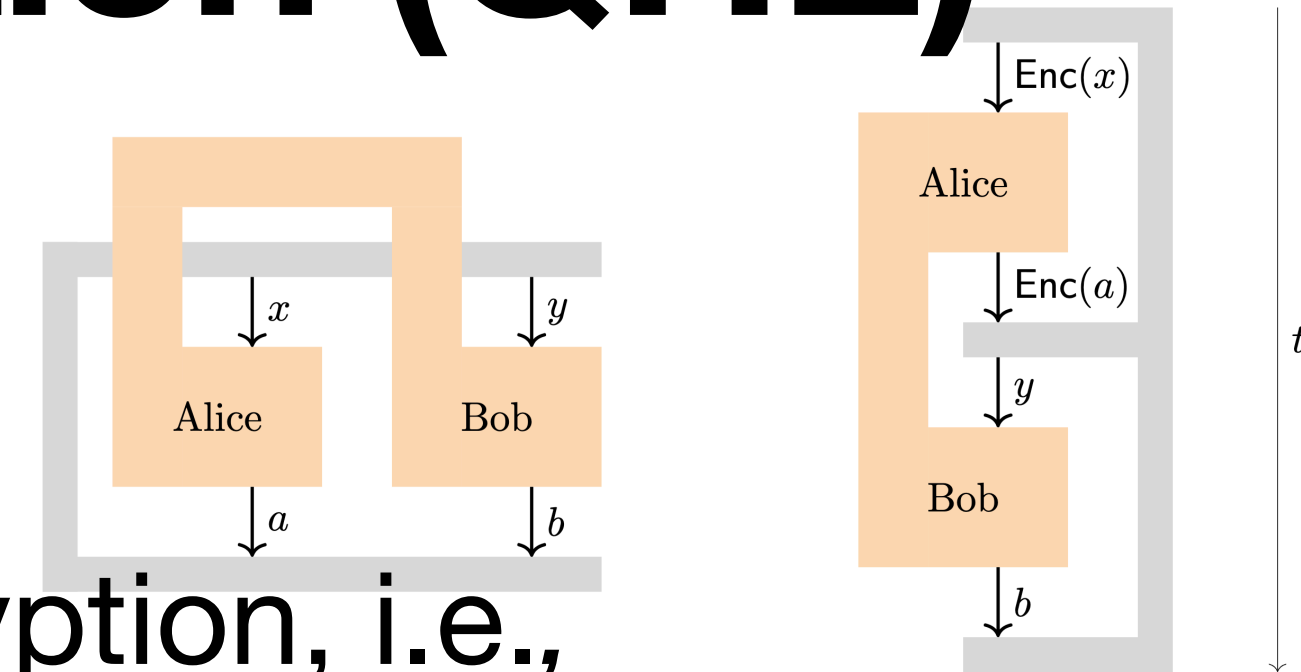


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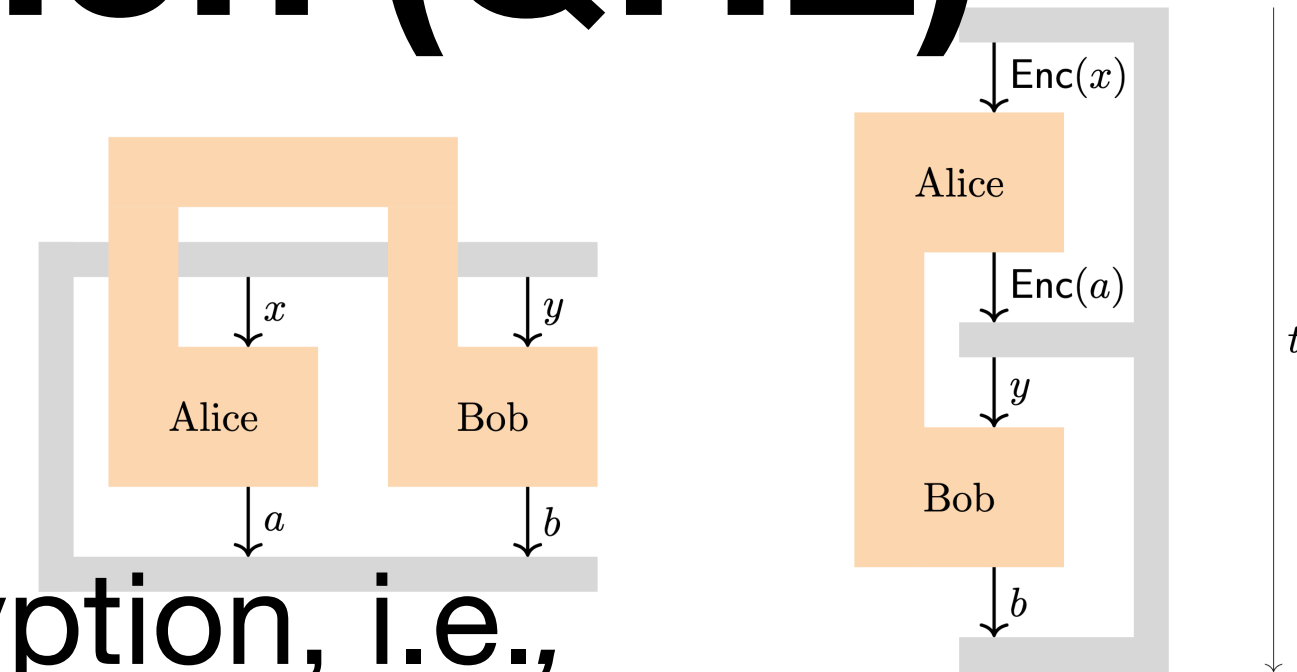
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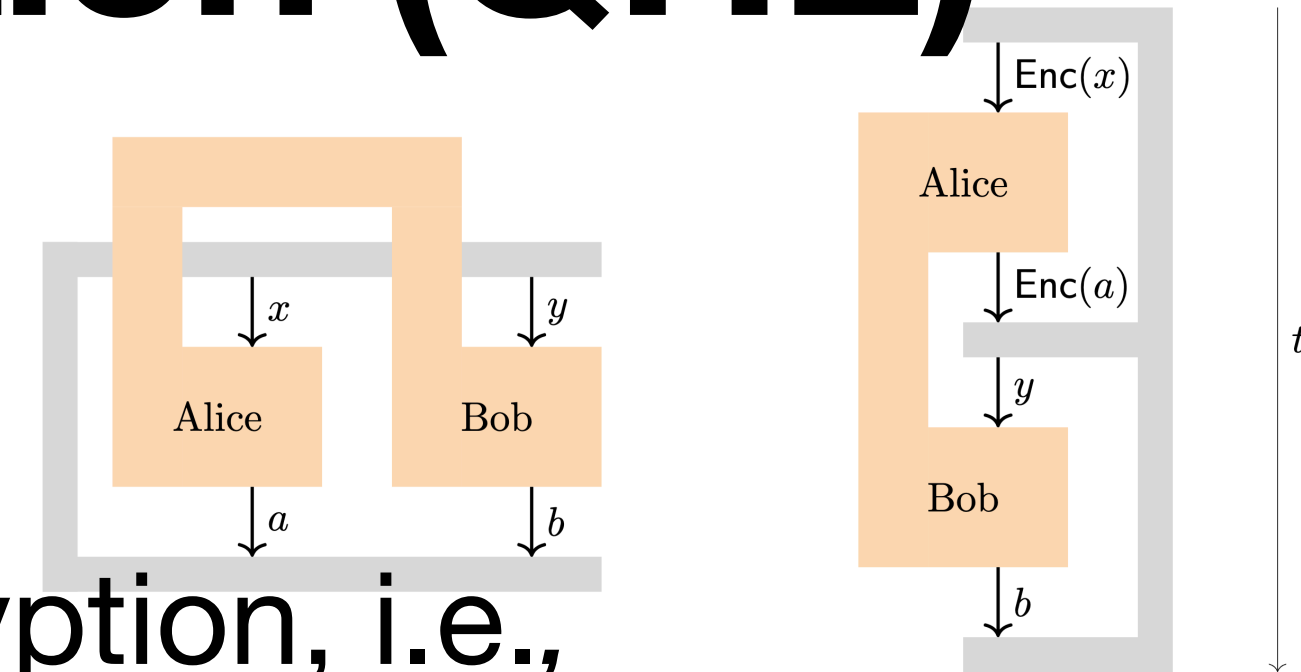
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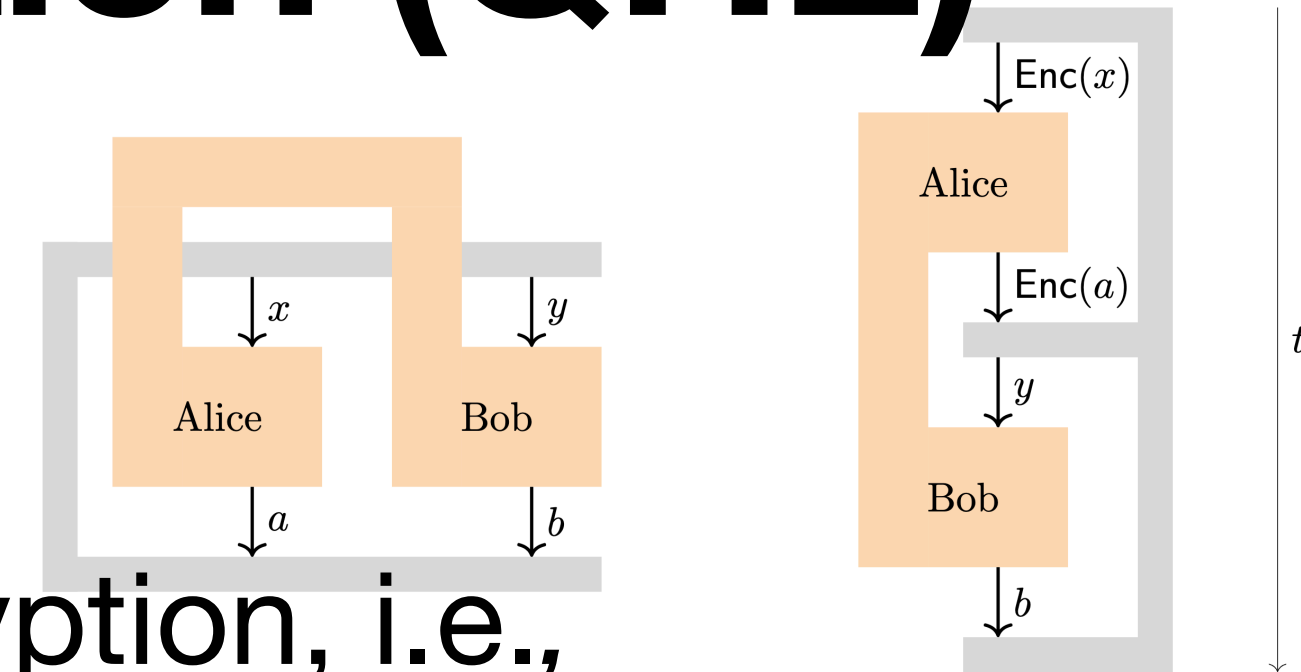
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 - λ -secure QHE safe against all λ -efficient provers' attacks

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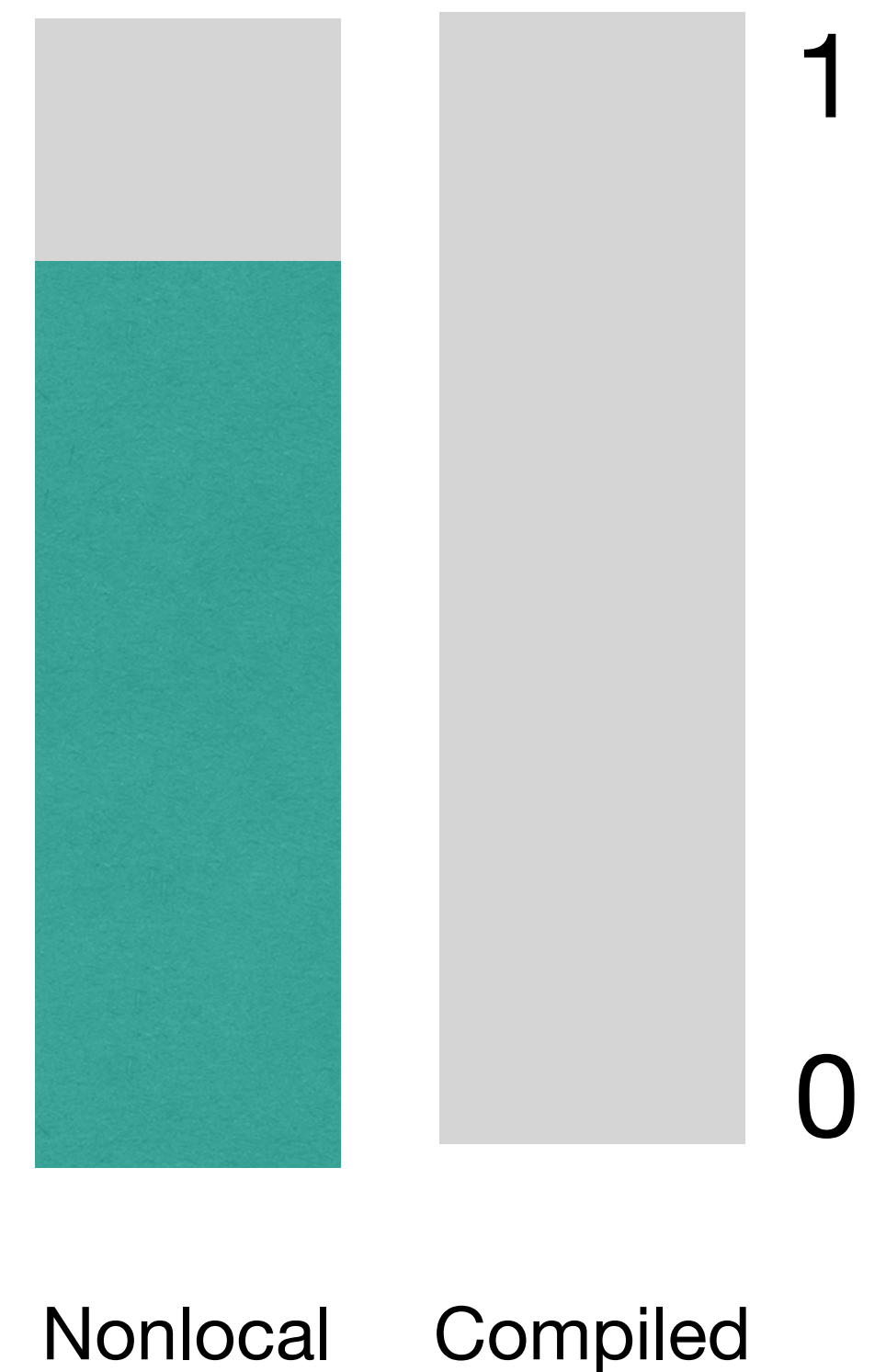
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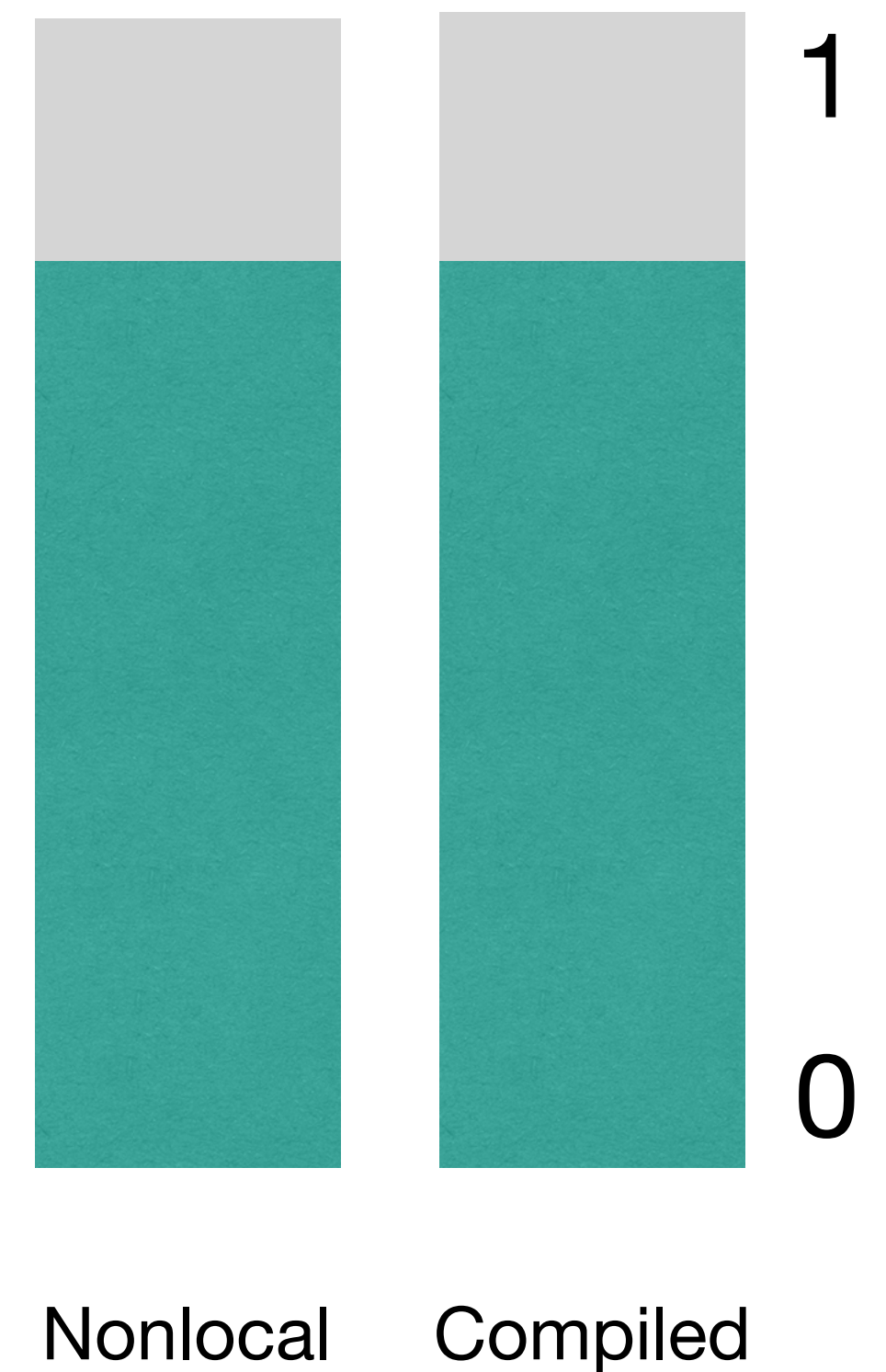


[KLVY22] arXiv: 2203.15877

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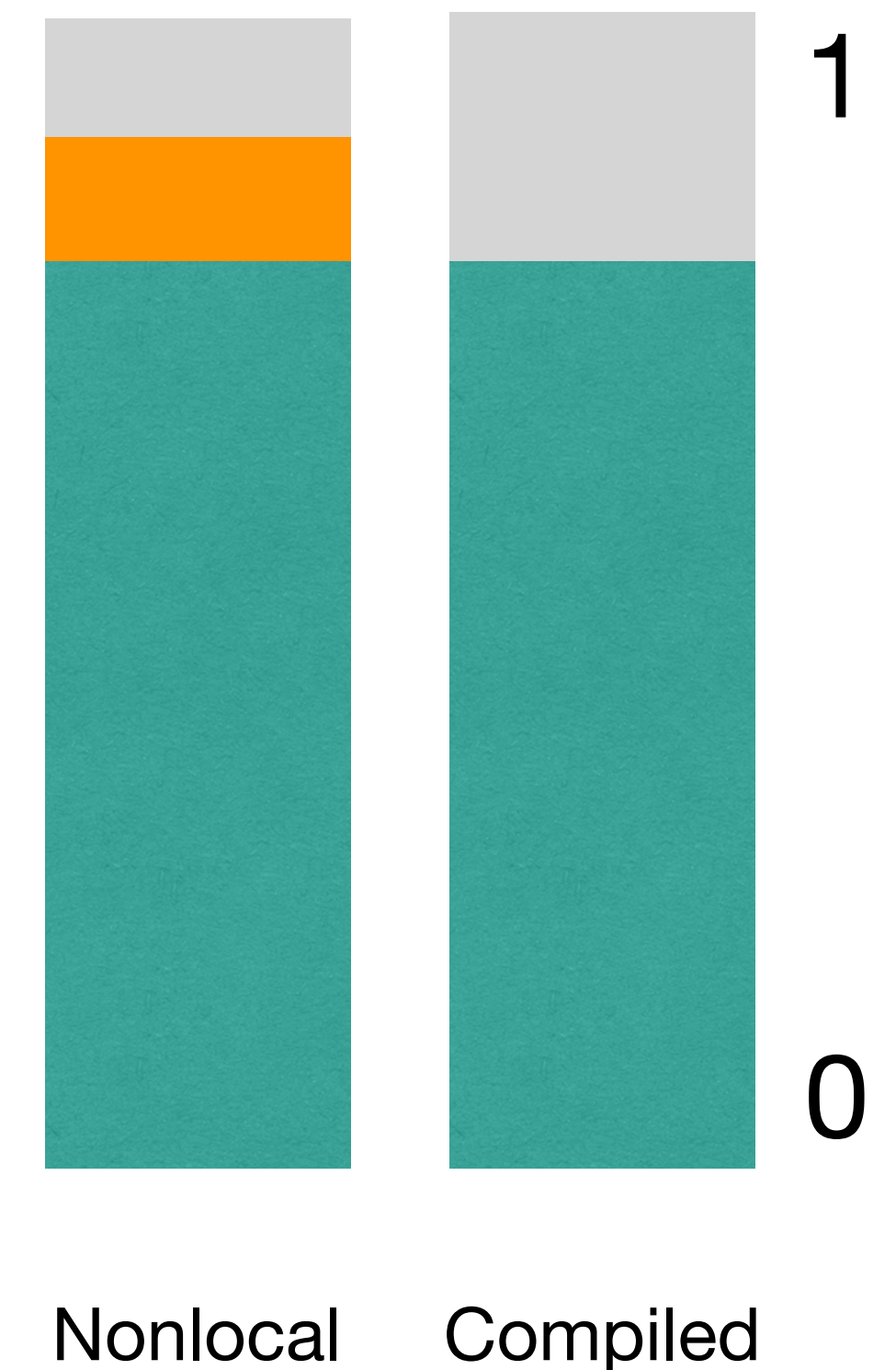


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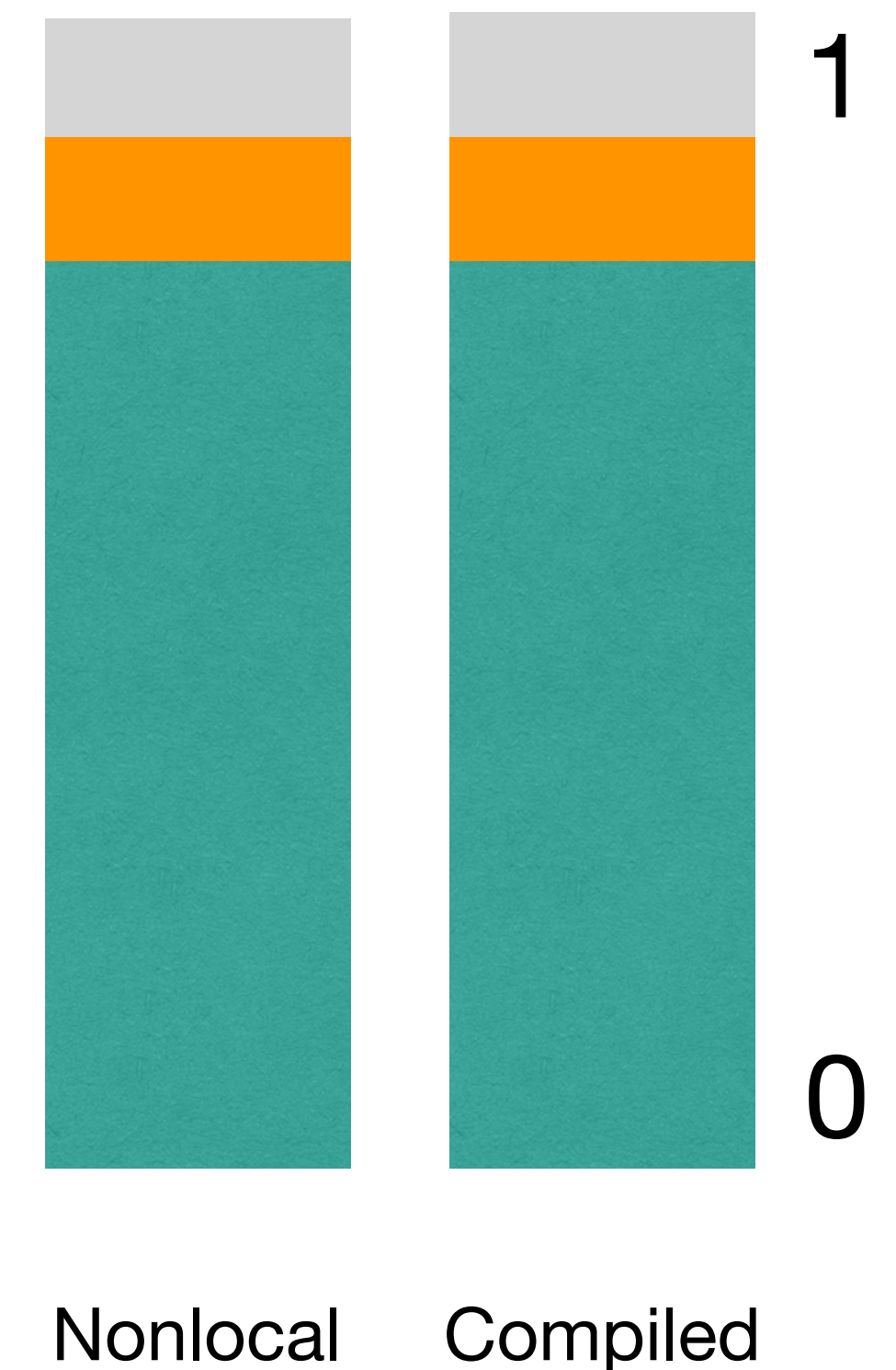


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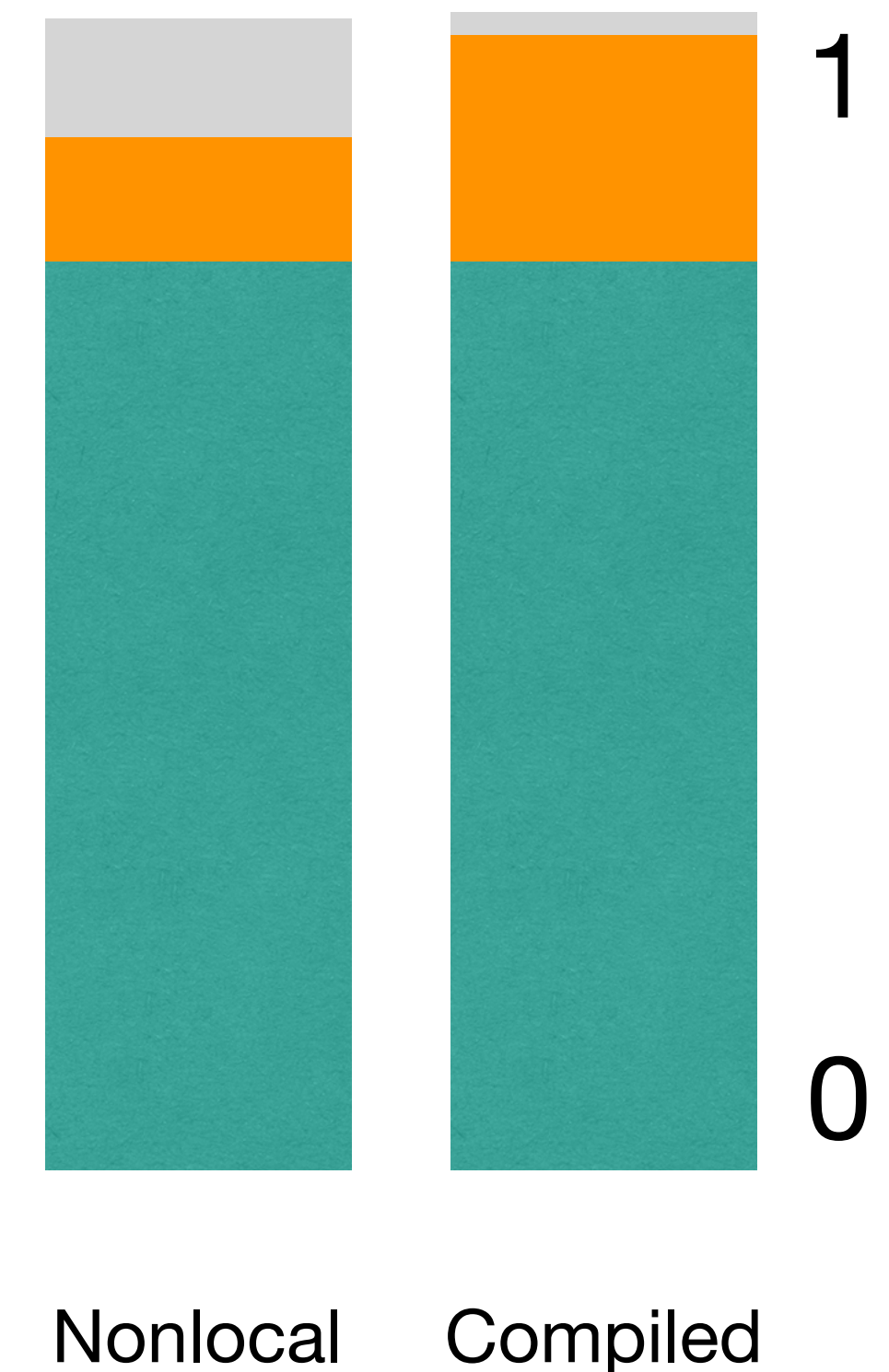


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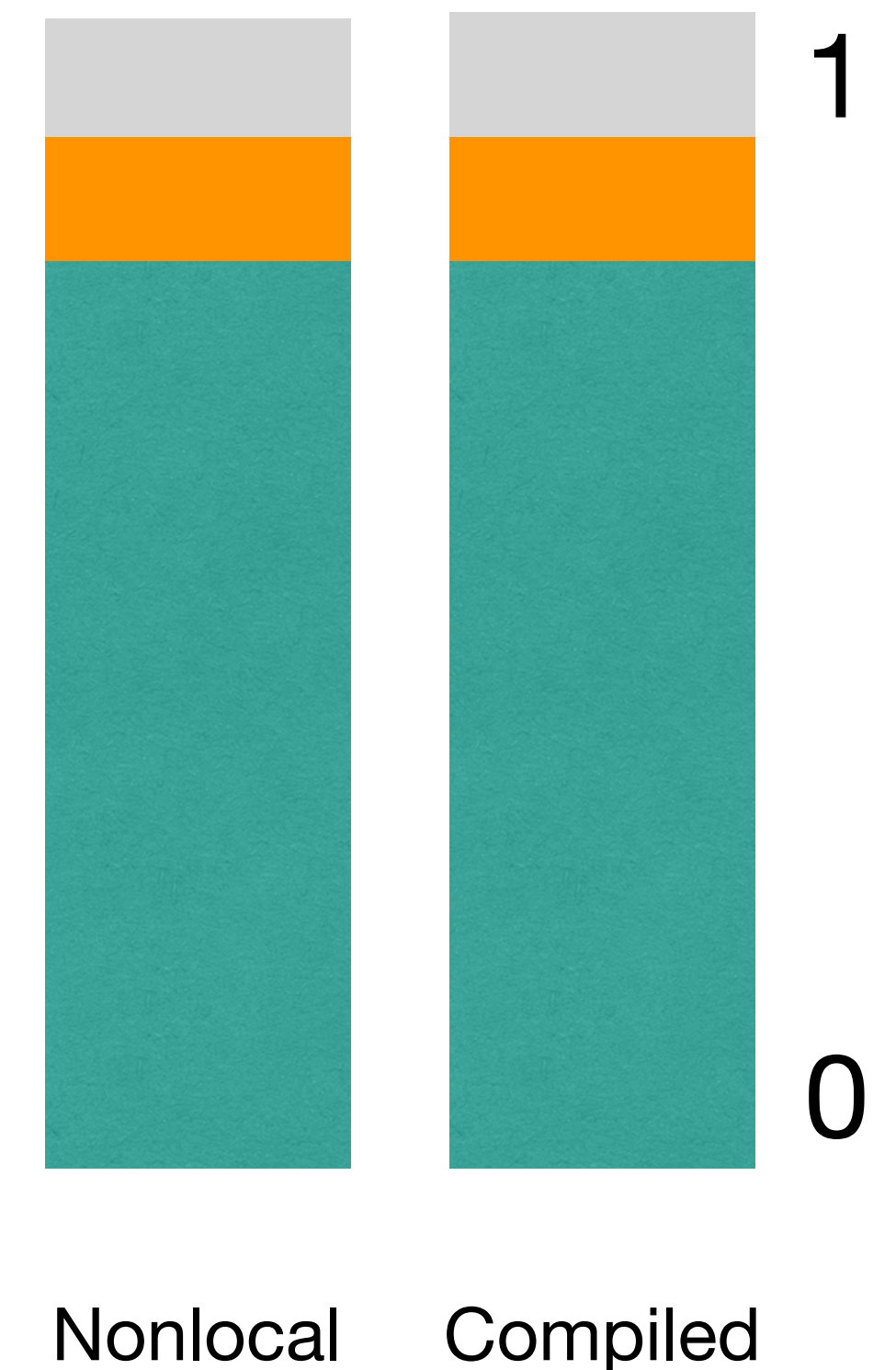


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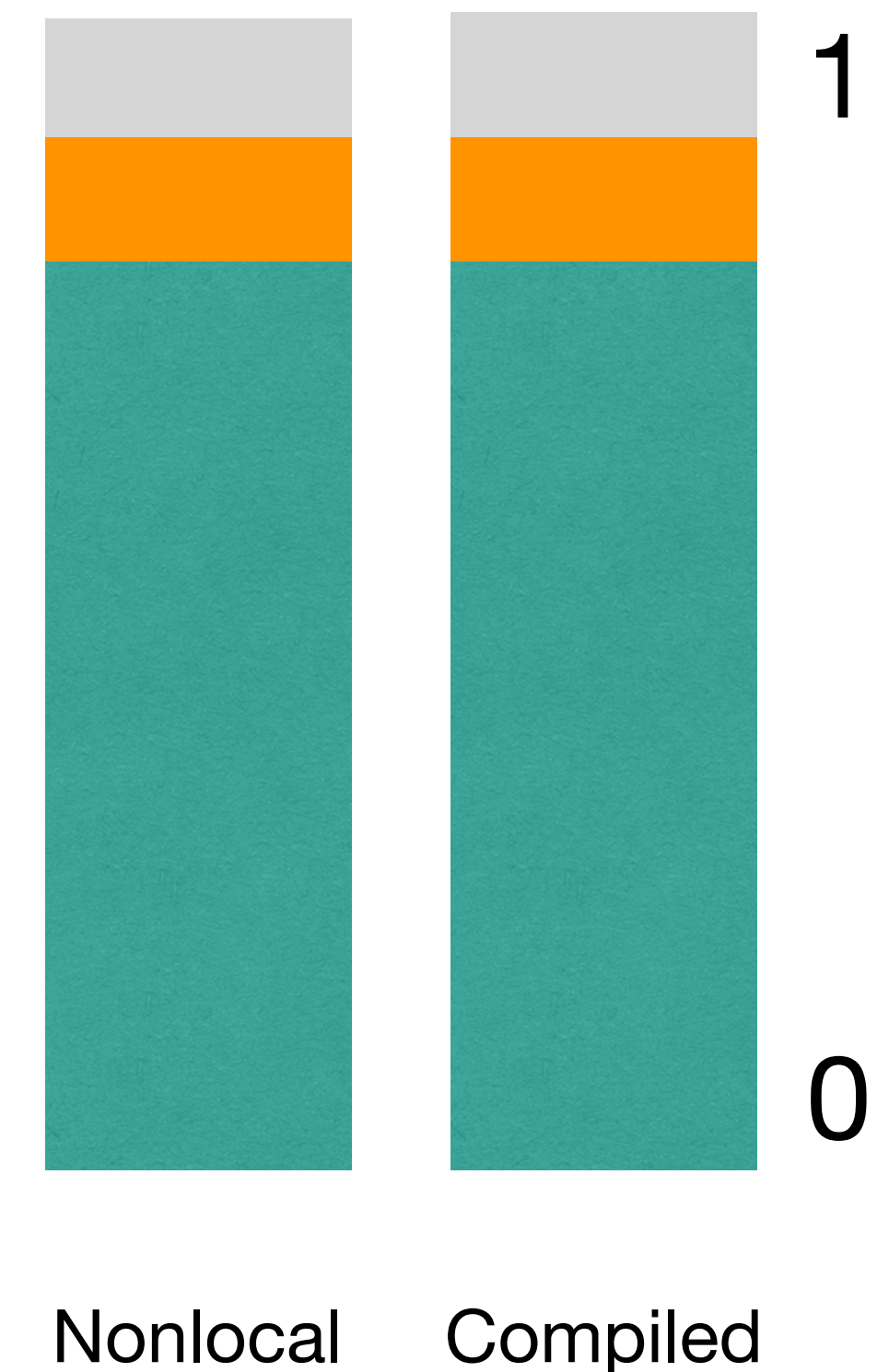


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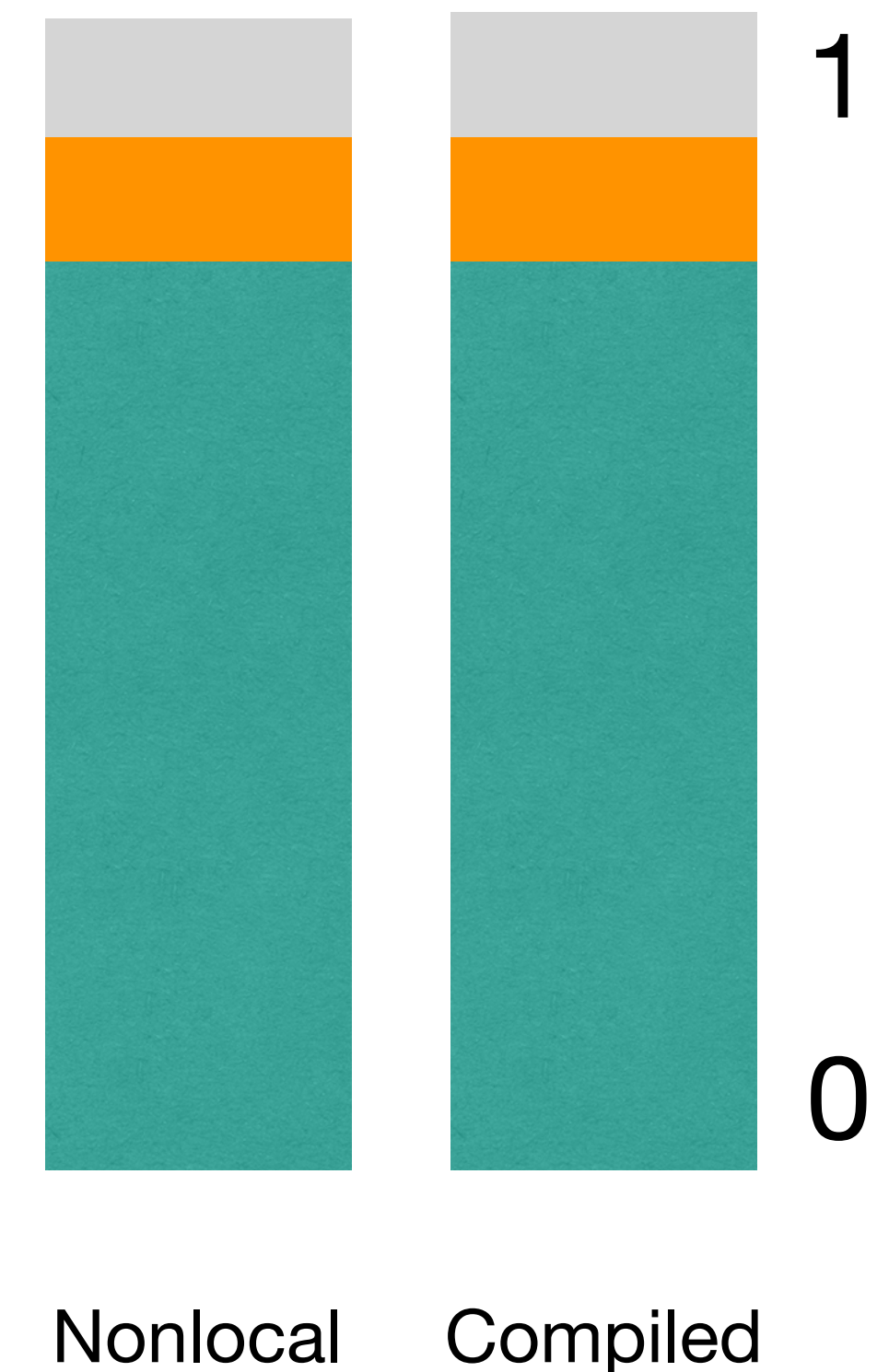


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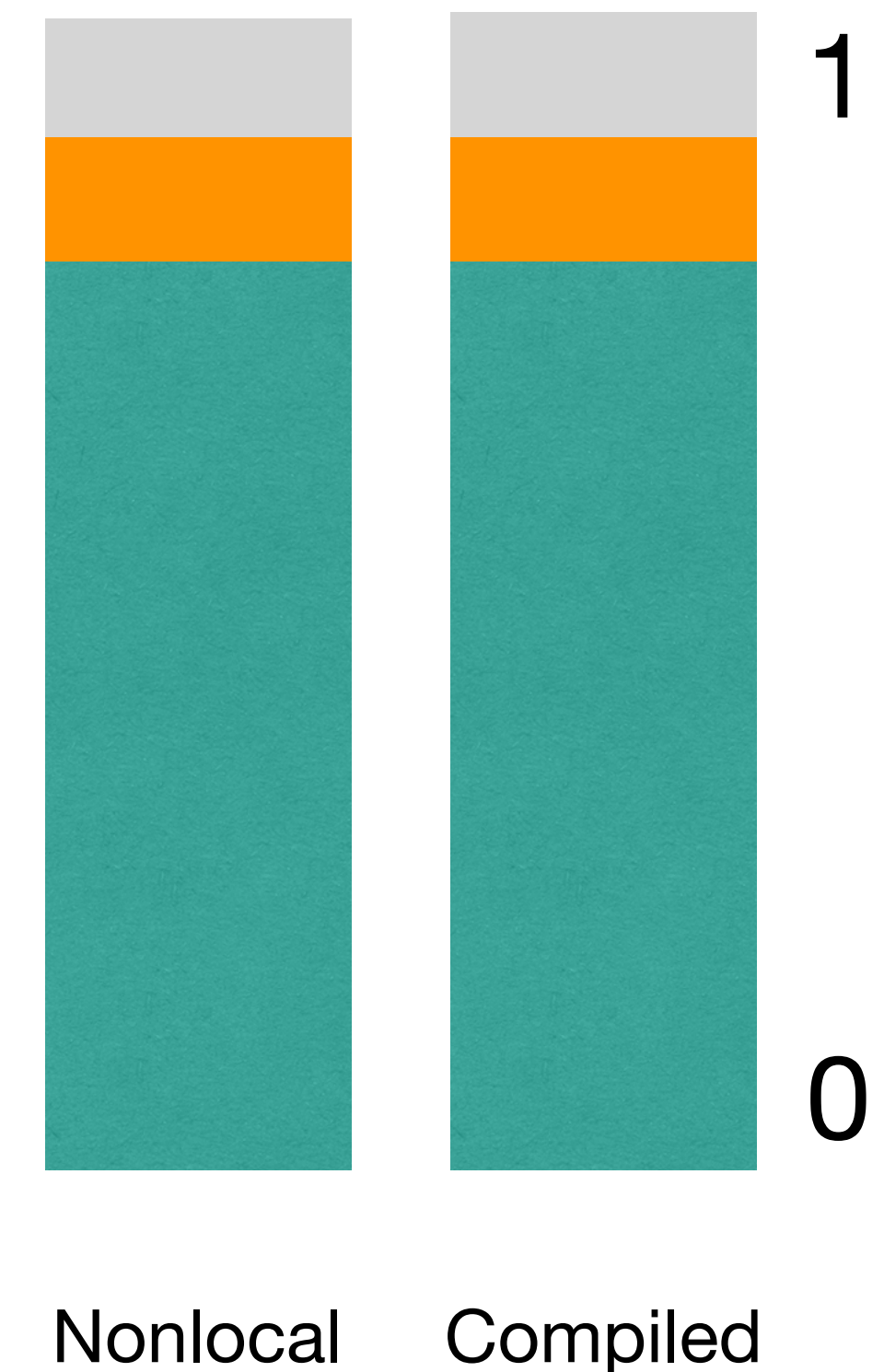
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5. Our three works [BLJŠ25, KPRSTXZ25, BKLRŠTX25]:
quantitative quantum soundness for all multipartite games



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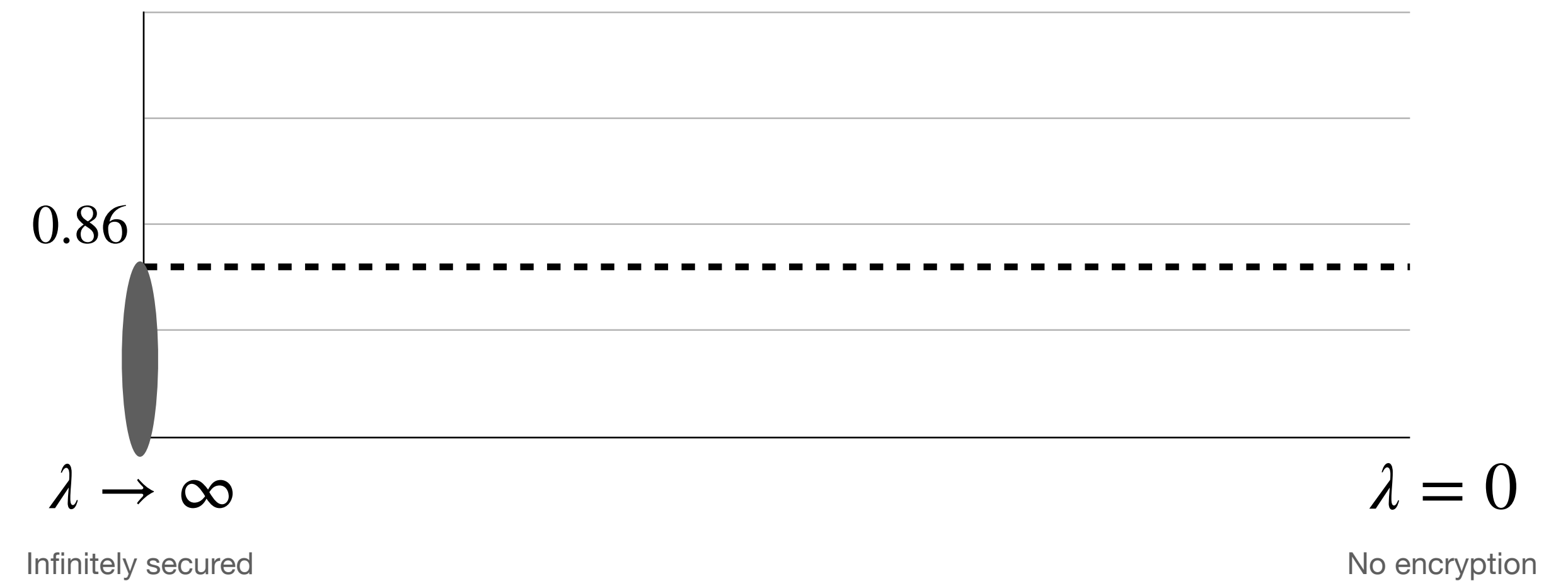
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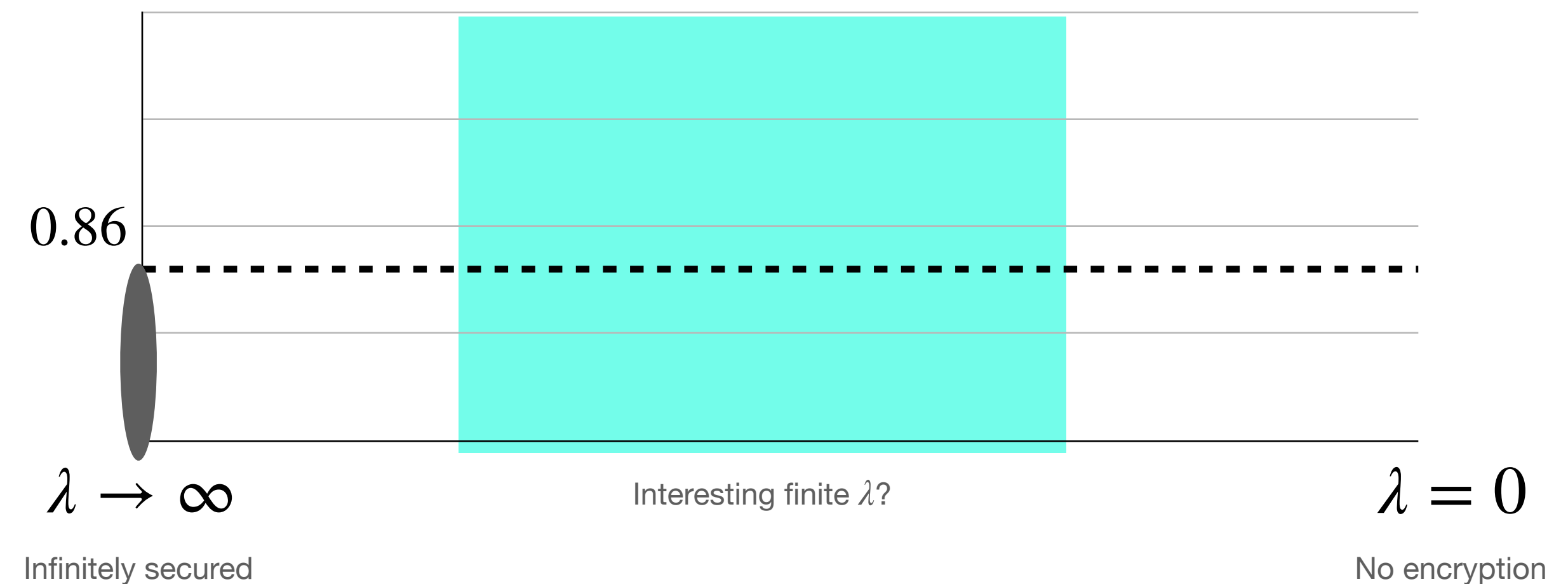
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- In practice, λ is finite, how sound can the compiled score still be?

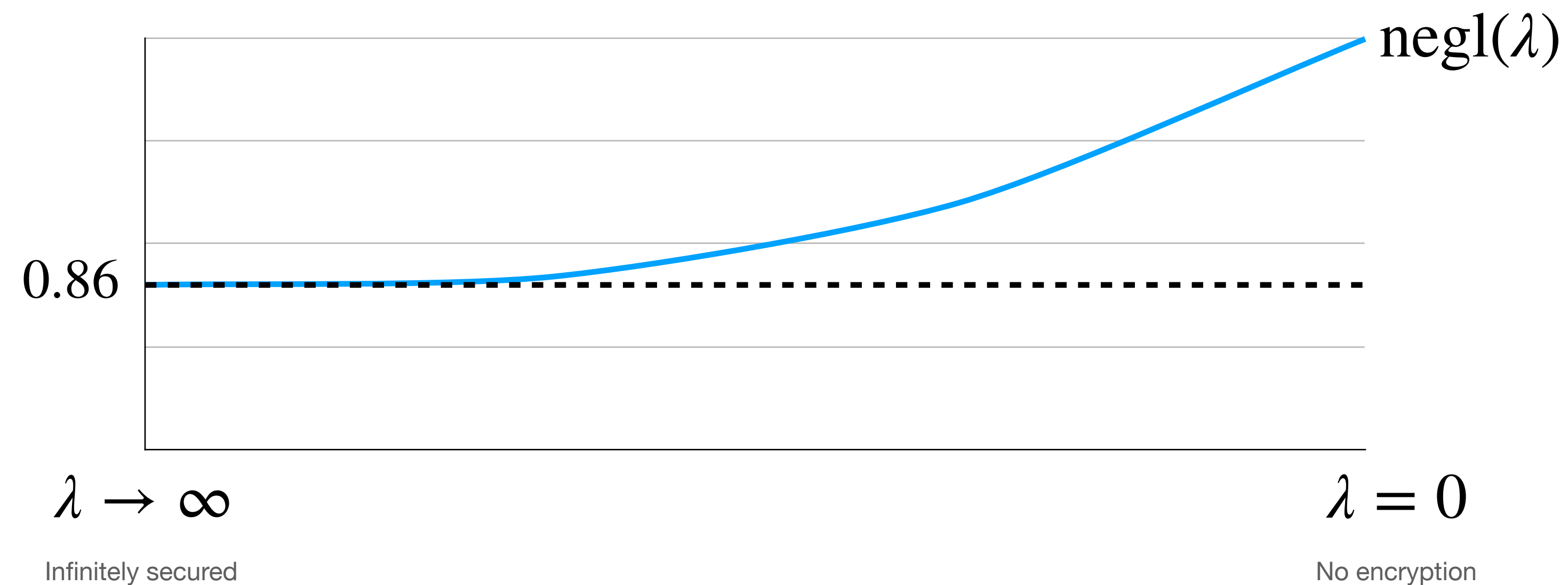
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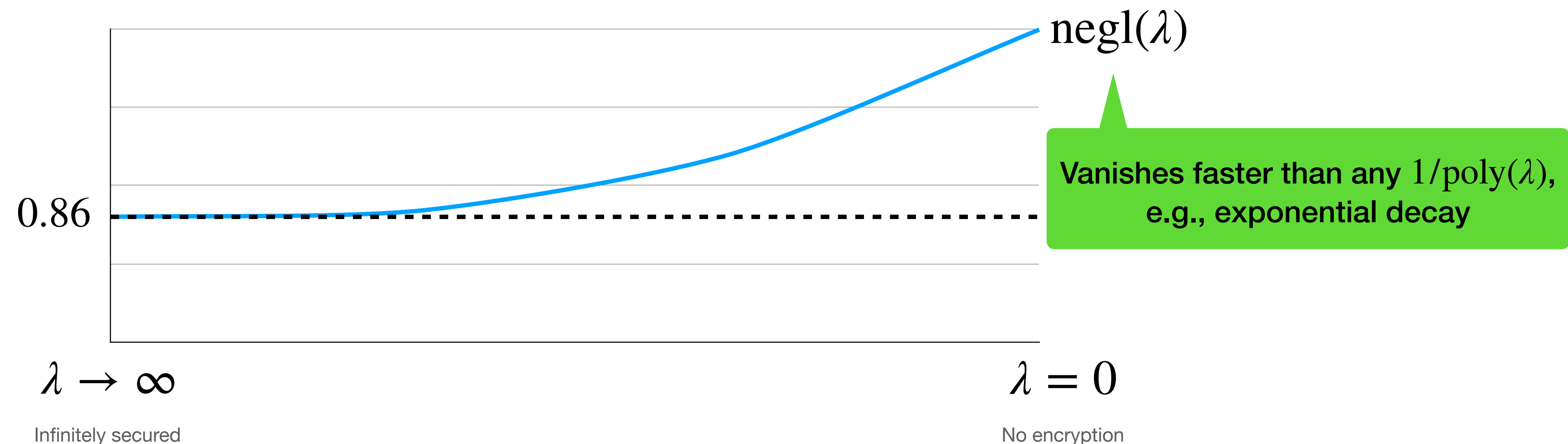
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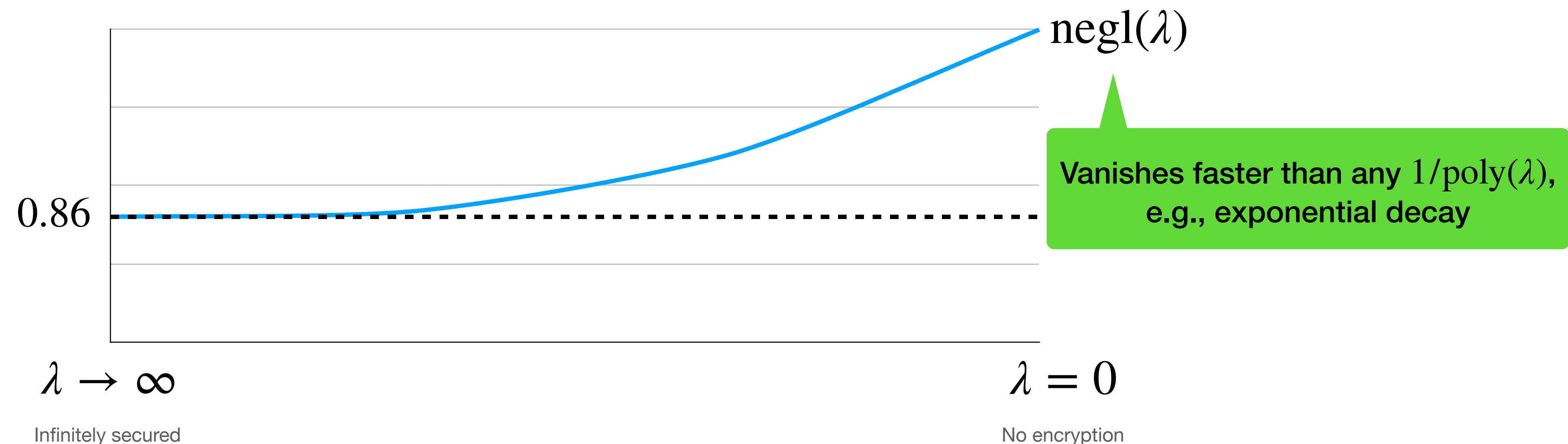
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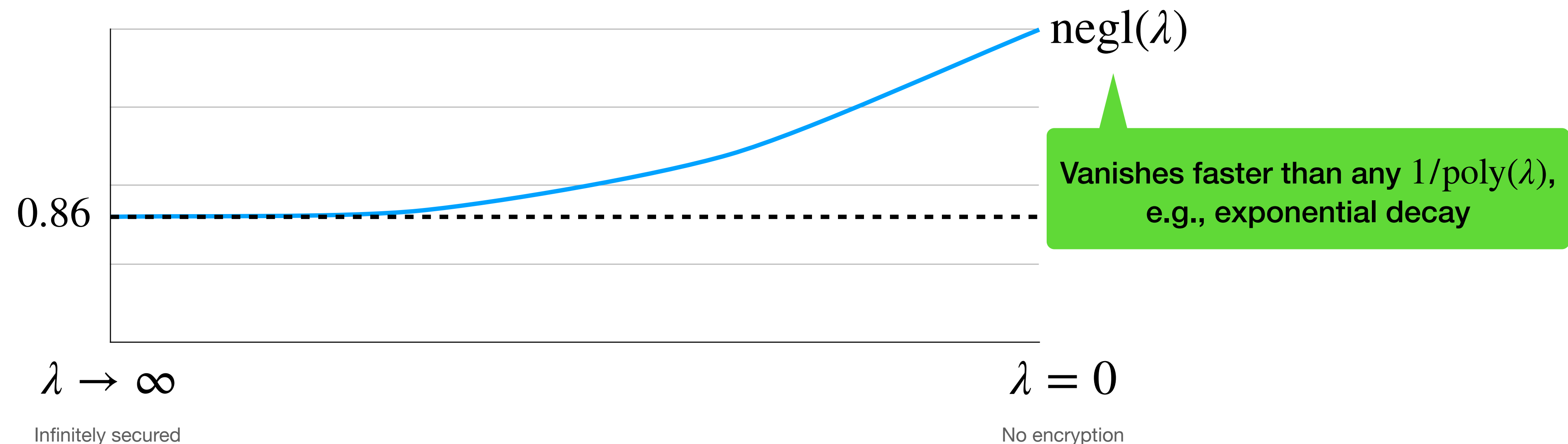
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Nonetheless, we have bounds for all games.
-> Has to get technical for 1 slide :/

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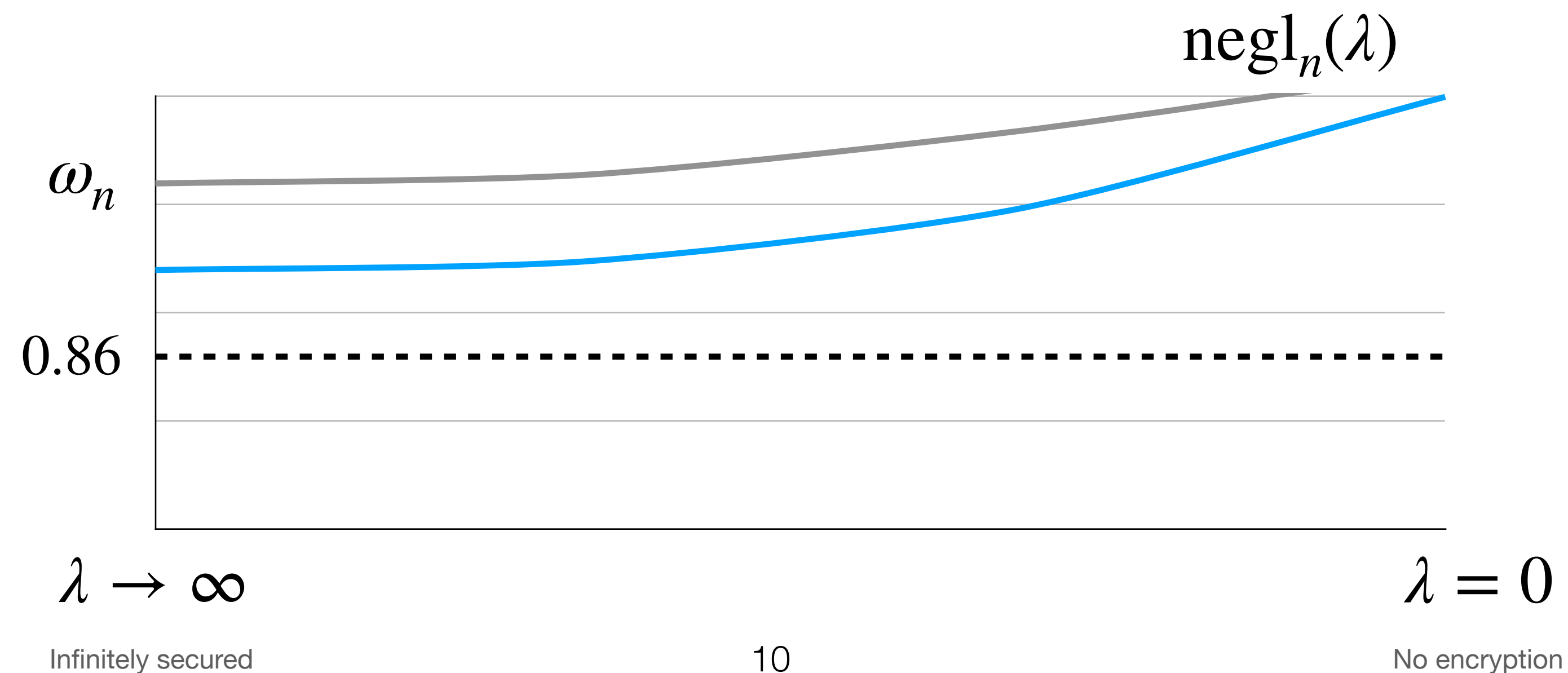
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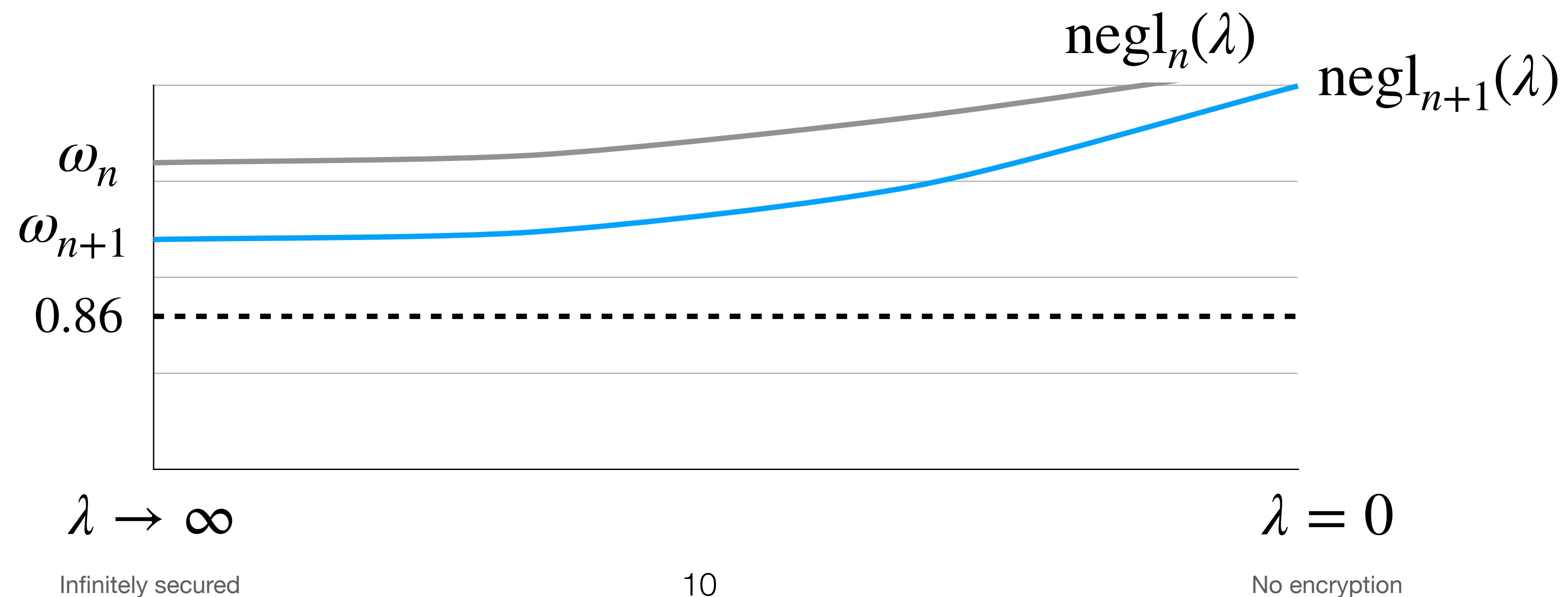
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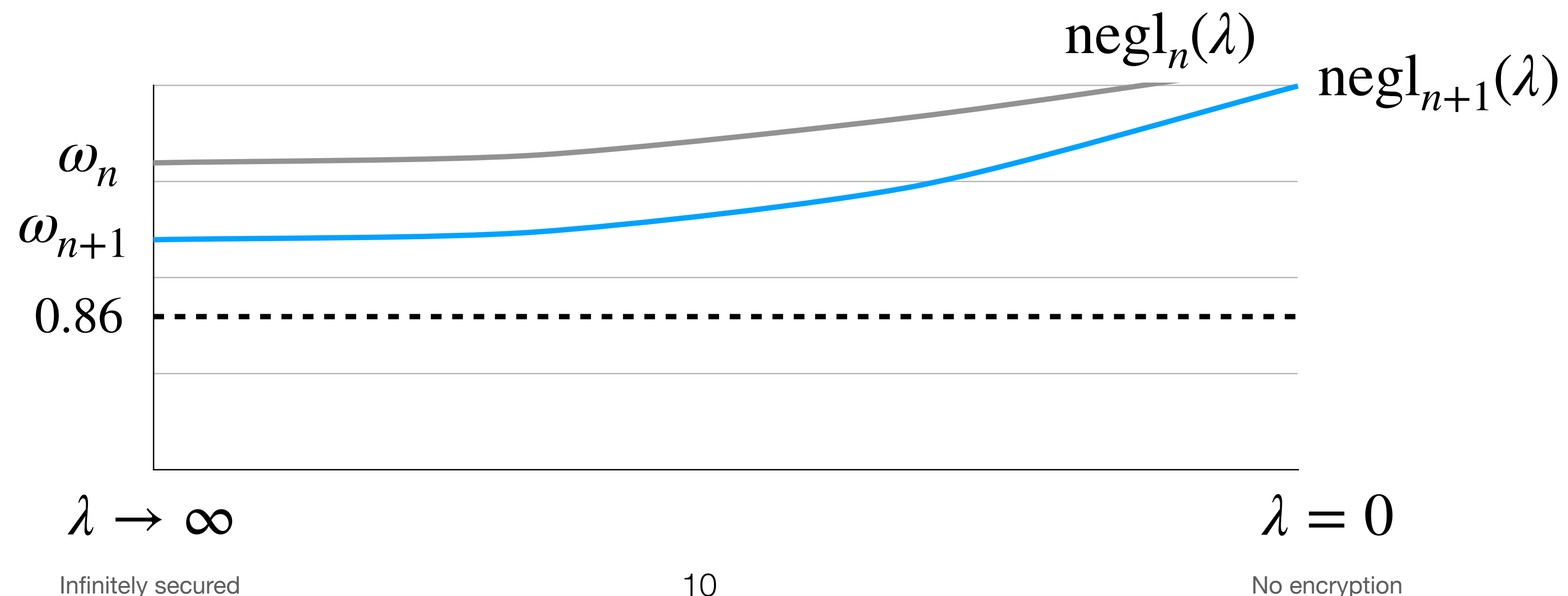
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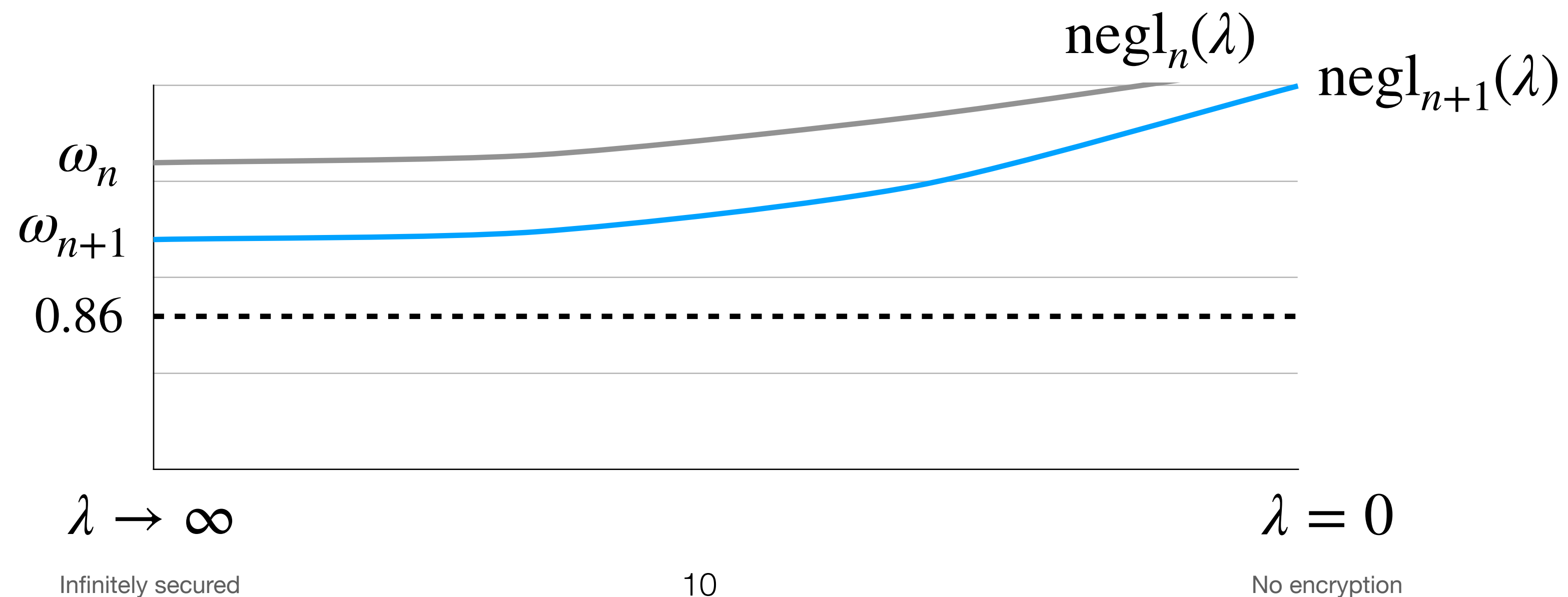
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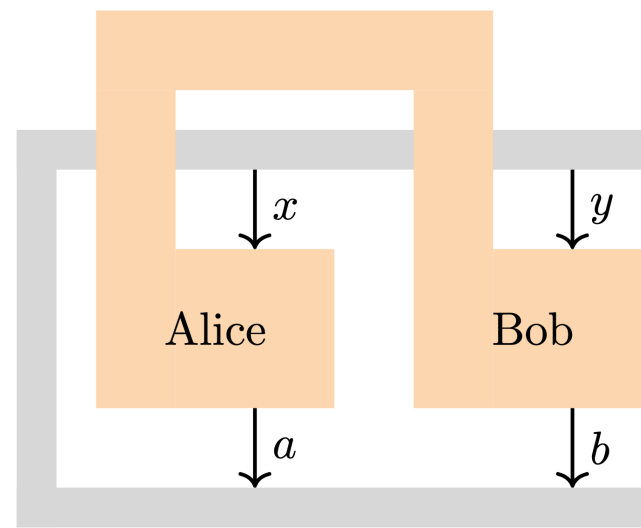
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- More generally, replace other information theoretical assumptions with cryptography?

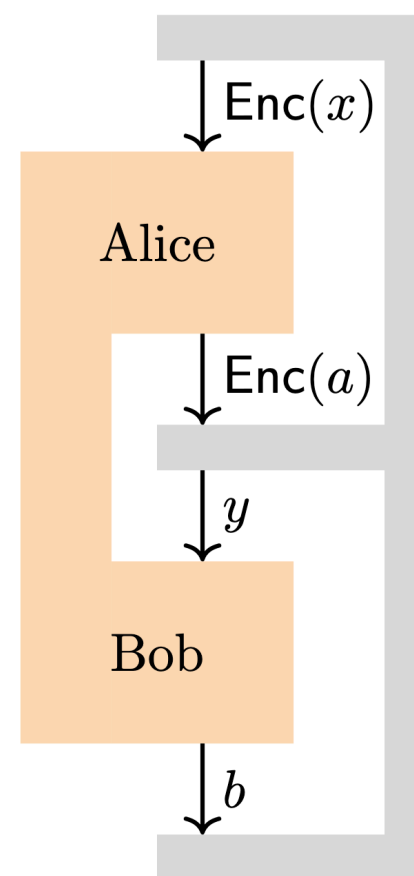
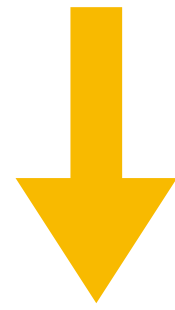
The three papers

- [BLJŠ25] arXiv: 2507.12408, accepted by SODA 2026. *Bounding the asymptotic quantum value of all multipartite compiled nonlocal games.* M. Baroni, D. Leichtle, S. Janković, I. Šupić
- [KPRSTXZ25] arXiv:2507.17006. *Quantitative Quantum Soundness for Bipartite Compiled Bell Games via the Sequential NPA Hierarchy.* I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, X. Xu, Y. Zhao
- [BKLRŠTX25] arXiv:2509.25145. *Quantitative quantum soundness for all multipartite compiled nonlocal games.* M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, X. Xu

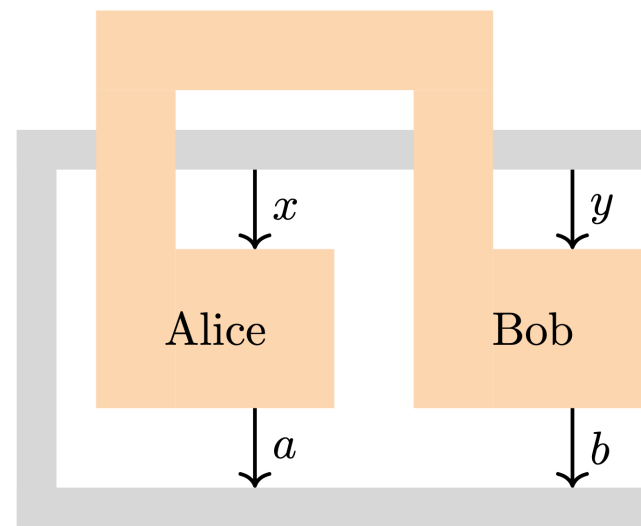
From bipartite to multipartite: [KMPSW24]



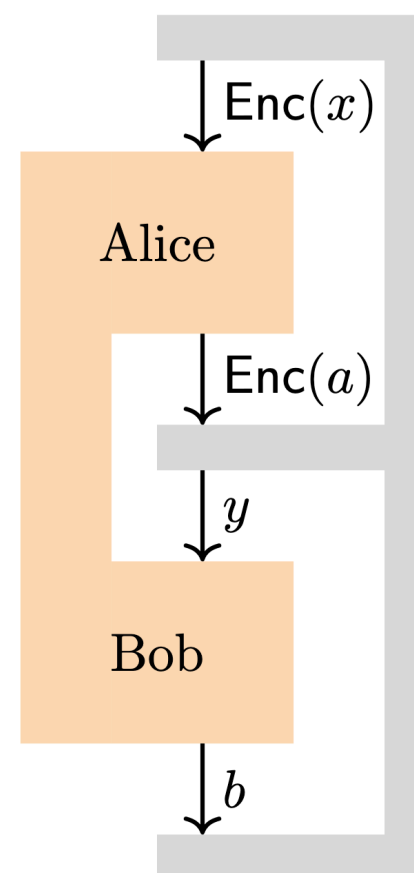
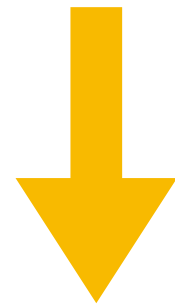
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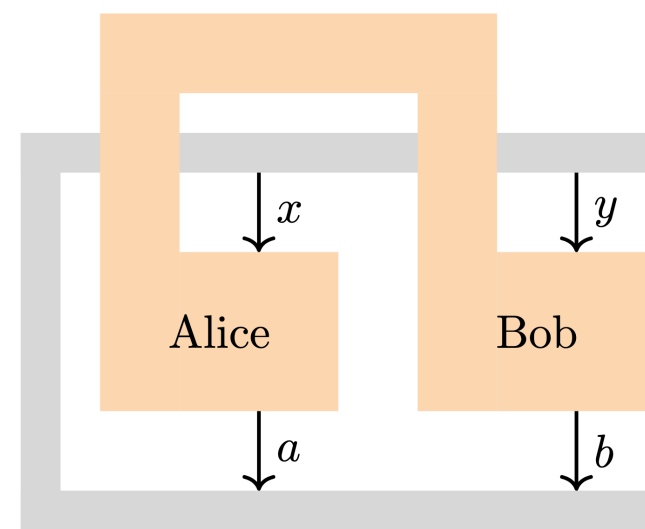


Sequential game

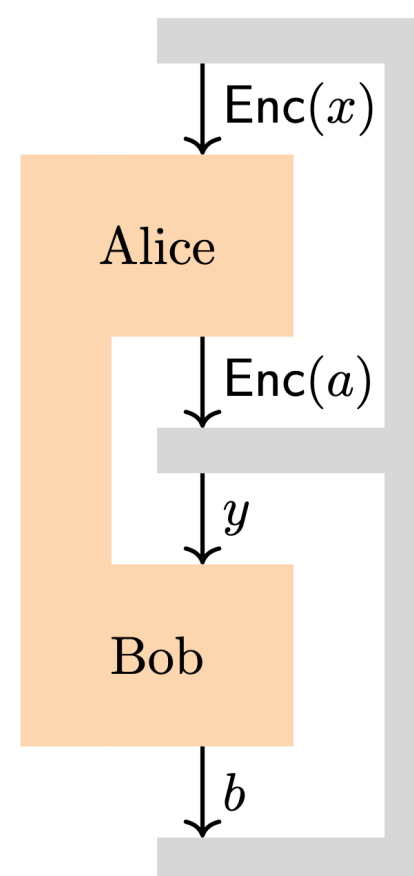
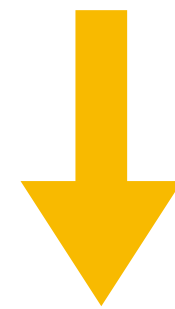


+ Secure against efficient provers
$$\left| \sum_a \rho_{a|x} - \sum_a \rho_{a|x'} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

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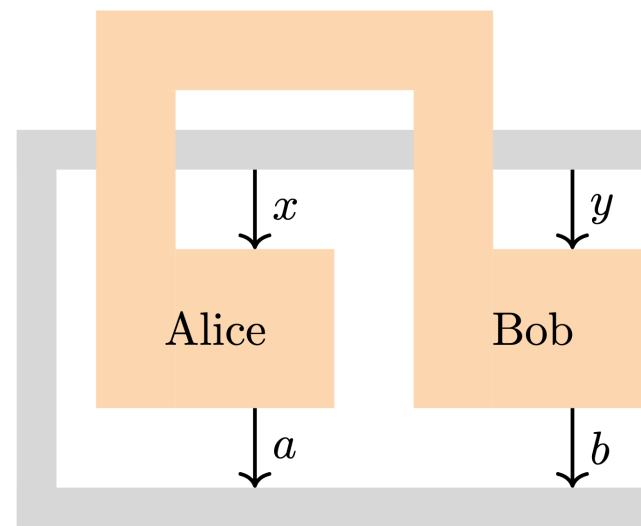
Sequential game



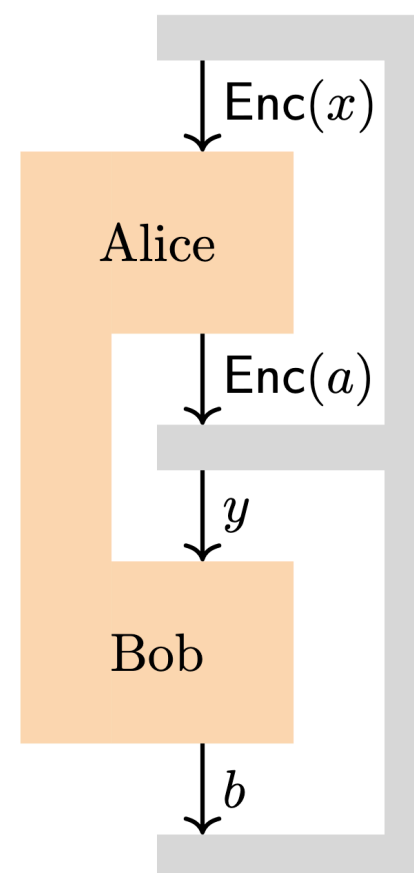
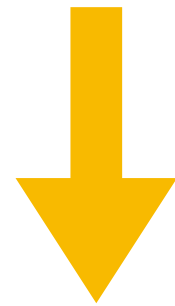
+ Secure against efficient provers

$$\left| \sum_a \rho_{a|x} - \sum_a \rho_{a|x'} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

From bipartite to multipartite: [KMPSW24]



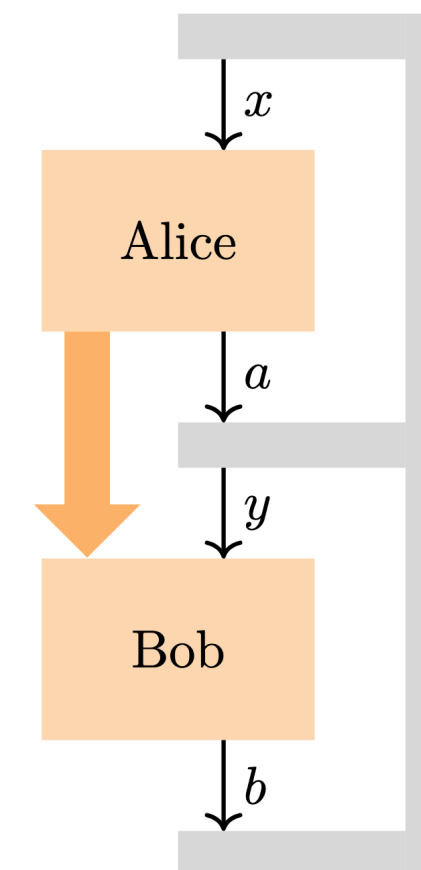
KL VY compiler



Sequential game



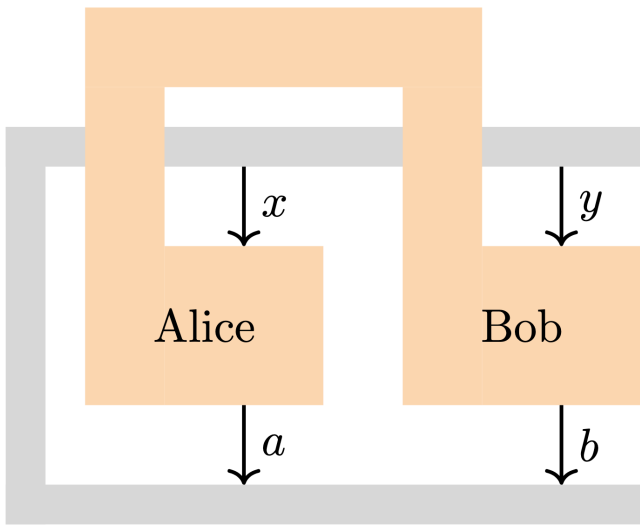
$\lambda \rightarrow \infty$



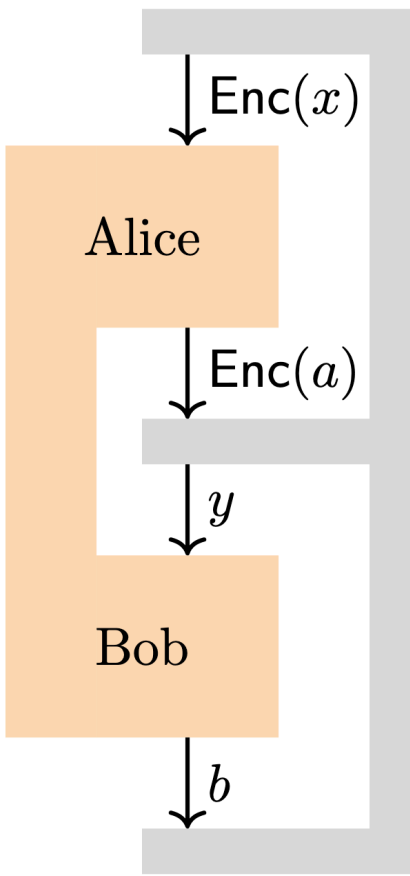
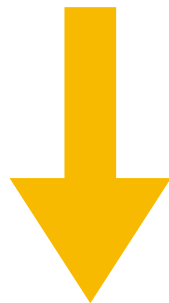
+ Secure against efficient provers

$$\left| \sum_a \rho_{a|x} - \sum_a \rho_{a|x'} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

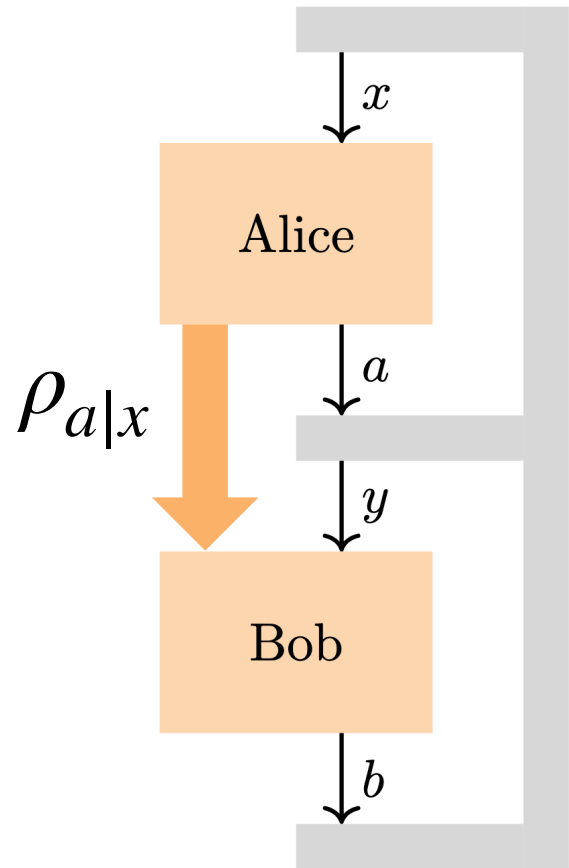
From bipartite to multipartite: [KMPSW24]



KL VY compiler



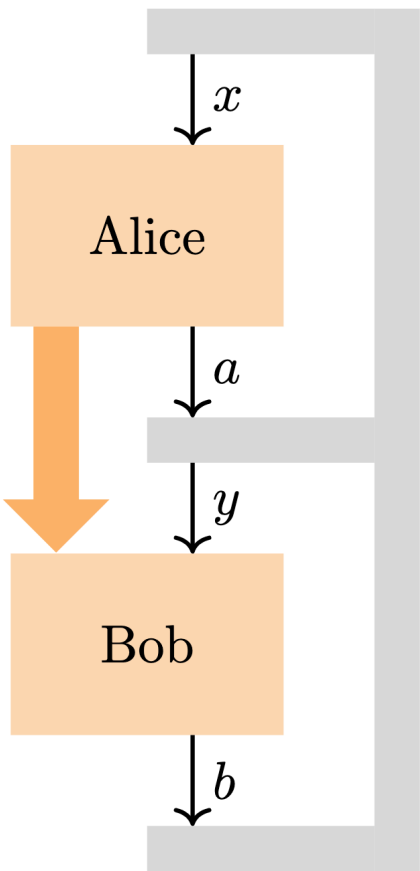
Sequential game



+ Preparation
equivalences

$$\sum_a \rho_{a|x} = \sum_a \rho_{a|x'} \quad \forall x, x'$$

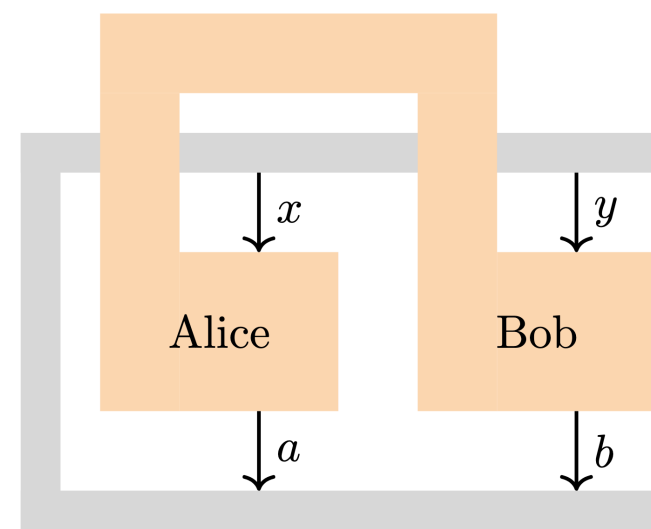
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+ Secure against efficient
provers

$$\left| \sum_a \rho_{a|x} - \sum_a \rho_{a|x'} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

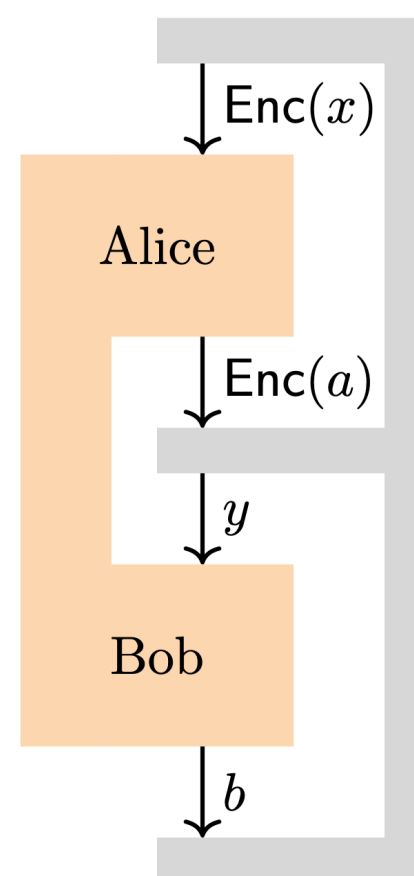
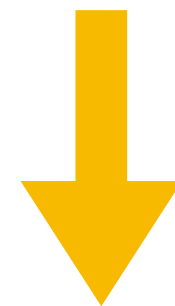
From bipartite to multipartite: [KMPSW24]



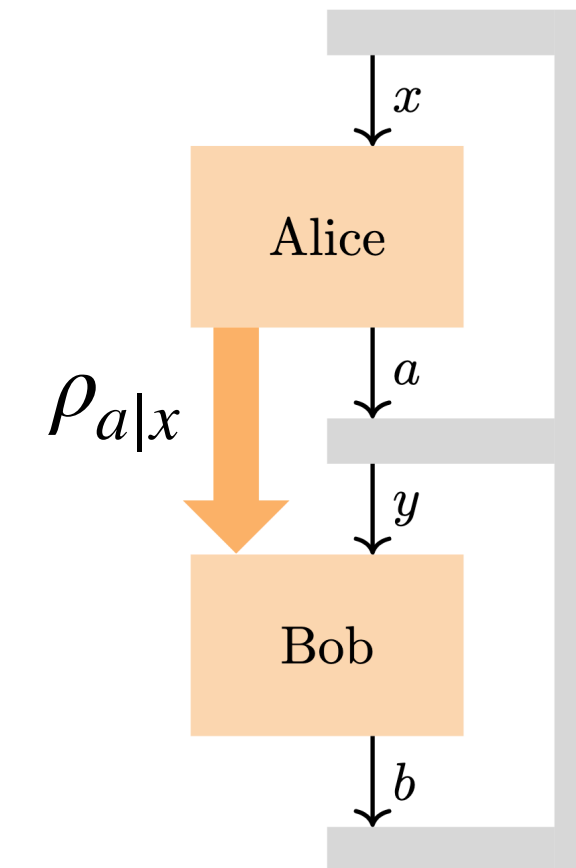
Non-local game



KLTY compiler



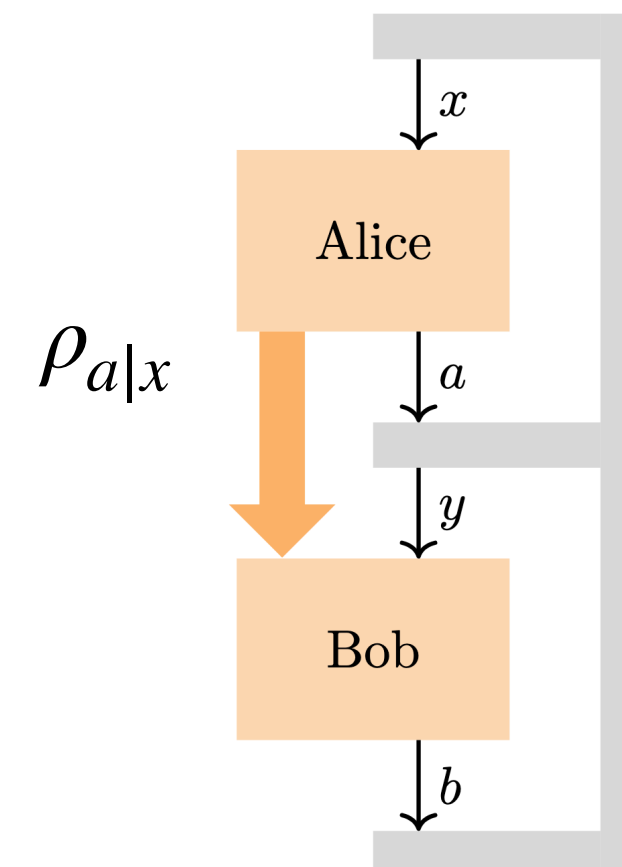
Sequential game



+ Preparation equivalences

$$\sum_a \rho_{a|x} = \sum_a \rho_{a|x'} \quad \forall x, x'$$

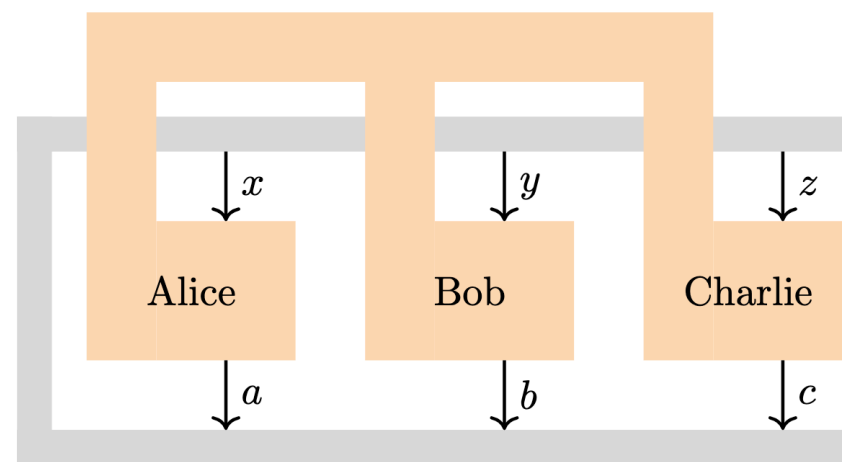
$\lambda \rightarrow \infty$



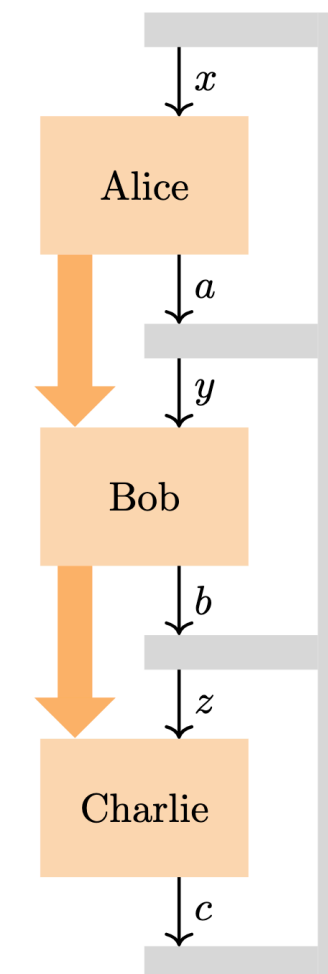
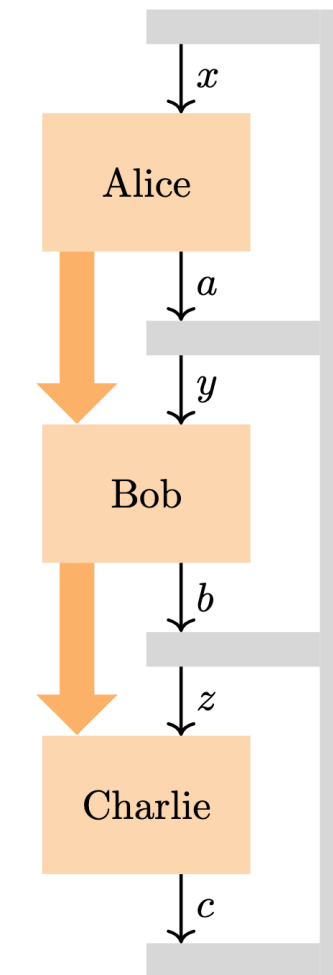
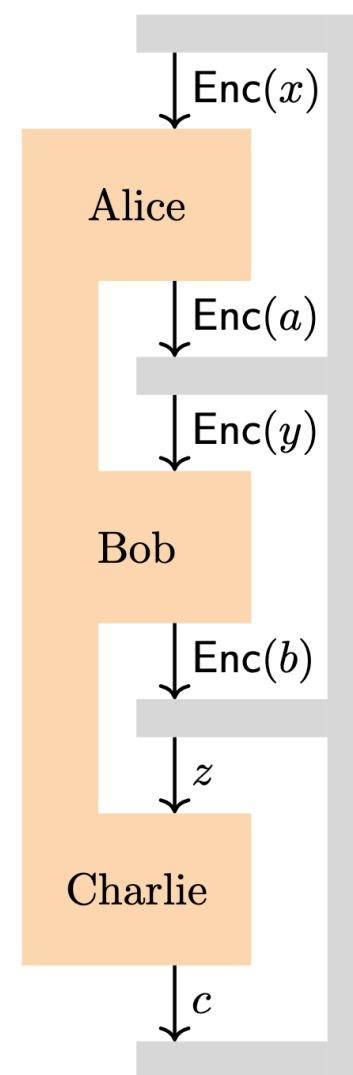
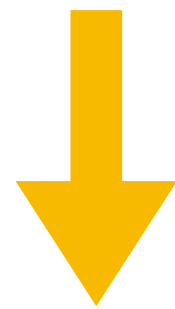
+ Secure against efficient provers

$$\left| \sum_a \rho_{a|x} - \sum_a \rho_{a|x'} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

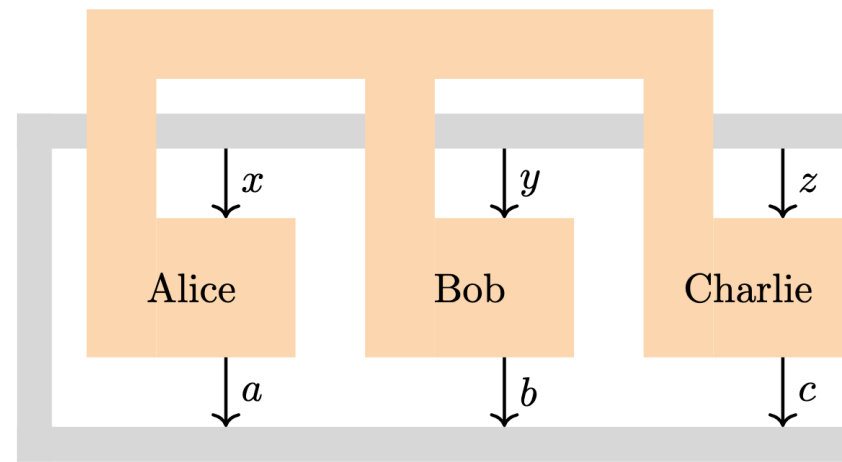
From bipartite to multipartite, naively



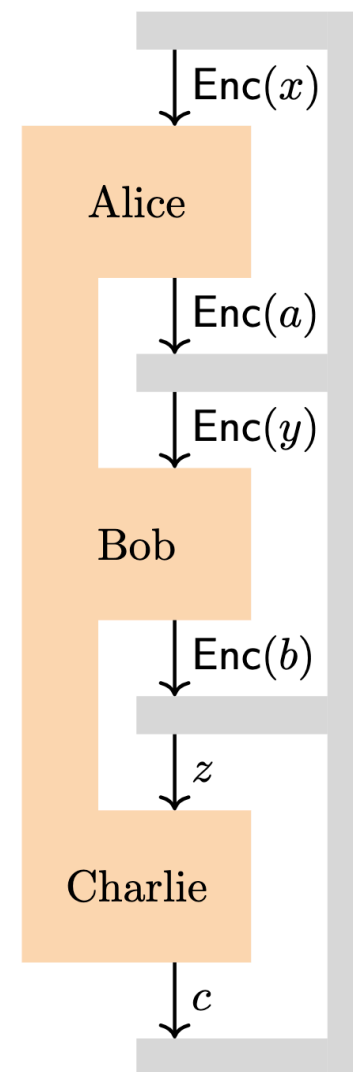
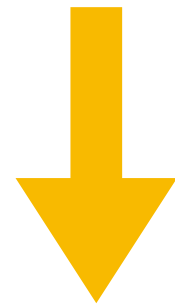
KLVY compiler



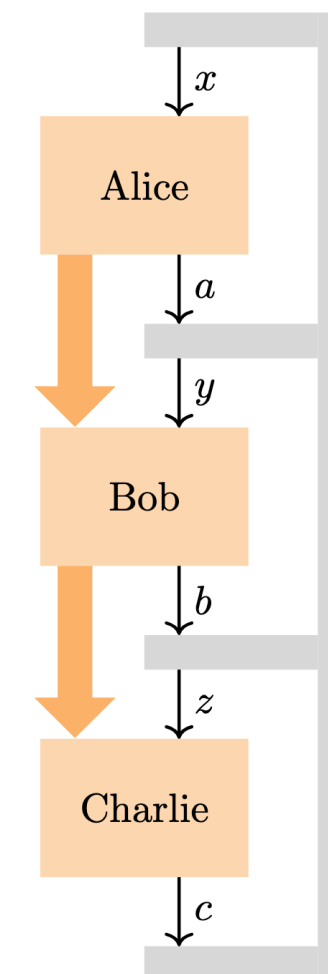
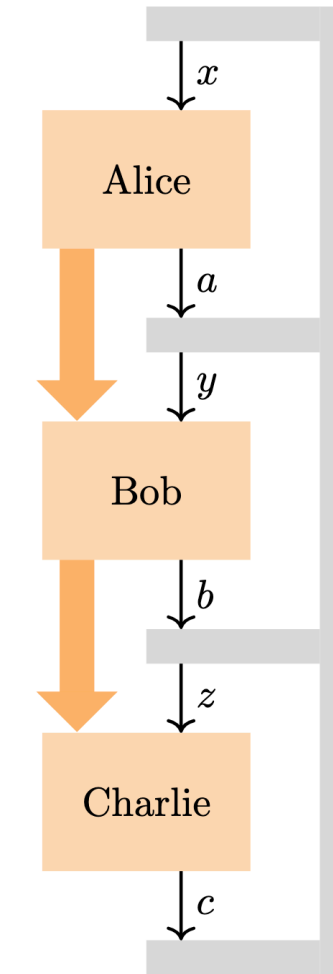
From bipartite to multipartite, naively



KLVY compiler



Sequential game

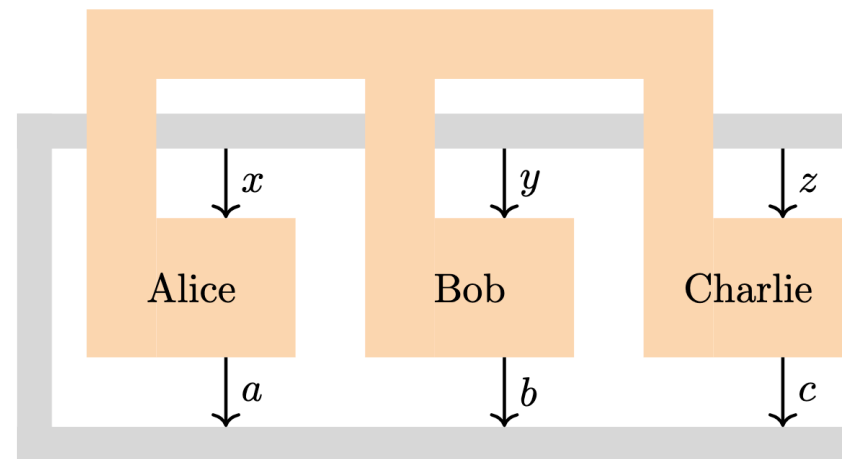


+ Secure against efficient provers

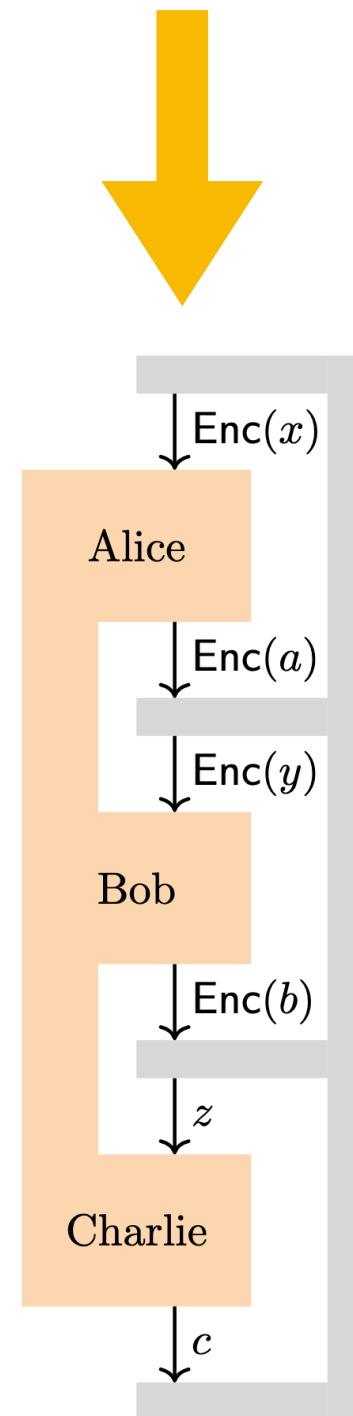
$$\left| \sum_a \rho_{ab|xy} - \sum_a \rho_{ab|x'y} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

$$\left| \sum_b \rho_{ab|xy} - \sum_b \rho_{ab|xy'} \right| \leq \text{negl}(\lambda) \quad \forall y, y'$$

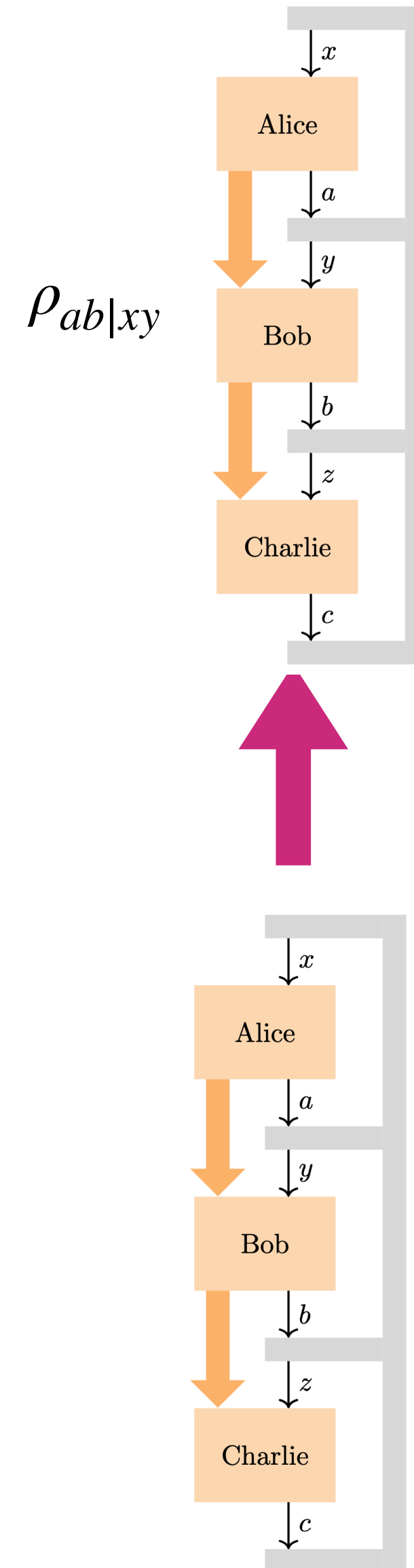
From bipartite to multipartite, naively



KL VY compiler



Sequential game



+ Preparation equivalences

$$\sum_a \rho_{ab|xy} = \sum_a \rho_{ab|x'y} \quad \forall x, x'$$

$$\sum_b \rho_{ab|xy} = \sum_b \rho_{ab|xy'} \quad \forall y, y'$$

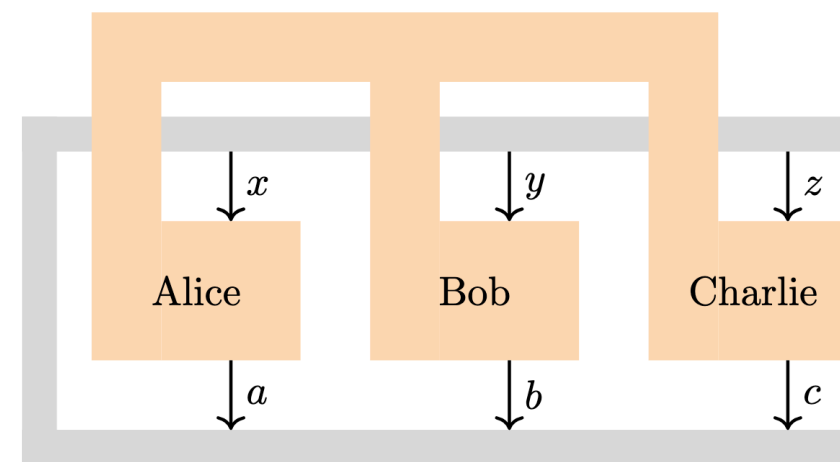
$$\lambda \rightarrow \infty$$

+ Secure against efficient provers

$$\left| \sum_a \rho_{ab|xy} - \sum_a \rho_{ab|x'y} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

$$\left| \sum_b \rho_{ab|xy} - \sum_b \rho_{ab|xy'} \right| \leq \text{negl}(\lambda) \quad \forall y, y'$$

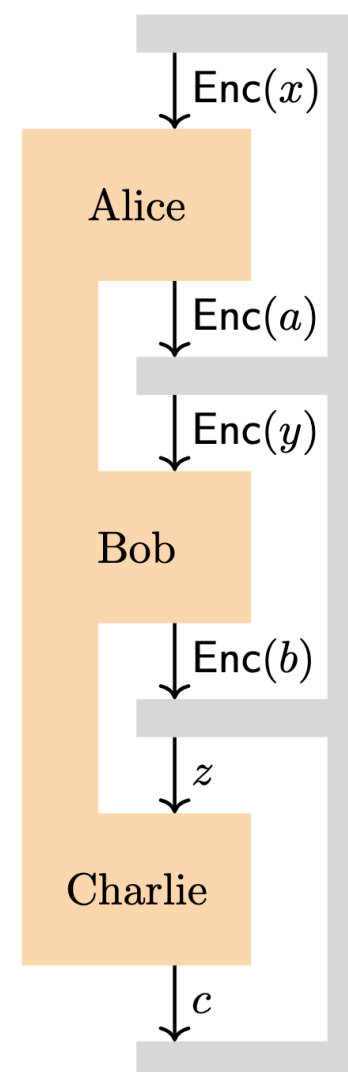
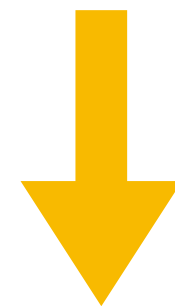
From bipartite to multipartite, naively



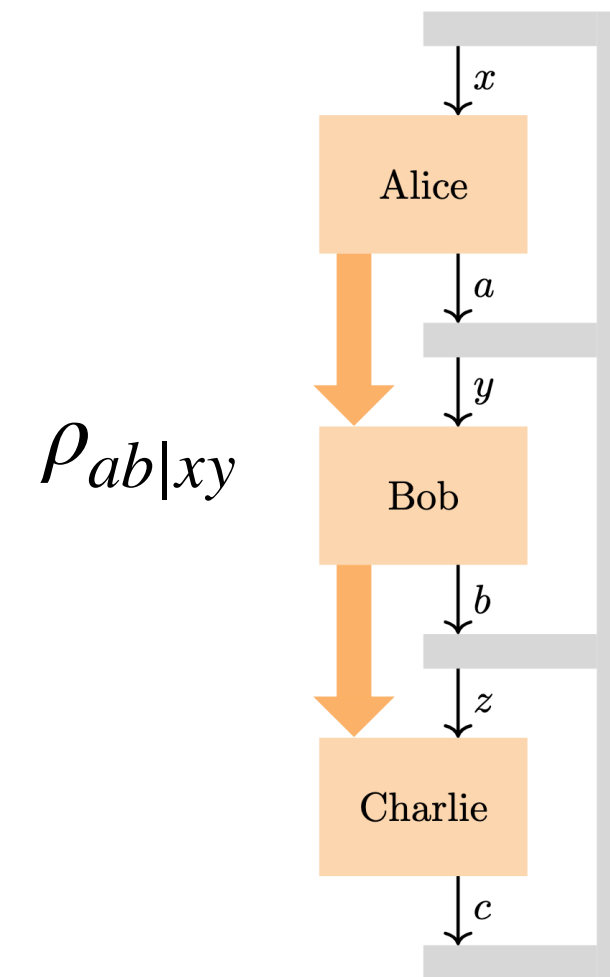
Non-local game



KL VY compiler



Sequential game



+ Preparation equivalences

$$\sum_a \rho_{ab|xy} = \sum_a \rho_{ab|x'y} \quad \forall x, x'$$

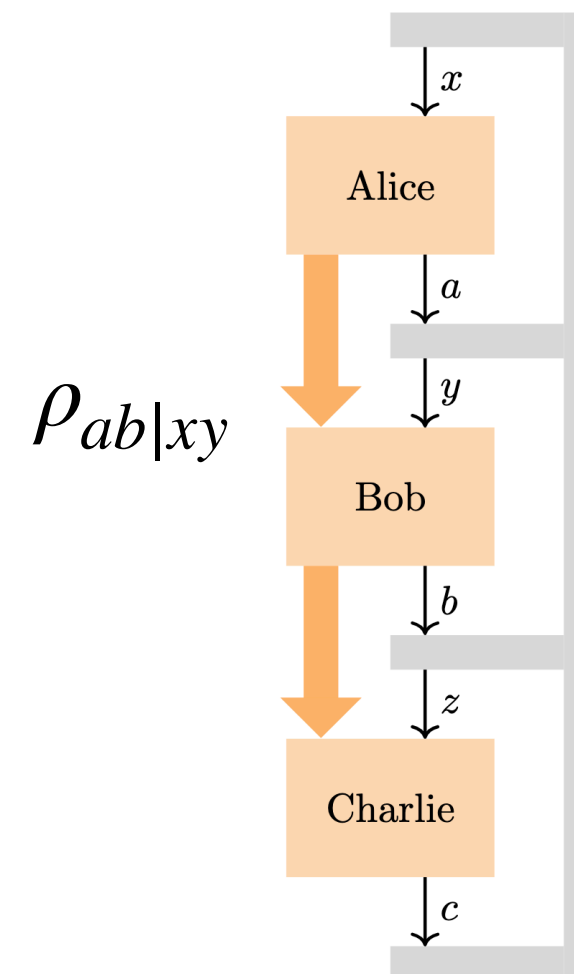
$$\sum_b \rho_{ab|xy} = \sum_b \rho_{ab|xy'} \quad \forall y, y'$$

$$\lambda \rightarrow \infty$$

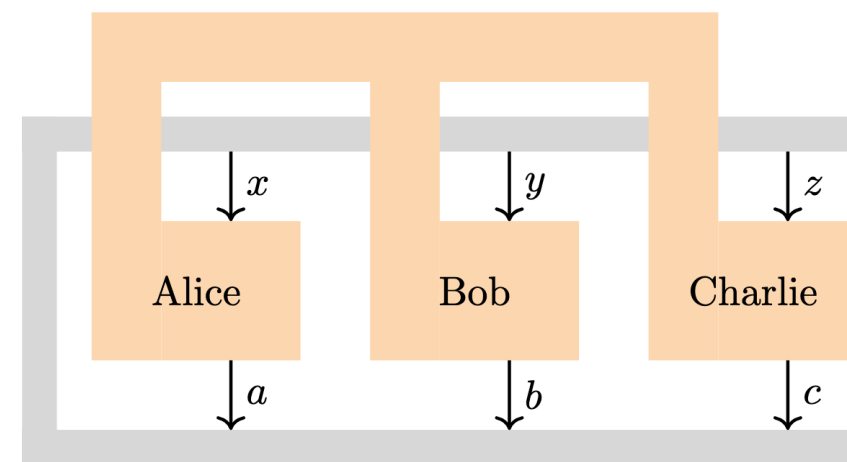
+ Secure against efficient provers

$$\left| \sum_a \rho_{ab|xy} - \sum_a \rho_{ab|x'y} \right| \leq \text{negl}(\lambda) \quad \forall x, x'$$

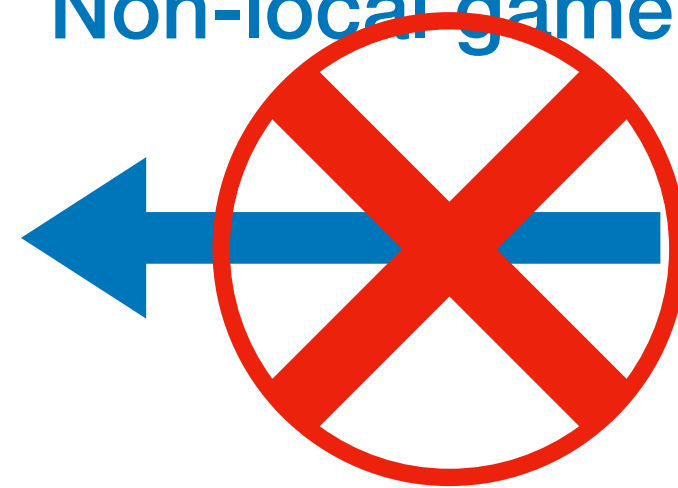
$$\left| \sum_b \rho_{ab|xy} - \sum_b \rho_{ab|xy'} \right| \leq \text{negl}(\lambda) \quad \forall y, y'$$



From bipartite to multipartite, naively

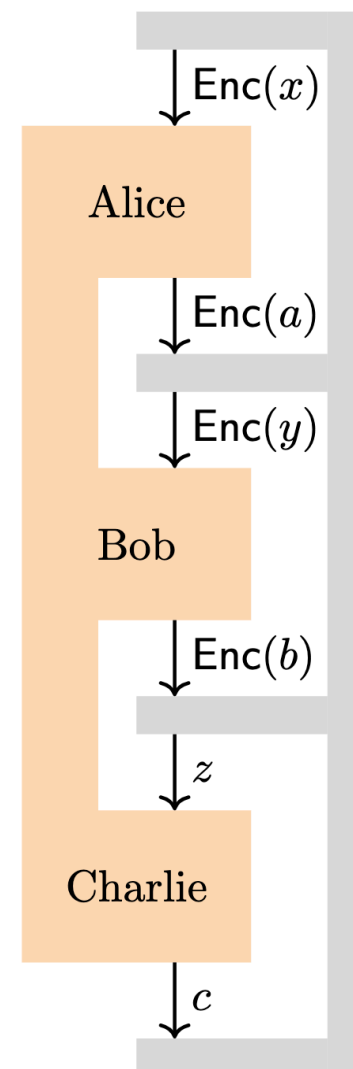
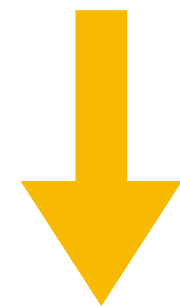


Non-local game

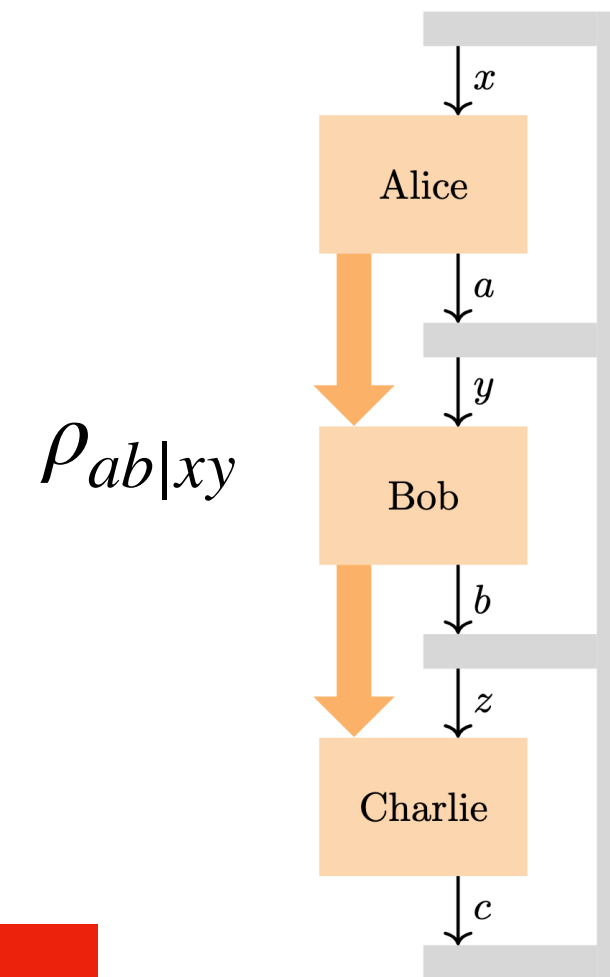


Schrödinger's picture, too restrictive

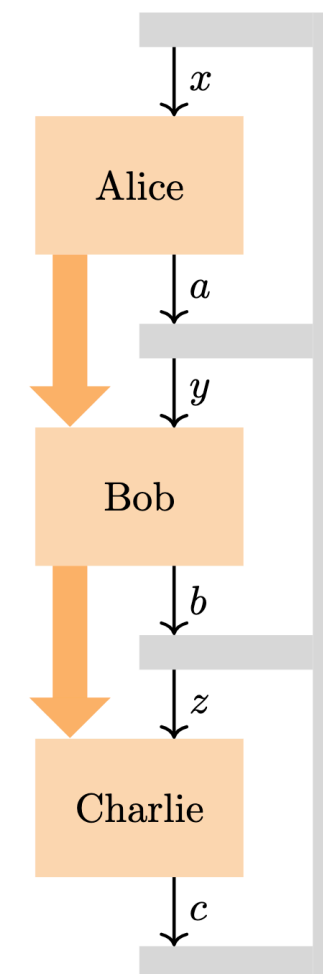
KLTY compiler



Sequential game



$\rho_{ab|xy}$



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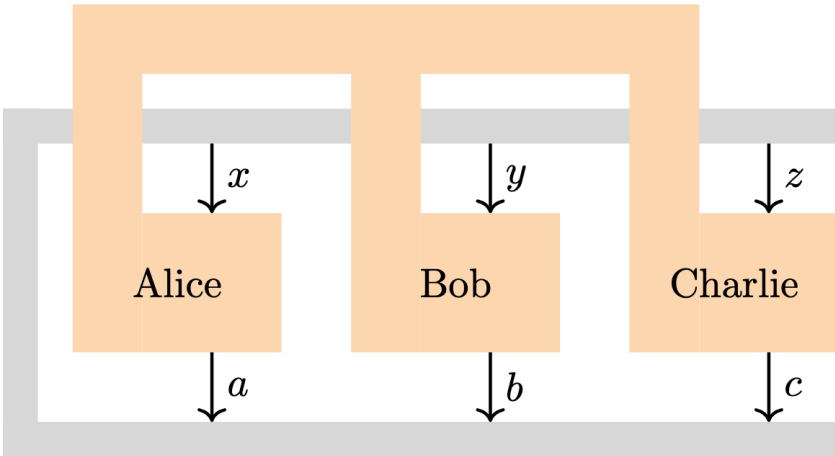
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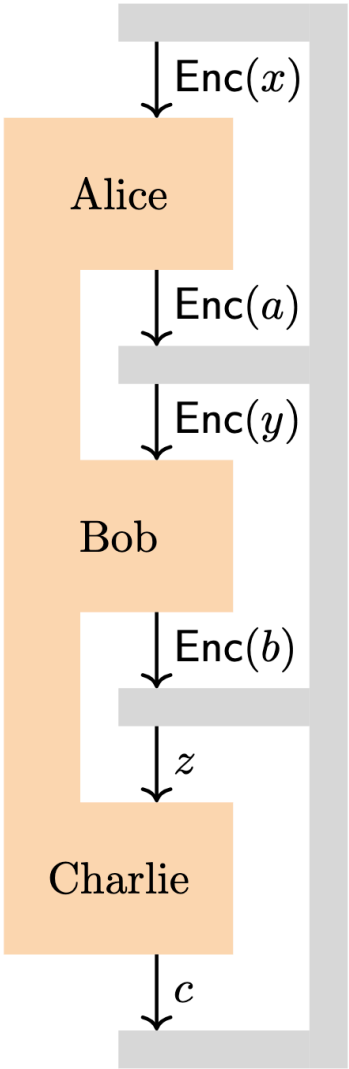
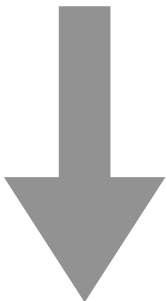
From bipartite to multipartite: [BLJŠ25]



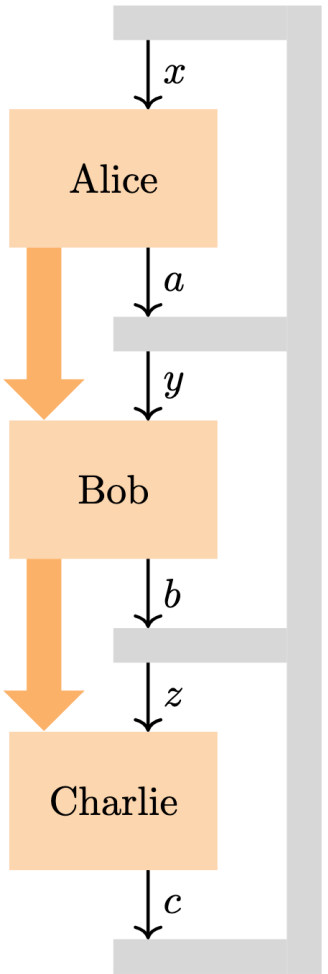
Non-local game



KLVY compiler

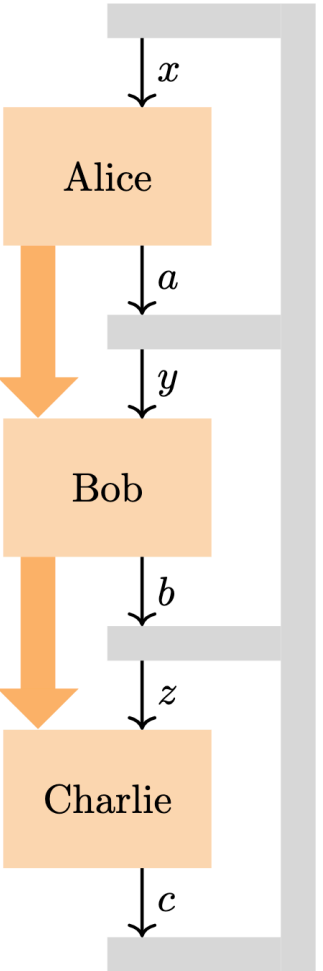


Sequential game

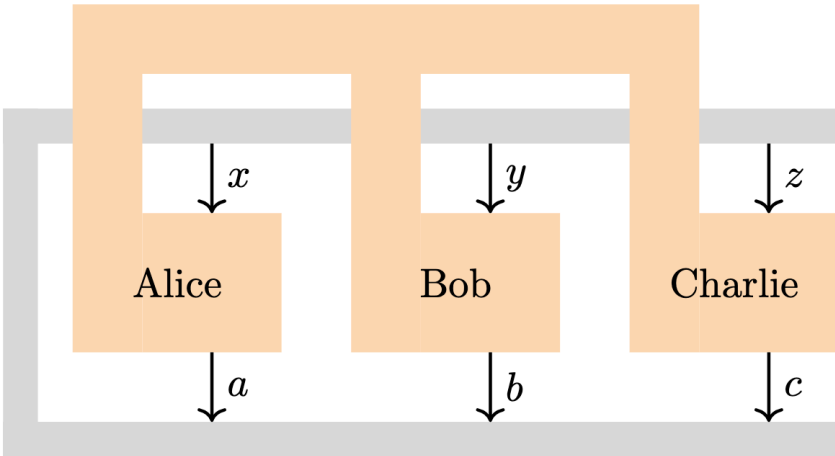


+ Preparation equivalences

$\lambda \rightarrow \infty$



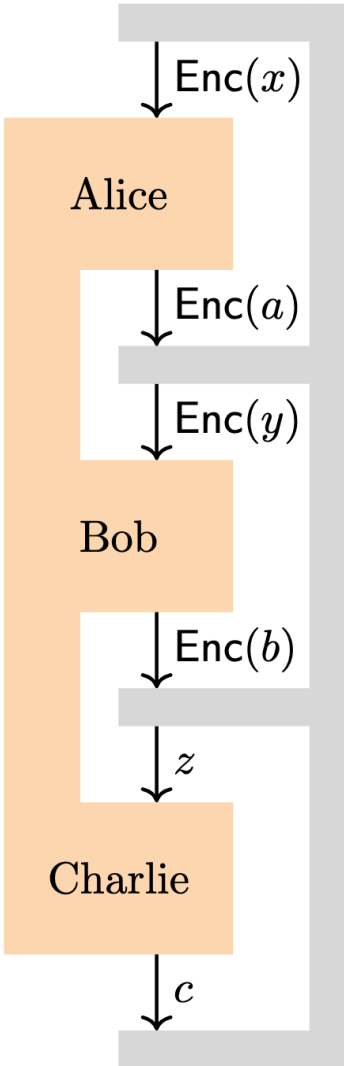
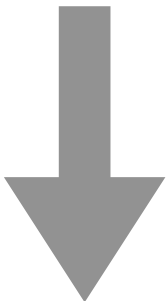
From bipartite to multipartite: [BLJŠ25]



Non-local game



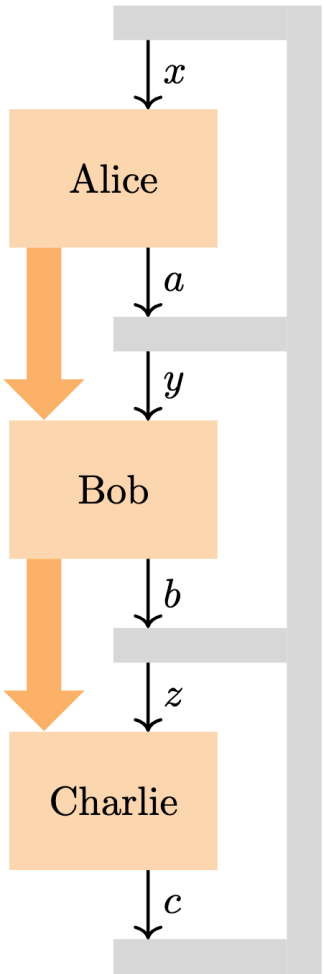
KLVY compiler



Sequential game

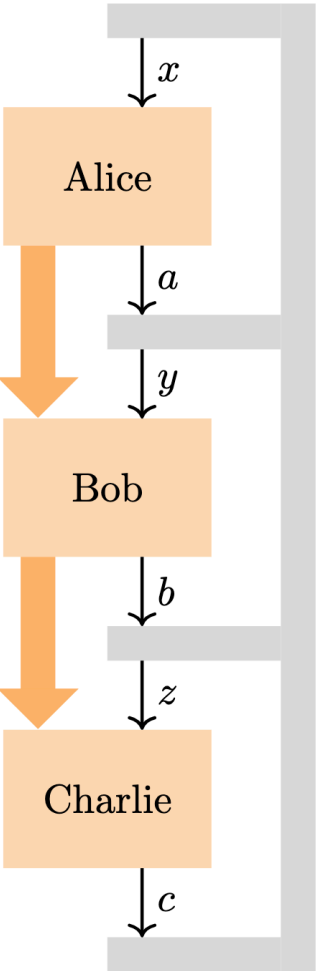


New mapping

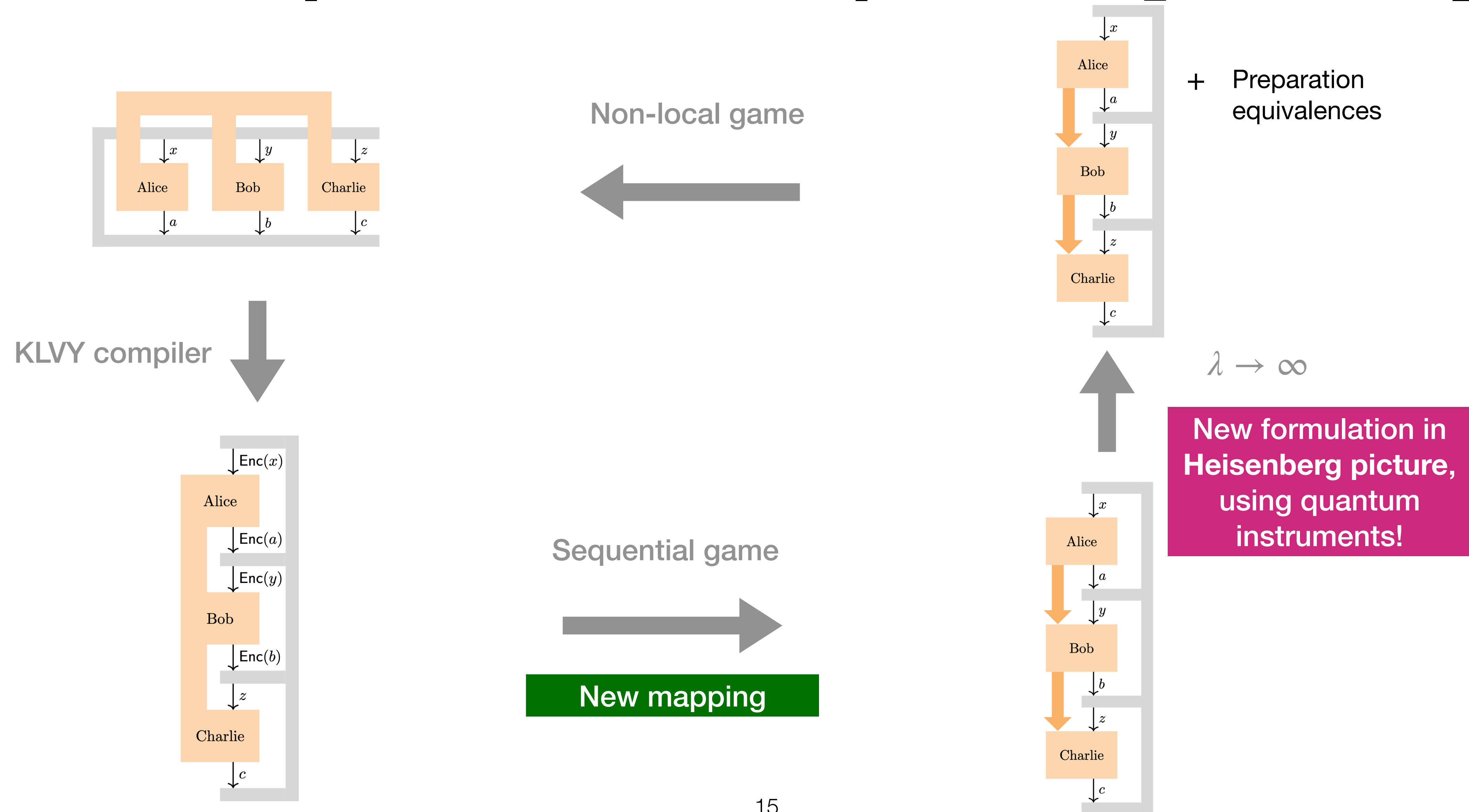


+ Preparation
equivalences

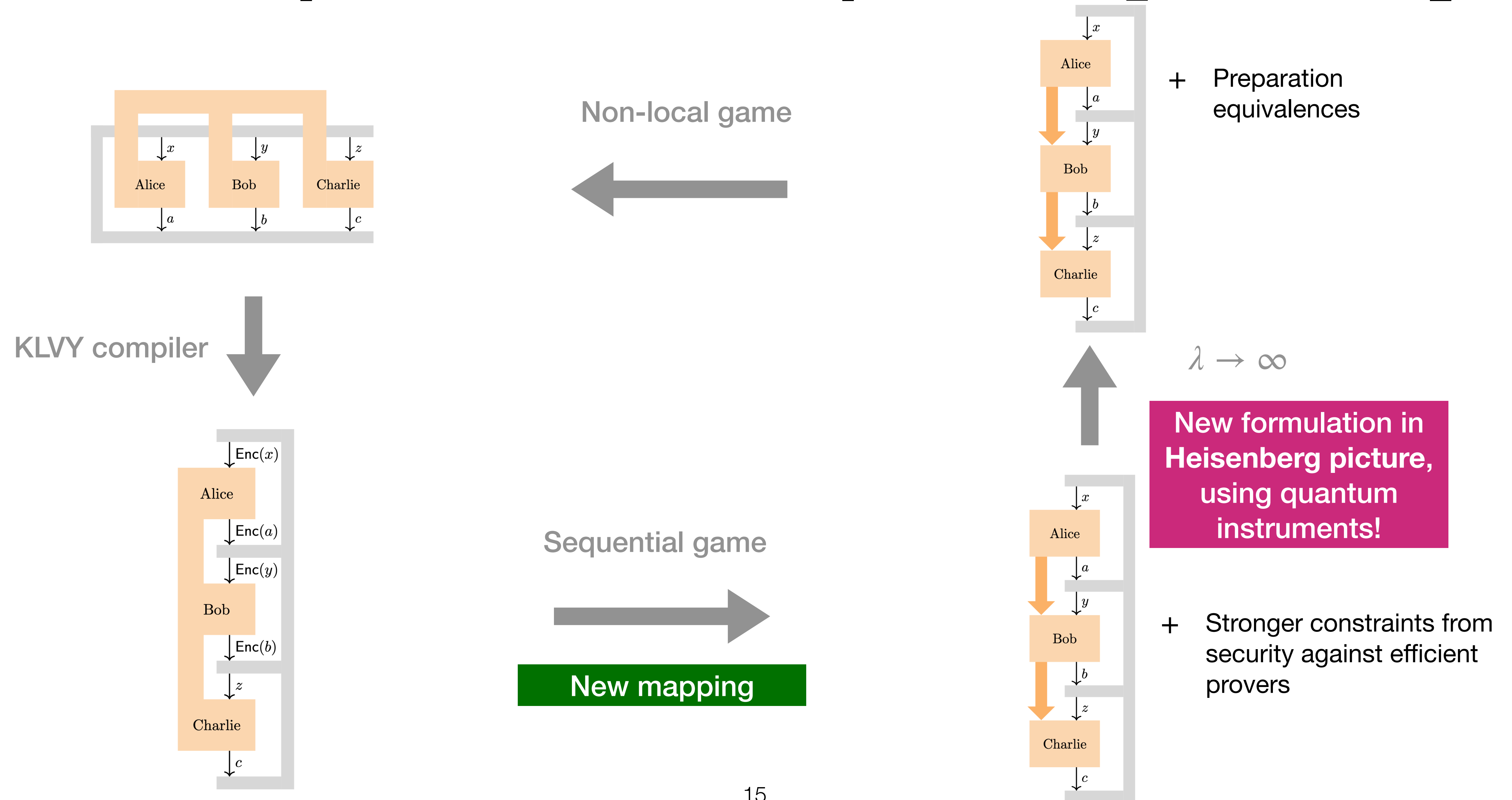
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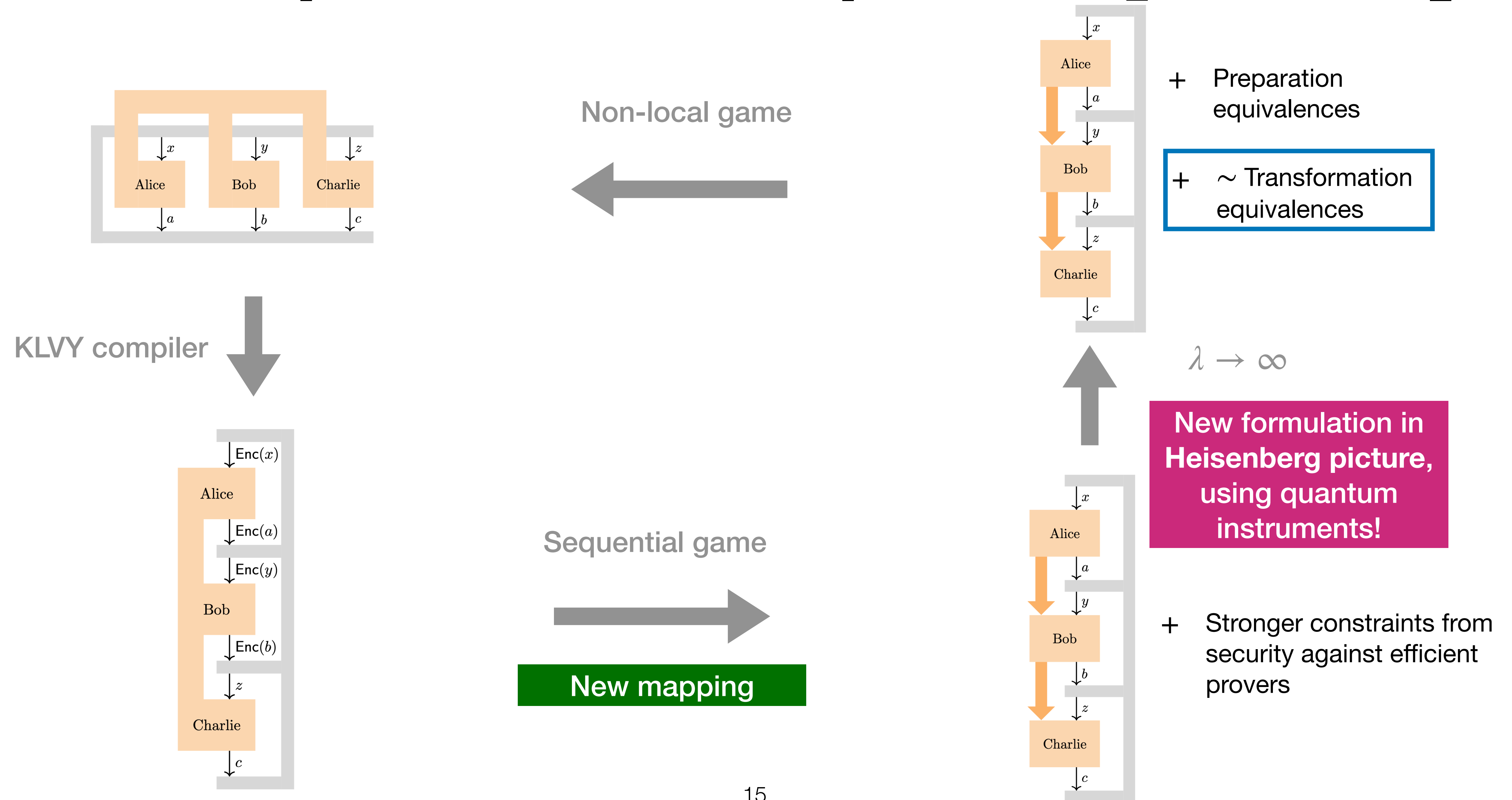
From bipartite to multipartite: [BLJŠ25]



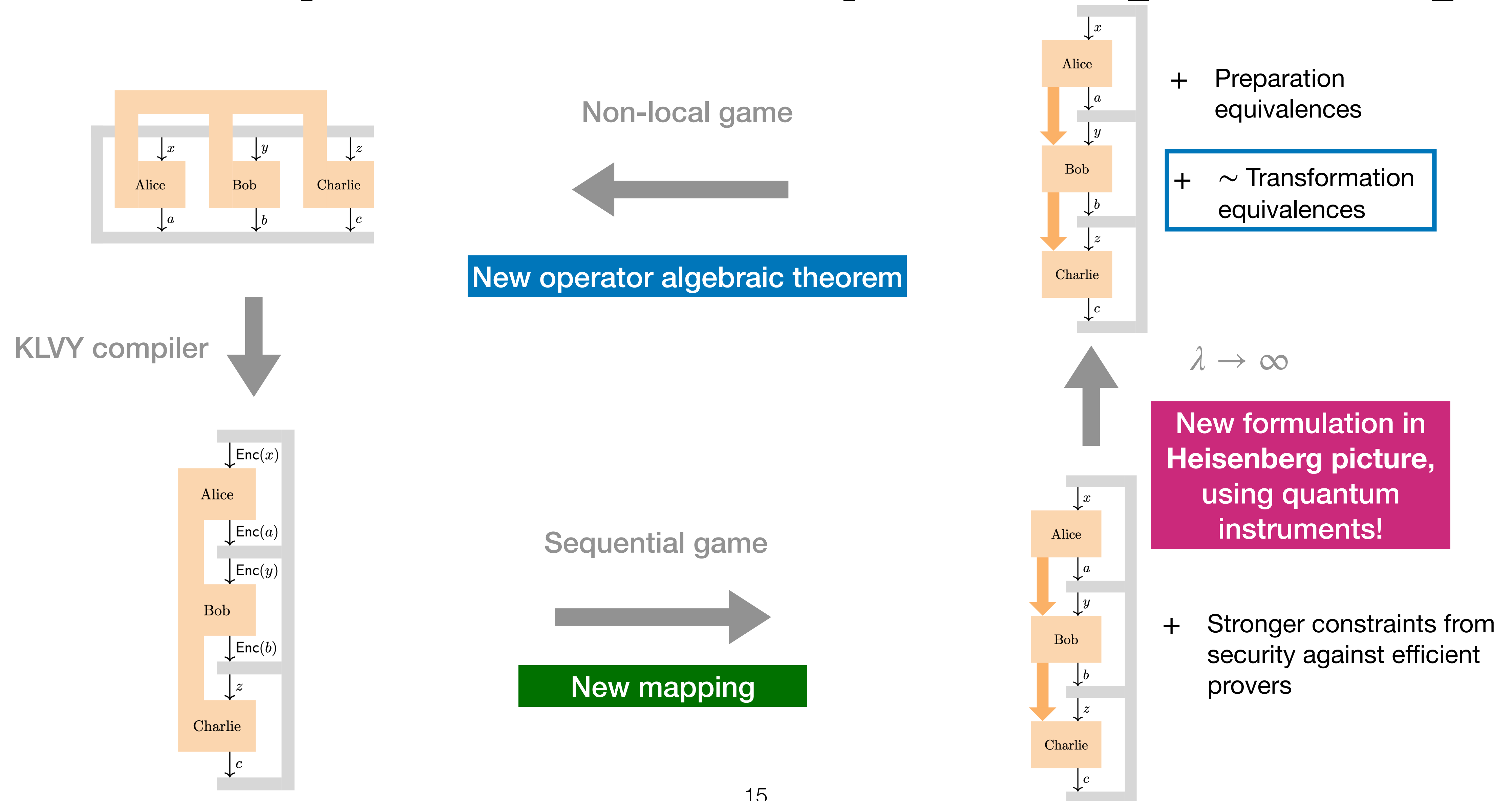
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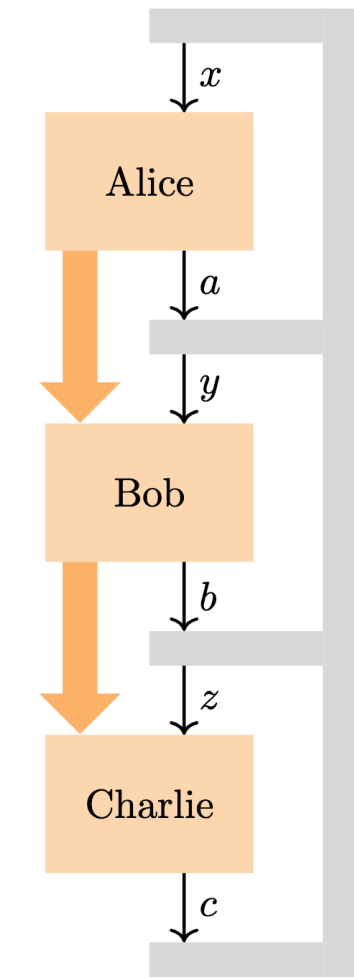
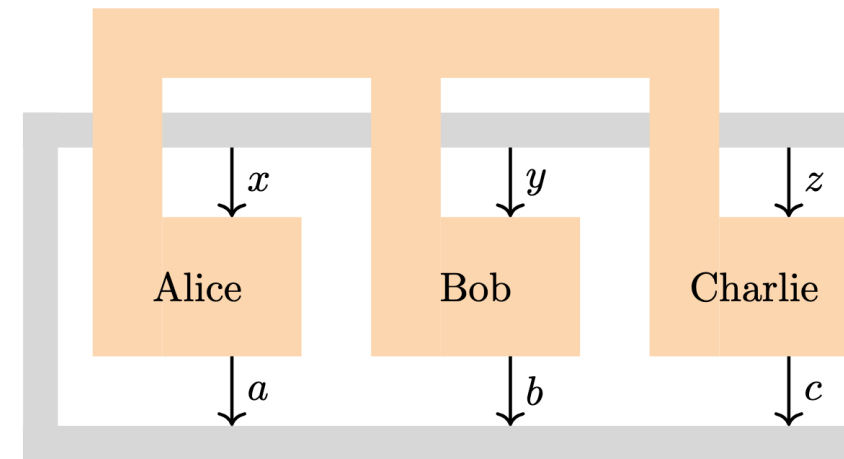
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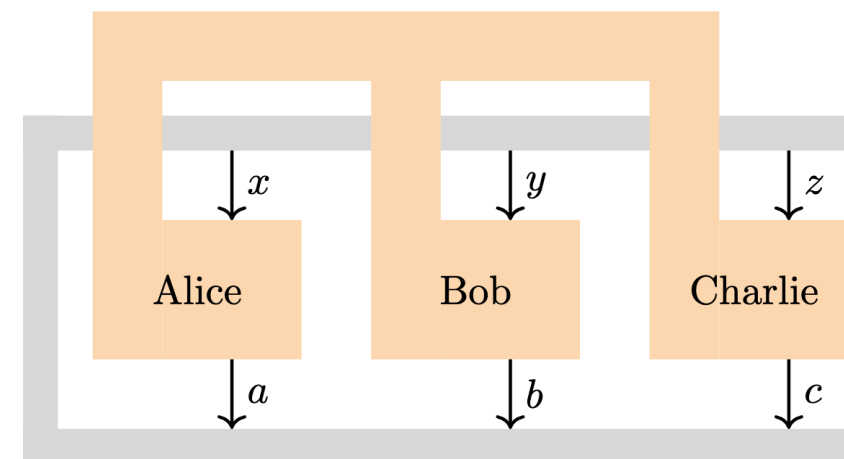
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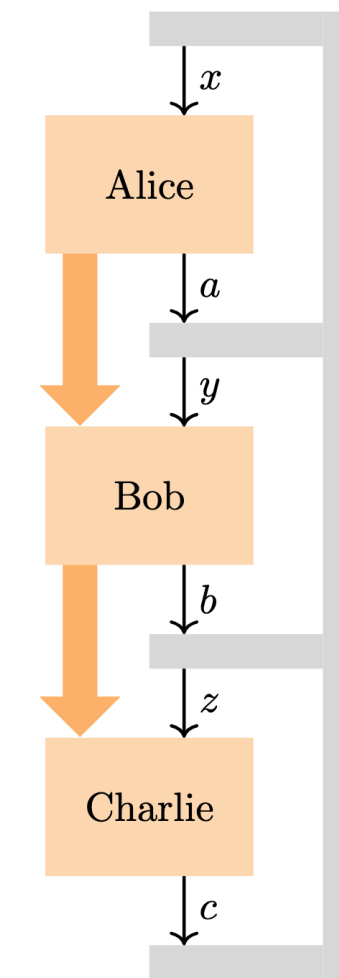
A sequential NPA-like hierarchy: [KPRSTXZ25, BKLRŠTX25]



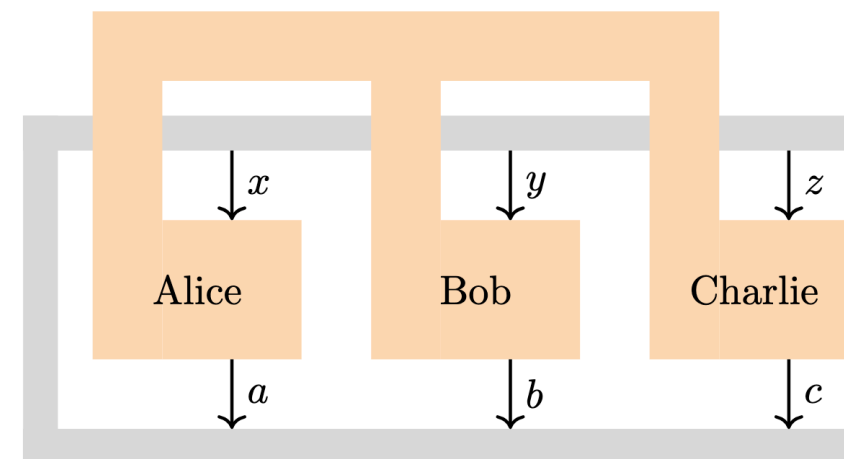
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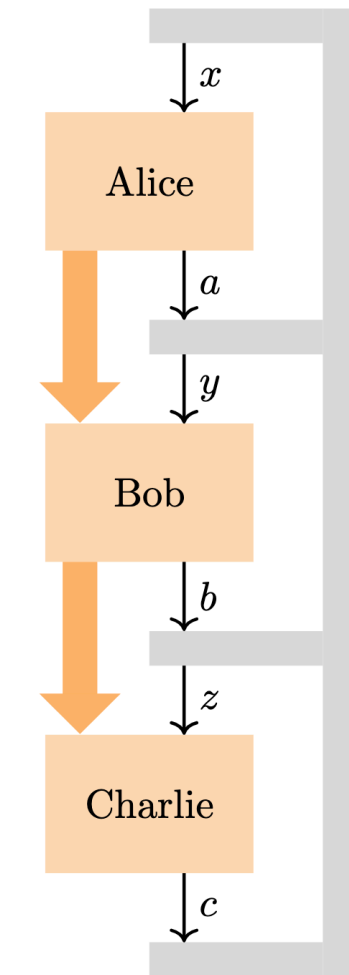
- NPA hierarchy = hierarchy of semidefinite programs, indexed by n



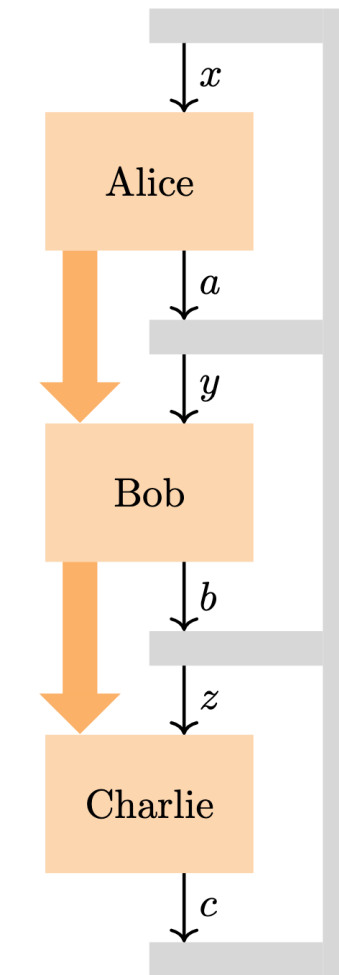
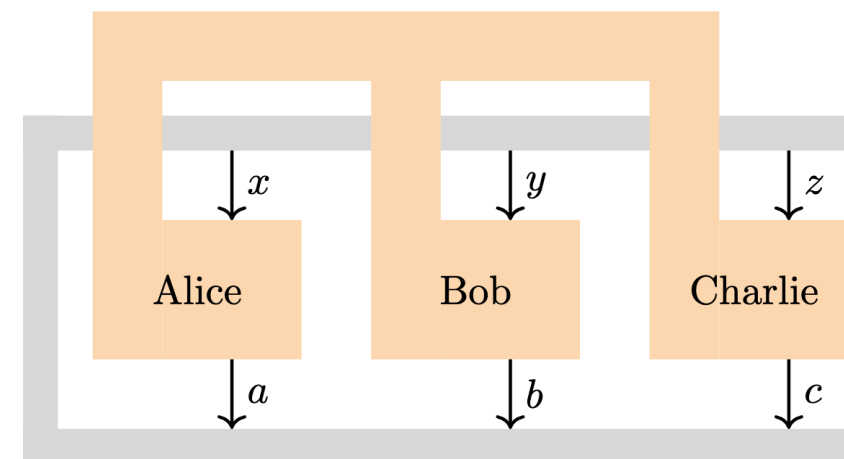
A sequential NPA-like hierarchy: [KPRSTXZ25, BKLRŠTX25]



- NPA hierarchy = hierarchy of semidefinite programs, indexed by n
- At each n , gives upper bound on quantum scores $\omega_{\text{NPA}}^n(\mathcal{G})$



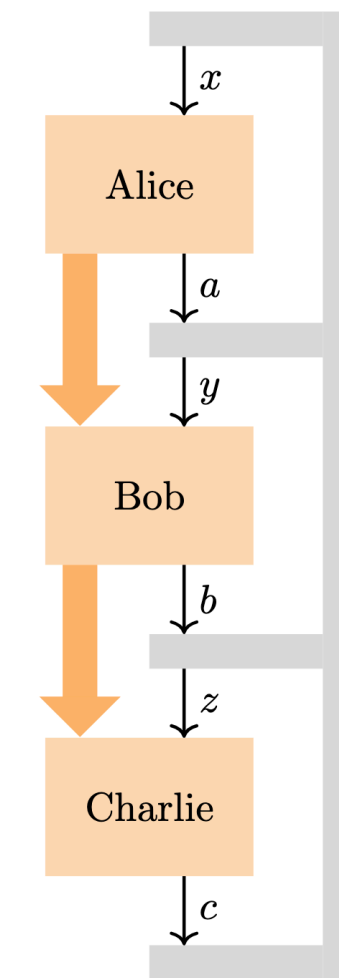
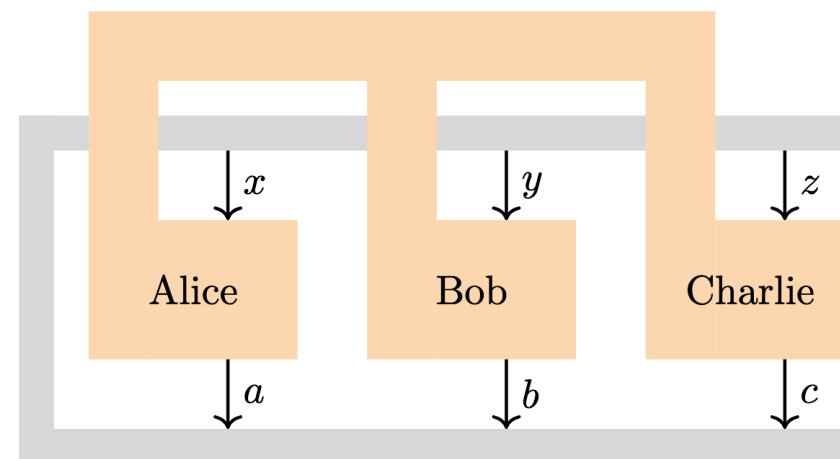
A sequential NPA-like hierarchy: [KPRSTXZ25, BKLRŠTX25]



- NPA hierarchy = hierarchy of semidefinite programs, indexed by n
- At each n , gives upper bound on quantum scores $\omega_{\text{NPA}}^n(\mathcal{G})$
- Increasingly tight and convergent to quantum score

$$\omega_{\text{NPA}}^1(\mathcal{G}) \geq \omega_{\text{NPA}}^2(\mathcal{G}) \geq \dots \geq \omega_{\text{qc}}(\mathcal{G})$$

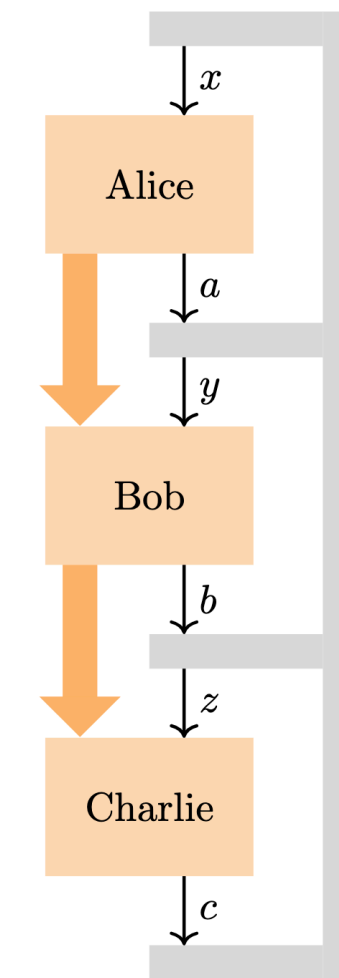
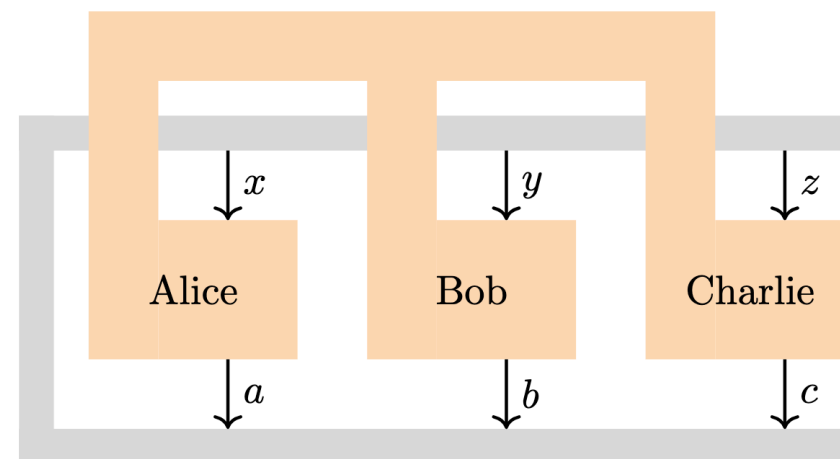
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- Our works, NPA-like hierarchy for sequential scenarios

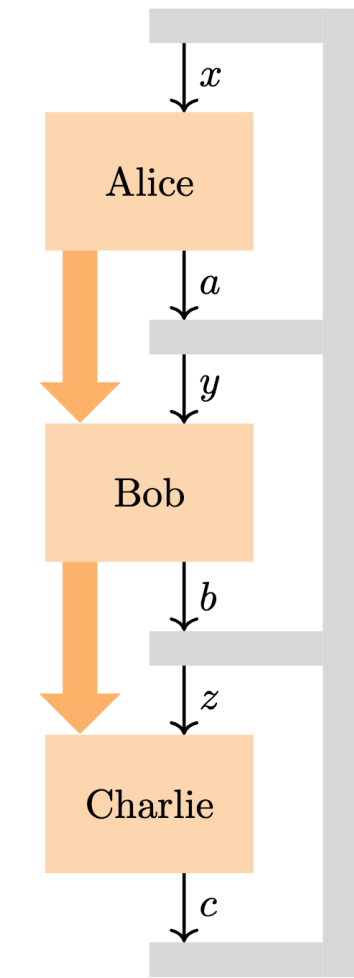
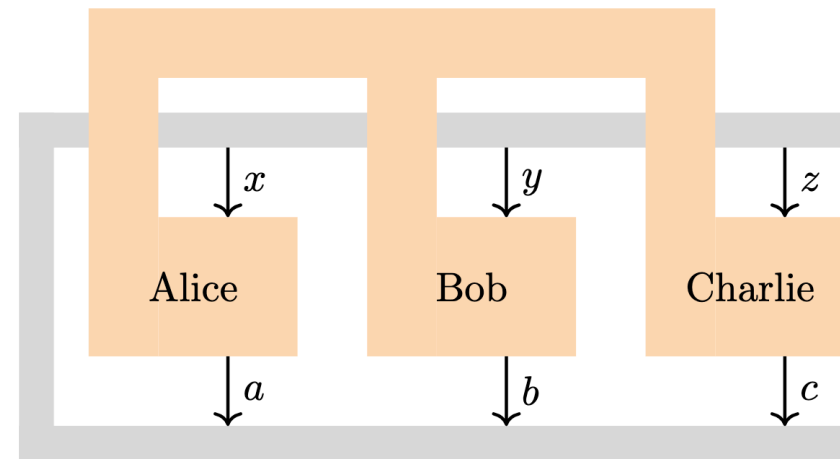
A sequential NPA-like hierarchy: [KPRSTXZ25, BKLRŠTX25]



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A sequential NPA-like hierarchy: [KPRSTXZ25, BKLRŠTX25]



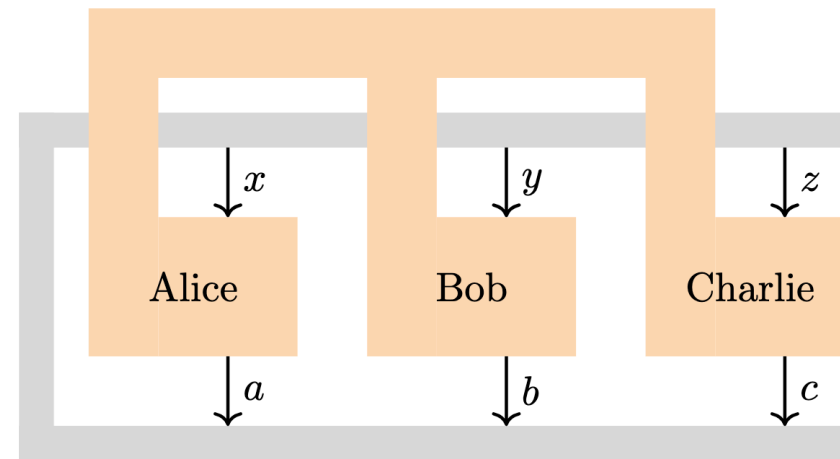
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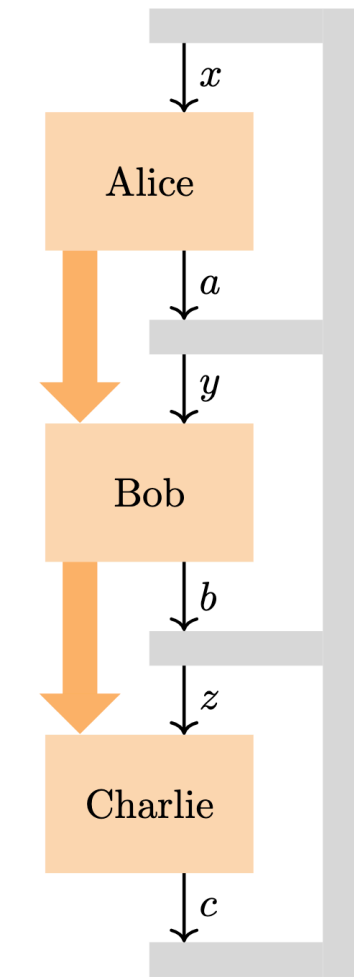
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$$\omega_{\text{seqNPA}}^1(\mathcal{G}) \geq \omega_{\text{seqNPA}}^2(\mathcal{G}) \geq \dots \geq \omega_{\text{qc}}(\mathcal{G})$$

A sequential NPA-like hierarchy: [KPRSTXZ25, BKLRŠTX25]



Very natural in the
Heisenberg picture!

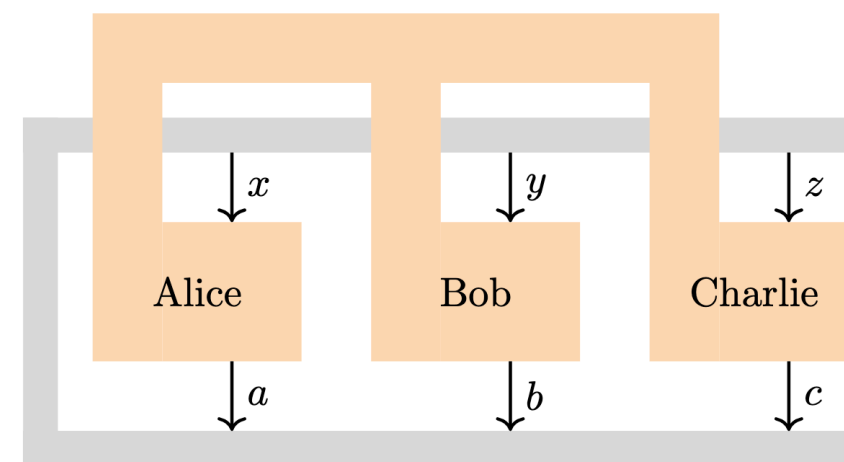


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- Encoding quantum instruments/ completely positive maps (first of its kind)
- Increasingly tight and convergent to quantum score

$$\omega_{\text{seqNPA}}^1(\mathcal{G}) \geq \omega_{\text{seqNPA}}^2(\mathcal{G}) \geq \dots \geq \omega_{\text{qc}}(\mathcal{G})$$

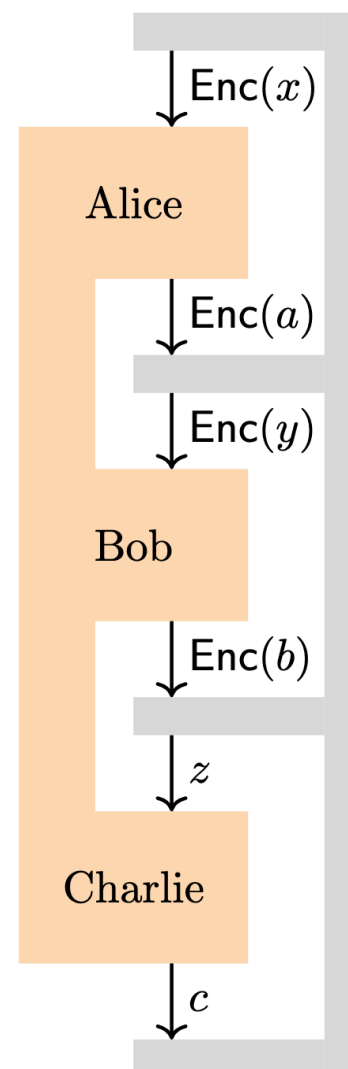
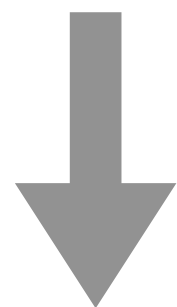
Asymptotic to quantitative: [KPRSTXZ25, BKLRŠTX25],



Non-local game



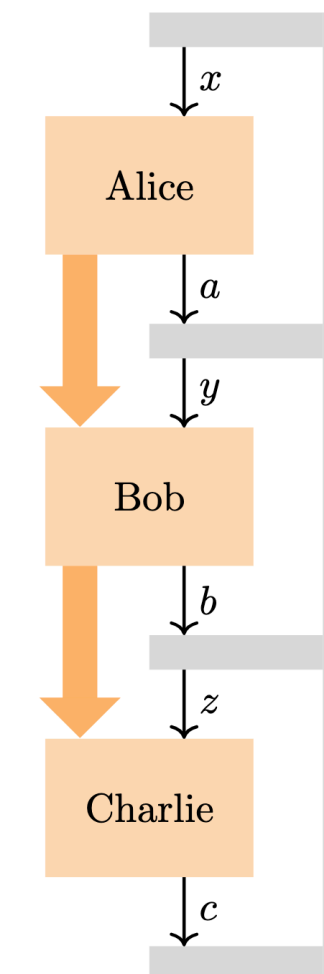
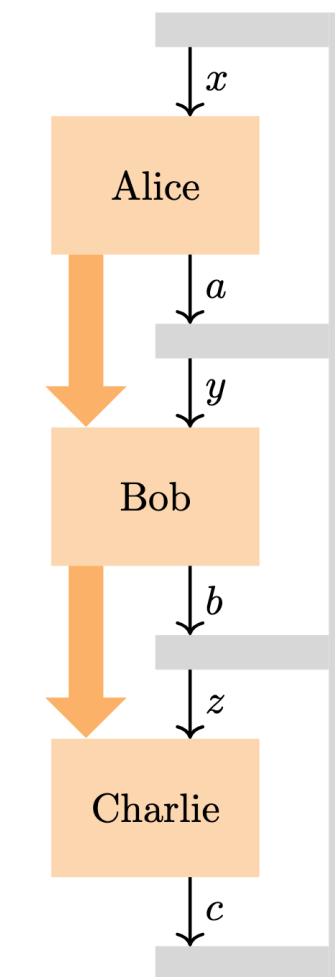
KLVY compiler



Sequential game

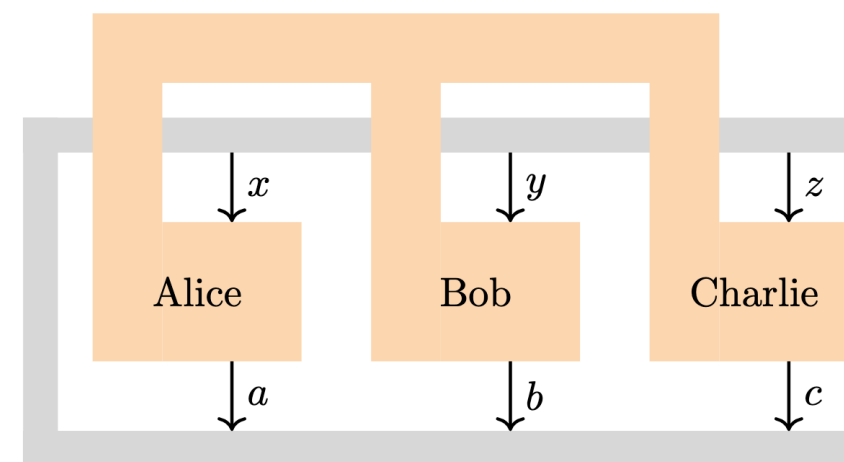


New mapping



+ Stronger constraints from security against efficient provers

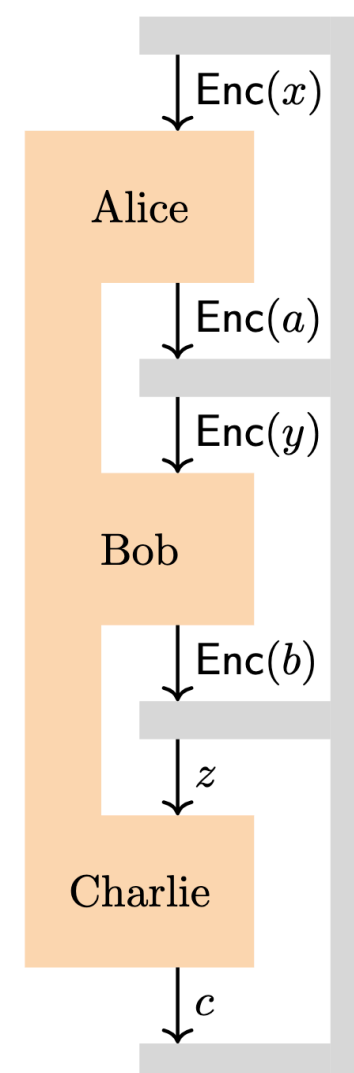
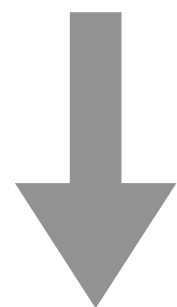
Asymptotic to quantitative: [KPRSTXZ25, BKLRŠTX25],



Non-local game



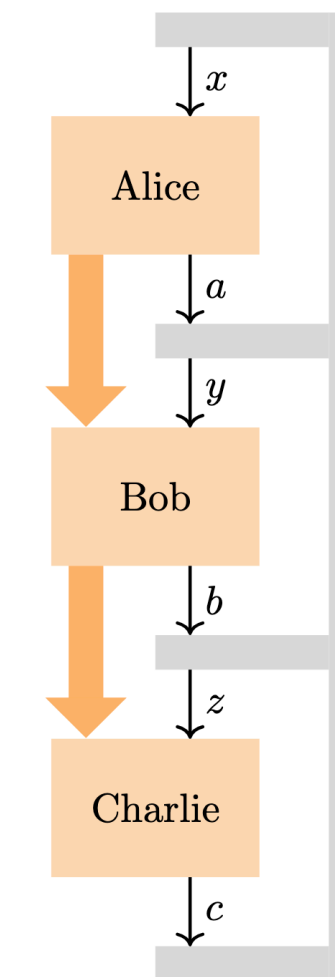
KLVY compiler



Sequential game

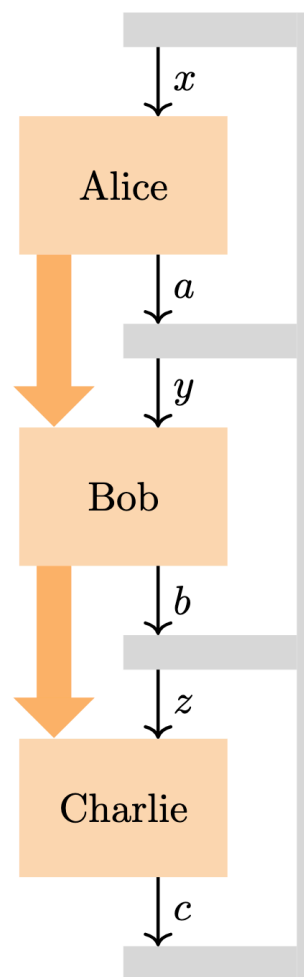


New mapping



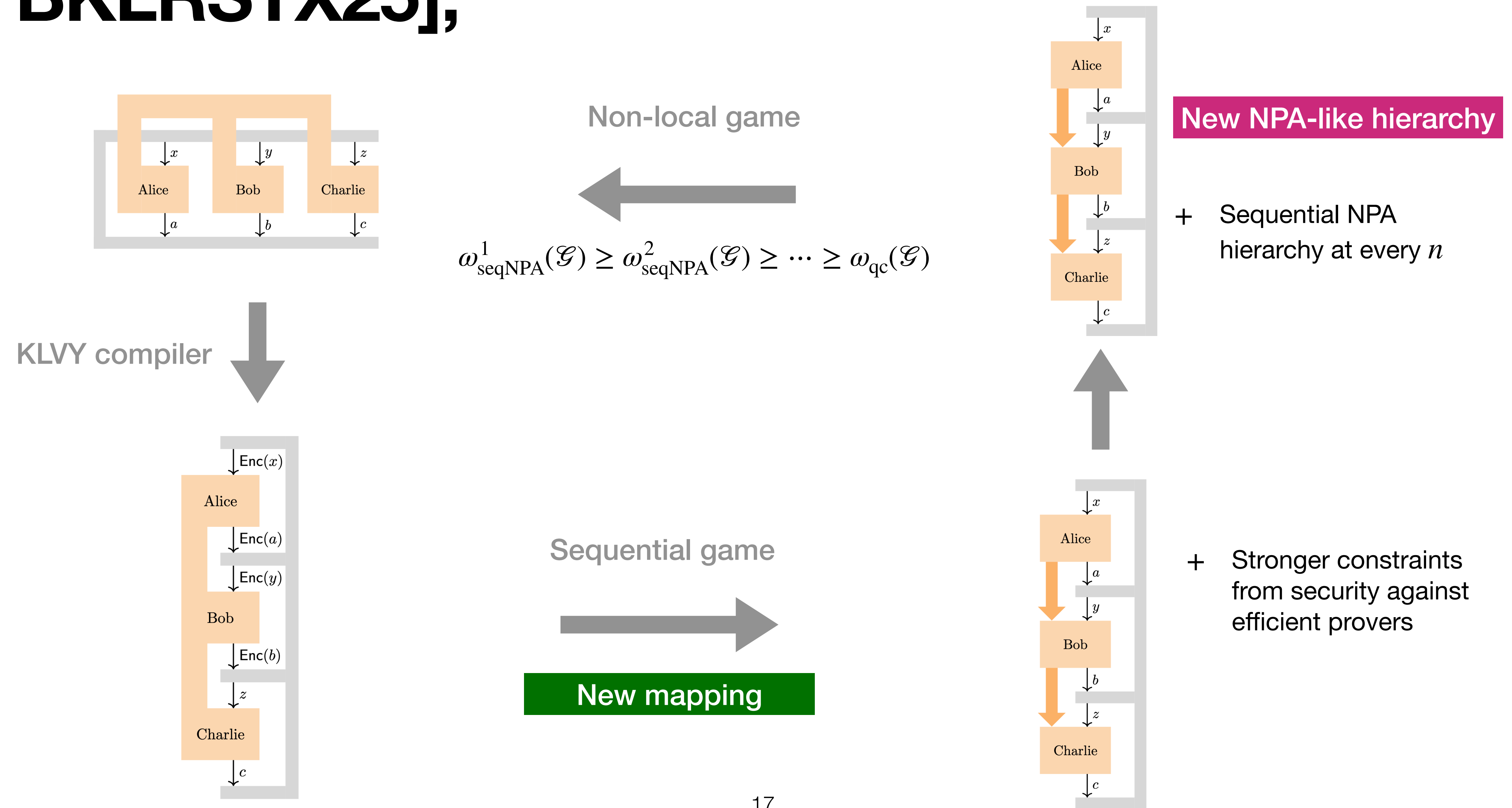
New NPA-like hierarchy

+ Sequential NPA hierarchy at every n

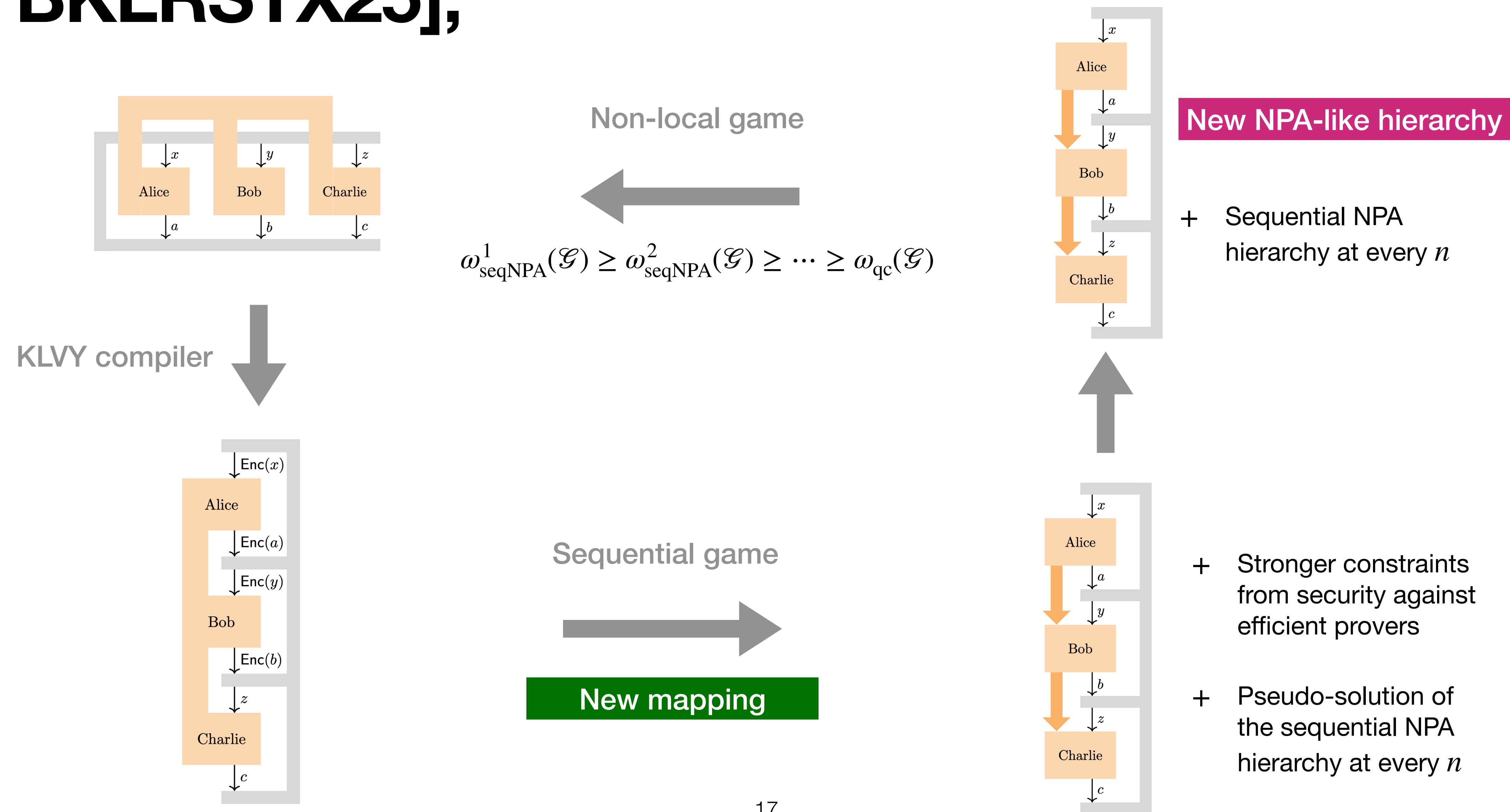


+ Stronger constraints from security against efficient provers

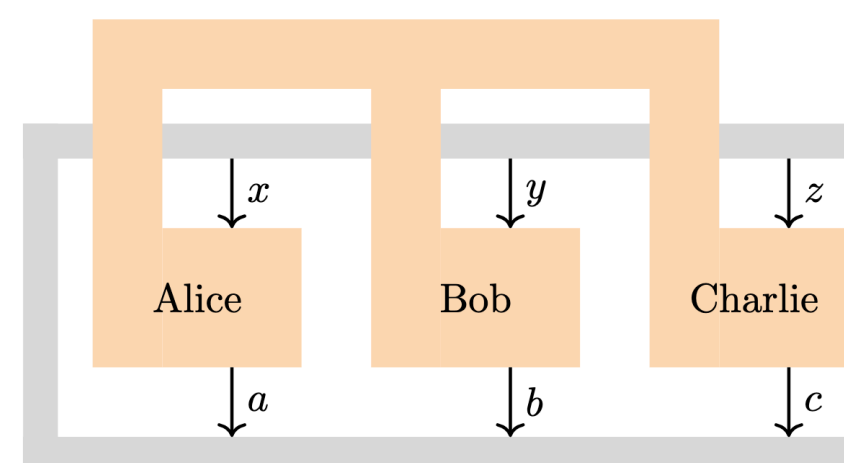
Asymptotic to quantitative: [KPRSTXZ25, BKLRŠTX25],



Asymptotic to quantitative: [KPRSTXZ25, BKLRŠTX25],



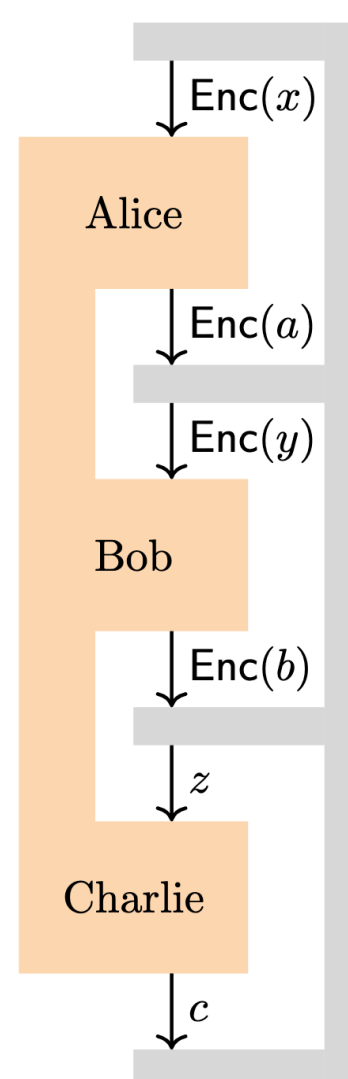
Asymptotic to quantitative: [KPRSTXZ25, BKLRŠTX25],



Non-local game

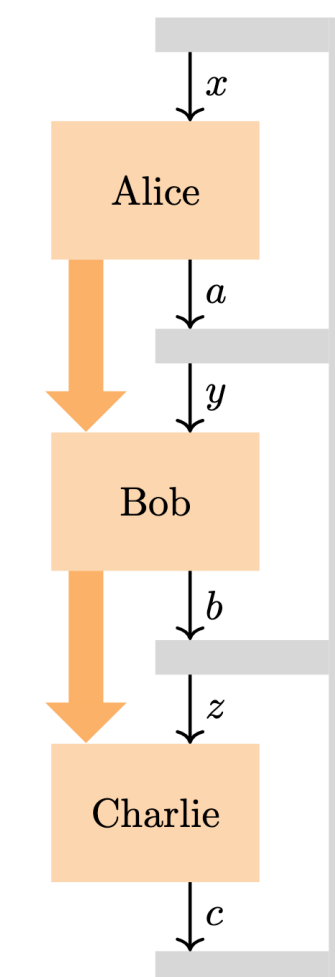
$$\omega_{\text{seqNPA}}^1(\mathcal{G}) \geq \omega_{\text{seqNPA}}^2(\mathcal{G}) \geq \dots \geq \omega_{\text{qc}}(\mathcal{G})$$

KLVY compiler



Sequential game

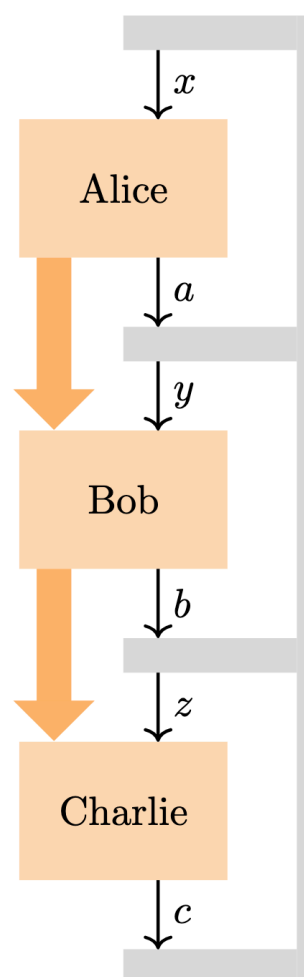
New mapping



New NPA-like hierarchy

+ Sequential NPA hierarchy at every n

New geometric argument:
 $\text{negl}_{S,n}(\lambda)$ -close



+ Stronger constraints from security against efficient provers

+ Pseudo-solution of the sequential NPA hierarchy at every n

Asymptotic to quantitative: [KPRSTXZ25, BKLRŠTX25],

